Adaptive Visual Noise Cancellation for Attention Guidance on a Consumer Passthrough Headset

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A Dissertation

Presented to the University of Dublin, Trinity College

in partial fulfilment of the requirements for the degree of

Master of Science in Computer Science (Augmented and Virtual Reality)

Supervisor: John Dingliana

August 2026

Declaration

I, the undersigned, declare that this work has not previously been submitted as an exercise for a degree at this, or any other University, and that unless otherwise stated, is my own work.

Shreyansh Soni

August 13, 2026

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Extended-reality headsets are normally used to add information to the real world. This dissertation asks the opposite question: can a consumer video-see-through (VST) headset subtract, dim, mute or re-grade the visual periphery and guide attention towards, and hold it on, a user-chosen region of the real world? That question has two halves. The first is whether such a system can be built at all under the constraints of consumer hardware. The second is whether, once built, it actually helps people focus on a real-world task.

The technical half confronts a defining constraint of the Meta Quest 3 platform: the operating system owns the passthrough layer, which applications may neither read nor spatially modify. The dissertation contributes a three-tier design space of diminished reality (DR) under this compositing constraint, and a working reference implementation spanning it. A world-locked *focus window* is rendered as an absence in an overlay sphere, so native passthrough passes through it at full quality, while the periphery carries the manipulation. Two peripheral treatments are carried to confirmatory evaluation: detection-gated salience re-grading (SignPop) and full peripheral occlusion (Hard Dark). A structured prior-art search found no published or shipped precedent for the system's hybrid composite of native passthrough and a shader-processed camera feed. Two on-device measurements validate the architecture's central trade-offs: the camera-free dark modes add no measurable GPU cost over an empty passthrough scene (95th-percentile frame time ≈3.3 ms against a 13.88 ms budget), and the native path through the window resolves a Landolt C acuity chart at least two 0.1-logMAR steps finer than the camera re-render it replaces.

The human half contributes a pre-registered, fully within-subjects user study, built to be falsifiable rather than merely hopeful: a workstation distractor-suppression paradigm carrying the single primary hypothesis, a driving-scene marker-detection paradigm with an equivalence-bounded situational-awareness safety test, and a decision table committed to before the data existed, under which every outcome maps to a pre-committed conclusion. Twenty people were tested, all completing the full two-block protocol. At the completed sample of nineteen workstation and seventeen driving pairs, the primary contrast and the pre-registered eccentricity band reach 79.8% and 89.0% achieved power, two-tailed and matched to the directional hypotheses. Peripheral dimming improved workstation accuracy by 3.07 points (95% CI +0.89 to +5.25), and the pre-registered trial-level test returns an odds ratio of 0.600 (p = .0025), a 40% reduction in the odds of an error. Peripheral awareness showed no measured loss, with the pooled detection measure rising rather than falling, though participants' own reports of awareness cost split close to evenly, and the detection benefit appeared in the 20–30° band that the system's own detection-gated geometry predicts, where the hit rate rose by 20.8 points. Two verdicts are reported as they came out rather than smoothed over. The primary effect, though significant and close to its planned sensitivity, is smaller than its pre-set smallest effect size of interest, so it is claimed as a real but bounded benefit rather than at the pre-set size. The two-sided equivalence statistic behind the safety hypothesis is not established, only because a benefit larger than the margin cannot be excluded, while the one-directional safety question it encodes is answered cleanly.

Acknowledgments

I thank my supervisor, John Dingliana, for direction that repeatedly sharpened the question this work was trying to answer, and for the specific pressure that turned a hopeful demonstration into a study that could fail.

I thank the twenty people who went through the long testing with the headsets, doing a repetitive task twice. The study owes them a huge debt. I thank my friend and roommate Sai Eeshwar Diwakar, together with Diksha Chottani and Yash Jain, who piloted the study before the protocol was frozen and sat through the versions that did not yet work to give me the initial feedback.

I thank the School of Computer Science and Statistics for the facilities and the hardware, and my family and friends for their support through the year.

Shreyansh Soni

University of Dublin, Trinity College
August 2026

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Chapter 1

Introduction

1.1 Motivation: we subtract for space and for sound, almost never for sight

We already delete distraction, and we have done it for a long time, but only in two of the three senses that matter for concentrated work.

For **space**, the instrument is the library carrel. Wooden walls on three sides. Nothing is added to help anybody read. The room is simply taken away until only the page is left, and the design has survived for centuries because it works.

For **sound**, the instrument is the noise-cancelling headphone. It does not make the audio louder. It just makes all the other sounds quieter. It is a product people buy, at a reasonable price. And its entire function is removal, with an architecture of a microphone, a processing stage and a speaker.

For **sight** we have nothing comparable today. What we do have is a description of the *state of mind* such an instrument would produce, and it is the oldest description of it we have, in the Mahābhārata[1]. Droṇa, the martial teacher of the Pāṇḍava and Kaurava princes, sets up a wooden bird in a tree and tells his students to aim for its eye. Before each releases the arrow, he asks what they see. One says the tree. Another the branches and the leaves. Then Droṇa turns to Arjuna and puts the question to him one thing at a time. Do you see the tree? No. The branches? No. Your brothers standing beside you? No. "I see only the eye of the bird." For that moment the tree, the branches, and everyone else standing there have not merely fallen out of Arjuna's interest. They have gone.

That is three thousand years of treating removal as the definition of visual focus,

[1]The Mahābhārata is an ancient Sanskrit epic of India. The archery lesson retold here is one of its best-known episodes.

and the tools we actually build for vision do the opposite. Every visual attention aid in common use adds something to the scene. This dissertation asks what the missing instrument looks like: the carrel and the headphone, applied to sight, on hardware people already own.

The need is not hypothetical. The modern visual environment is engineered against sustained attention. Open-plan offices place moving colleagues at the edge of vision. Shared homes put televisions and phone screens beside workspaces. Streetscapes compete for a driver's gaze with advertising designed by professionals to win it. The cost is well documented: task-irrelevant peripheral events capture attention involuntarily, and each capture carries a performance penalty on the task the viewer intended to do. In immersive settings the effect is directly measurable. Introducing peripheral visual distractors into a virtual classroom more than doubled commission errors, responses to stimuli that should have been ignored, on a sustained-attention task in a recent study of sixty-six participants (Ai et al., 2025), replicating decades of laboratory findings on involuntary attentional capture by peripheral motion and luminance change (Itti et al., 1998).

The extended-reality (XR) community's response has been additive in exactly the way the carrel and the headphone are not: arrows, halos, attention funnels and highlights layered onto the world to point at what matters (Biocca et al., 2006). This dissertation pursues the complementary and far less explored operation, *subtraction*. If the periphery pulls attention away, dim it, mute it, or re-grade it so that it pulls less. In the literature this family of operations is called diminished reality (DR), concealing or attenuating real-world content rather than adding to it (Mori et al., 2017; Cheng et al., 2022), and its application to distraction has recently been named *visual noise cancellation*, by direct analogy with the acoustic kind (Hong et al., 2024). The analogy is also architectural: a headset's camera, processing stage and display mirror a headphone's microphone, processing stage and speaker (Figure 1.1).

A video-see-through (VST) headset is, in principle, the ideal instrument for visual noise cancellation. Unlike optical see-through glasses, whose additive light engines physically cannot darken the world, a VST headset re-displays the world on an opaque screen, so every pixel the user sees is in principle available for modification. The Meta Quest 3, an affordable and widely owned consumer device with colour passthrough, would seem to make DR-based attention guidance deployable at scale for the first time. In principle.

1.2 The compositing constraint

In practice the premise that every pixel is available for modification is false on consumer hardware, and that falsity is the motivating technical problem of this dissertation.

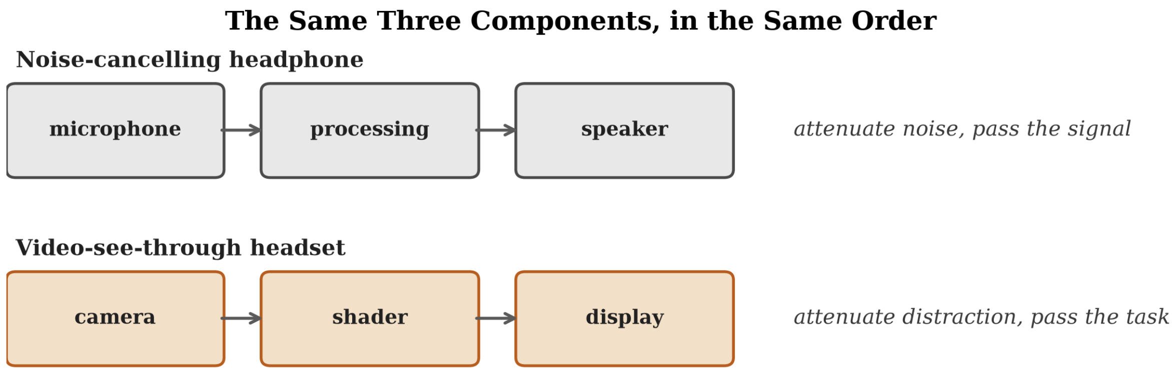

Figure 1.1: The noise-cancellation analogy as architecture: capture, process, re-present. The headphone attenuates noise and passes the signal; the headset attenuates distraction and passes the task.

On the Quest 3 the passthrough view of the world is composited by the operating system, not by the application. An application cannot read the passthrough layer, cannot shade it, and cannot spatially mask it. It may only draw *on top* of it, or apply a global colour transform through a narrow styling interface. Since early 2025 the platform has additionally exposed the raw forward camera feed to applications through Passthrough Camera Access, but at materially lower quality than the system passthrough the user normally sees: a monocular 1280×960 stream covering roughly 85–90° of field of view, against the system layer's full coverage of the headset's approximately 110° display with its correct per-eye reprojection. The one place where full pixel control exists is therefore also the place where image quality is worst. A recent psychophysical study shows the stakes plainly: no current VST headset reaches normal human visual acuity through its passthrough, and the Quest 3 degrades further in low light (Wang et al., 2026).

Subtractive attention guidance on this platform therefore poses a *compositing-rights* problem. The effect one wants, dim everything except the region that matters without degrading the region that matters, cannot be obtained by editing the image the user sees, because no application is permitted to edit it. Chapter 3 shows that what remains is a narrow but genuinely usable design space, organised into three tiers of access, and Chapter 4 presents a working system exploiting an architectural inversion at its centre. The region that matters is rendered as an *absence*, a transparent window in an overlay, so that the highest-quality image on the device, the native system passthrough, serves the focus region for free, while the manipulation is confined to the periphery where acuity is lower and precision matters less.

A mode is one named recipe for manipulating the periphery, and the system implements ten of them. Three matter for this dissertation. **Hard Dark** blacks the periphery out completely. **Soft Dark** dims it partially, leaving events visible through the veil. **SignPop** re-grades a camera copy of the periphery towards dim grey, sparing signal colours, and lifts a confirmed signal lamp or sign to full visibility. Figure 1.2 shows the two evaluated modes, Hard Dark and SignPop, alongside Soft Dark, as the user sees them. The most capable of these configurations composites native passthrough inside the window with a shader-processed camera feed outside it, a hybrid for which a structured prior-art search found no published or shipped precedent.

Figure 1.2: The two evaluated treatments and Soft Dark, seen from the user's viewpoint, from on-device recordings. SignPop is shown in the video test scene it is evaluated in; Soft Dark and Hard Dark are shown over live passthrough of a real room, each with the world-locked focus window revealing untouched passthrough.

1.3 One question, two halves

The dissertation asks a single question, with two distinct halves:

> **Can subtractive attention guidance be delivered on consumer XR hardware? And when it is delivered, does it actually help people focus?**

Part T (technical): can it be built? What degree of diminishment control is achievable on a consumer VST headset without OS-level access to the passthrough layer, and at what cost to image quality, field of view, latency and comfort? Part T is answered by the design space of Chapter 3 and the reference implementation of Chapter 4.

Part H (human): does it work? Under controlled distraction, does peripheral diminishment reduce the cost distractors impose on a focal task, and does salience re-grading enhance focal detection without unacceptably degrading peripheral awareness? Part H is answered by the pre-registered user study of Chapter 5.

Two methodological commitments connect the halves and should be named at the outset, because they are choices rather than accidents. First, the study's dynamic scenario runs as a *simulation*, pre-recorded driving footage presented on the system's rendering sphere rather than live passthrough of a street, following the closest published DR evaluations, which simulated diminishment inside virtual environments because controlled, repeatable, event-dense stimuli cannot be obtained from the uninstrumented real world (Cheng et al., 2022; McLaughlin et al., 2025). Second, detection is supplied by an *oracle* computed offline over the study footage rather than by a model running on the headset, so the study can ask whether an effect built on excellent detection helps attention without first engineering excellent detection on-device.

The two study blocks bracket the feasibility spectrum. The primary, workstation, block runs on *real passthrough with real physical distractors*, the actual artefact with no simulation. The secondary, driving, block evaluates the concept in the simulated deployment context.

1.4 Research questions and hypotheses

The dissertation's question breaks into three research questions. Each confirmatory research question is answered by one or more pre-registered hypotheses, a specific, testable prediction with a named condition and measure. *Pre-registered* here means the hypotheses, margins, tests and decision table were frozen and lodged with the supervisor in a written memo before data collection began (Appendix A, Sections A.3 and A.6). No external registry was used, and the claim made throughout is that these choices predate the data, not that a third party holds them. RQ1 carries the single primary hypothesis (H1). RQ2 carries two secondary hypotheses, a benefit prediction (H2a) and a safety prediction (H2b). RQ3 is exploratory and carries no hypothesis. The full pre-registered forms, including smallest effect sizes of interest, equivalence margins, statistical tests, and a decision table committing the dissertation to a conclusion under every outcome, are given in Chapter 5.

- **RQ1 (primary).** Does dimming the periphery reduce the performance cost that peripheral visual distractors impose on a focal task? The study tests this with the Hard Dark mode and a world-locked focus window, on a workstation task.
- **RQ2 (secondary).** Can salience re-grading improve detection of relevant peripheral events without degrading detection overall? The study tests this with SignPop on a driving scene: a benefit prediction where the re-grading has room to act, and a safety prediction that detection pooled across the whole scene stays within an acceptable margin of the untreated view. The safety prediction is tested by equivalence, so a failure to establish it would carry more information than an ambiguous null. As

it turned out, the two-sided statistic was not established, for a one-directional reason that Chapter 5, Section 5.7.2 and Chapter 6 report as a specification mismatch rather than a safety result.

- **RQ3 (exploratory).** After experiencing the system, what do participants report about perceived focus benefit, perceived awareness cost, and visual comfort?

The study is built so that every outcome maps to a pre-committed conclusion. Distractors are already known to impose a measurable cost (Ai et al., 2025, N = 66). The study takes that cost as given, asks whether the system reduces it, and pre-commits to the interpretation of every outcome, including failure.

1.5 Contributions

- **C1: Design space.** A taxonomy of what DR is and is not achievable under the compositing constraints of consumer VST hardware, with the ceilings and costs of each kind of access, characterised from implementation experience rather than speculation (Chapter 3).
- **C2: Reference implementation.** An open, working multi-mode DR attention-guidance system on Quest 3 passthrough. Its main technical elements are a world-locked focus window rendered as an absence in the overlay, a world-direction sampling scheme that lets the monocular camera feed fuse binocularly, motion-coupled comfort suppression, and ten peripheral treatments spanning the design space of Chapter 3. Two of them, SignPop and Hard Dark, are carried to confirmatory evaluation, with documented failure cases and with the architecture's two central trade-offs, frame cost and legibility, measured on device (Chapter 4). The full source code is publicly available at `https://github.com/batunii/Arjuna`.
- **C3: Pre-registered user study, carried to analysis.** A within-subjects study designed so that any outcome is informative: one primary hypothesis powered close to its design target, a safety check with a pre-set margin on how much peripheral awareness may be lost, pilot checks that gate the design, and a decision table committing the dissertation to a conclusion under every outcome. It is reported with the full analysis of the nineteen workstation and seventeen driving pairs collected, read against that table (Chapter 5).

1.6 Scope: what this dissertation does not claim

Because the credibility of a small, carefully scoped study depends on the discipline of its claims, the non-claims are stated as prominently as the claims.

No driving or road-safety claims. The driving footage used in the secondary study block is a dynamic, high-event-rate monitoring *stimulus*. A seated participant passively watching passenger-perspective video licenses no inference about driving, drivers, or road safety. Such inference would require a driving simulator with vehicle control, which is out of scope.

No product claims. The study evaluates attention mechanisms under controlled distraction, not the desirability or readiness of a wearable product. Comfort findings from a 2026 headset transfer to future lighter hardware only as bounds, and are reported as such.

Oracle conditioning stays visible. SignPop's finding depends on detection, and specifically on offline oracle-quality detection rather than live on-device detection. The gap between the two was never measured, so every place this dissertation reasons about a deployed detector-driven system, that reasoning is conditioned on the oracle assumption and marked accordingly. Oracle performance is never substituted for live performance without saying so.

Two technical measurements, not a benchmark campaign. Chapter 4, Section 4.11 reports two on-device measurements: per-mode GPU frame cost against the display budget, and legibility of the native passthrough path against the camera re-render. No latency, world-locking stability, or thermal-endurance measurement was performed, so where those properties are argued, they are argued from the structure of the system rather than from a measured number.

Analysed sets smaller than the tested sample. The pre-registered twenty participants completed the full two-block protocol. The workstation block lost one pair, P1's, whose arm assignment could not be verified from the session ledger, and the driving block lost three pairs to its response-validity and comprehension exclusions. The final counts are nineteen usable pairs for the workstation block and seventeen for the driving block, at 79.8% and 89.0% achieved power (two-tailed) for the primary contrast and the pre-registered eccentricity band. The pre-committed decision table is applied to this sample, and where a verdict is weaker than the headline number suggests, the weaker verdict is the one reported.

Mode-in-context claims, not mode-generalised ones. Each mode was tested in one scenario: Hard Dark in the workstation block, SignPop in the driving block. Mode and scenario are therefore confounded by design, and nothing in the study separates them.

Every confirmatory claim is a claim about a mode in its context, never about the mode in general.

No attribution to any single effect. A SignPop detection fires a bundle of six visual changes at once, so the 20–30° benefit is attributed to the bundle and not to any one of its components (Chapter 4, Section 4.5.5).

Soft Dark carries no participant data. Of the three modes that matter for this dissertation, only Hard Dark and SignPop were placed in front of a participant. Soft Dark is design-space material throughout, and no empirical claim is made about it.

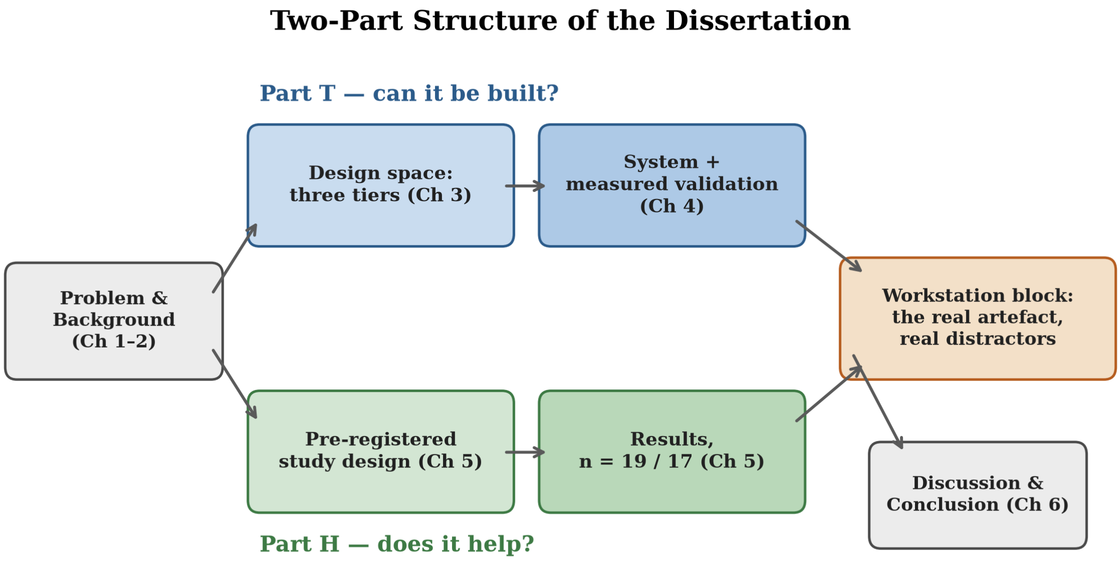

Figure 1.3: The two-part structure of the dissertation.

Figure 1.3 sets out how the two parts of the dissertation fit together, with the workstation block bridging them on real hardware.

Chapter 2

Background and Related Work

This chapter follows one line of reasoning rather than surveying bodies of literature side by side. Is the problem real, what has already been tried, has removing things instead of adding them been tried and did it work, can any of this be built on hardware people actually own, and how should the result be tested. The review is selective on that basis: it includes work that establishes the problem, backs a value this system uses, or supplies a method the evaluation adopts, and it leaves work that merely neighbours the topic to the surveys that cover it.

2.1 Why the Periphery Costs Something

The computational account of visual attention underpinning this dissertation is the salience-map model of Itti et al. (1998): attention is drawn to locations whose centre–surround contrast, in intensity, colour opponency, or orientation, differs most from their neighbourhood. Three working assumptions, adopted from this model, recur throughout the design. First, the model treats *contrast as the master variable*, so any manipulation raising local contrast should attract gaze and any manipulation flattening it should repel gaze. Second, salience is channel-specific and the channels are not equal: in every head-to-head comparison in the guidance literature, luminance manipulations outperform chromatic ones (Bailey et al., 2009; Grogorick et al., 2017). This is why every mode in this system dims before it does anything else. Third, salience operates *pre-attentively*, so capture by a high-contrast peripheral event happens before, and regardless of, the observer's intentions. On this account a periphery whose salience has been flattened should stop issuing capture events, which is the mechanism the dark modes aim to exploit. Duan et al. (2022) extend the account to augmented reality and show that the opacity of AR content directly modulates where attention lands, which makes opacity a dose rather than a style and the attenuation axis of Chapter 3 an axis with a middle rather than an on/off switch.

Whereas salience explains *how* the periphery captures attention, Cognitive Load Theory explains why that capture is costly. Extraneous load, the processing demanded by information irrelevant to the task, competes for the same limited working-memory resources as the task itself. Buchner et al. (2022) review the evidence that AR can reduce, not only add, cognitive load when used to structure or filter what the user must process. This is the theoretical justification for the whole project: peripheral diminishment is an extraneous-load intervention. It does not make the user smarter or the task easier. It removes competing processing demands at the perceptual source.

That framing also identifies the project's central risk, which is why this dissertation pairs a benefit hypothesis with a safety hypothesis rather than testing benefit alone. Attention capture by peripheral events is not a design flaw of the human visual system. It is a safety mechanism. Suppressing it trades focus against situational awareness, a tension documented empirically in the DR literature (Murph et al., 2021; McLaughlin et al., 2025) and formalised in Chapter 5 as an equivalence-tested safety hypothesis rather than an afterthought. The evaluation is designed to measure benefit and cost together.

What was missing until recently was the distractor-cost baseline in immersive settings. Ai et al. (2025, $N = 66$) supply it: peripheral visual distractors in a virtual classroom significantly increase both commission errors ($1.33 \rightarrow 3.15$) and omission errors on a Go/No-go continuous performance task, while leaving mean reaction time unchanged. That single result does double duty here: it establishes that *errors, not latency*, are the sensitive currency of distraction cost, fixing the primary dependent variable of Chapter 5, and it quantifies the effect the vignette must reduce for the primary hypothesis to be supported.

2.2 Guiding Attention by Adding

The techniques for steering attention in a mediated view form a spectrum from imperceptible to overt, and what matters here is what the additive approach costs.

The subtle gaze direction family originates with Bailey et al. (2009): a brief luminance modulation in the viewer's periphery attracts a saccade, and the cue is terminated the moment gaze moves toward it, so the viewer never foveates the modulation and remains unaware of being steered. Luminance modulation outperformed warm–cool chromatic modulation, an early instance of the luminance-dominance regularity described in Section 2.1. Two results from this family set hard constraints on the present design. Grogorick et al. (2018) compared five guidance methods in an immersive dome with $N = 102$: local magnification achieved the best guidance-per-noticeability ratio, while whole-periphery blur was the least subtle and actively irritated 59 of 102 participants. Their conclusion that no method remains completely unnoticed, together with the blur-irritation finding, decided two things here. Whole-periphery blur is not carried forward as an evaluated mode, and the design accepts noticeability rather than chasing imperceptibility. Second, Erickson et al. (2023) found hue shifts the most confusing of hue, saturation and value, which is why every mode manipulates saturation and luminance and none shifts hue.

Previous work suggests slow ramps can escape the noticeability trade-off. Hata et al. (2016) showed that a visual change introduced gradually enough stays below the awareness threshold while still biasing gaze, because the guidance threshold sits below the awareness threshold. This is the perceptual justification for the system's formation animation, where effects ramp in over seconds rather than switching on.

Where subtle gaze direction adds a transient stimulus, saliency *modulation* re-grades the image itself. This is the direct technical lineage of this system's colour- and saliency-gated re-grading mode (ColorPop). The foundational demonstration is Veas et al. (2011), who raised salience in a focus region and suppressed it in context regions of video imagery using an Itti-style conspicuity model. Modulated video was statistically indistinguishable from unmodulated video for naive viewers yet produced faster first fixations on target regions, which remains the strongest evidence that video-mediated reality can be re-graded imperceptibly with measurable attentional consequences. A passthrough headset is exactly video mediation. Kalkofen et al. (2007) contribute the focus-and-context vocabulary used throughout: keep a focus region maximally legible while context is compressed but not destroyed.

The modern anchor is Sutton et al. (2022), saliency modulation of the *real world* viewed through AR glasses, implemented as saturation adjustment plus sigmoidal contrast remapping and deliberately avoiding hue shifts. Modulation drew significantly more fixations to target regions but, an honest and important negative, was *not* imperceptible. Against an overt circle cue, modulation's advantage was that it preserved natural scene exploration where explicit markers glue gaze to themselves. This dissertation's ColorPop implements Sutton et al.'s exact sigmoidal contrast recipe on its salient-colour channel, inheriting the philosophy: accept visibility, preserve exploration, re-grade rather than annotate. The critical difference is platform. Sutton et al. built a bench-mounted optical see-through rig whose additive optics can only add light. The present system runs on consumer video see-through hardware with full subtractive control of its rendered layers.

An eye-tracked comparison in cinematic VR speaks to which *diminishment* channel works. Comparing area darkening, context-based darkening and desaturation for steering attention in 360° video, area darkening guided attention most effectively while desaturation alone had almost no guidance effect (Wang et al., 2020). It is a single result from one small-venue study rather than a consensus finding, but it corroborates this project's mode hierarchy: the dark modes are the guidance workhorses, and desaturation appears only as one ingredient inside ColorPop's compound re-grading, never as a mode of its own.

At the overt end, the Attention Funnel (Biocca et al., 2006) establishes that aggressive guidance pays in speed what it costs in scene engagement. Its failure mode is *attentional tunnelling*, excessive allocation of attention to guided content at the cost of neglecting the surroundings, isolated experimentally as a function of task demand (Syiem et al., 2021). That is the risk most relevant to Hard Dark, and the reason its evaluation measures peripheral awareness rather than assuming it.

2.3 Guiding Attention by Removing

Mori et al. (2017) provide the canonical survey and taxonomy of diminished reality, the family of techniques that conceal, eliminate, or see through real objects in a mediated view. Their survey establishes both the field's legitimacy and its historical centre of gravity: most classical DR work concerns object removal, inpainting a region so the object appears gone, which is computationally demanding and semantically targeted. The present work sits in a different corner of that taxonomy. It does not remove objects but *attenuates regions*, trading semantic precision for robustness, generality and real-time feasibility on mobile hardware.

Cheng et al. (2022) supply the field's most direct study of how users *want* reality diminished. Across two studies (N = 16 and N = 12, conducted inside VR simulations of DR because no deployable DR hardware existed) they found participants preferred partial opacity reduction over complete removal, and wanted context-preserving diminishment, outlines or residual structure indicating that something is there even when its detail is suppressed. Both findings are load-bearing. The preference for partial attenuation fixes the opacity ceiling of this system's partial peripheral occlusion mode (Soft Dark), and the context-preservation finding motivates treating Hard Dark's total blackout as the *extreme* of the design space rather than its default. Equally load-bearing is their method: they could only simulate DR inside VR, where this dissertation runs on the real passthrough hardware their participants could only imagine.

The most recent conceptual frame for this family is *visual noise cancellation*: by analogy with acoustic noise cancellation, a head-worn display actively moderates the visual environment, attenuating noise while passing signal (Hong et al., 2024). The analogy is productive because it imports an architecture. Acoustic cancellation requires a microphone, a processing stage, and a speaker, which is precisely the camera → shader → display pipeline of a video-see-through headset. Against the prior art surfaced by the structured search of Section 2.5.1, no software-defined visual noise cancellation system on consumer video-passthrough hardware was found, and this dissertation can be read as one, where each mode is a cancellation profile and the profiles differ in what they classify as noise. The analogy also imports a useful discipline from its original: noise-cancelling headphones are evaluated on measured attenuation *and* on what they cost the wearer in awareness, which is exactly the hypothesis pairing of Chapter 5.

The empirical case that diminishment helps focus is recent but consistent. McLaughlin et al. (2025) provide the closest published analogue to this dissertation's primary hypothesis: across two within-subject experiments, universal DR-style attenuation of distractors, the condition structurally closest to this project's peripheral vignette, produced fewer errors ($\beta = -0.37$, $p = .021$) and lower subjective workload than context-aware removal. Their trial-level mixed-effects analysis is adopted directly by this dissertation's analysis plan, and their measurement of awareness of off-task objects, finding the focus/awareness trade-off real but manageable, is the empirical ancestor of this work's safety hypothesis. Lee and Kim (2025) approach the same goal from the object side: AR-camouflaging a single distracting object restored working-memory performance to a level statistically comparable with physically removing it (N = 60). Murph et al. (2021) document the same benefit–risk structure in a medical assembly context, where DR lowered cognitive workload but risked situational-awareness loss, and in later work describe gradually reintroducing distractions as tolerance grows (Murph et al., 2022), which is the source of this system's temporal fade transitions.

Two 2026 threads confirm both the demand for DR-for-attention and its immaturity. A co-design study with fifteen students with ADHD developed the concept of attentional diminishment but built no working prototype (Noh et al., 2026), and a CHI 2026 study on Apple Vision Pro obscured specific individuals to alleviate social discomfort, which is overlay-based diminishment on consumer hardware but targeted occlusion rather than peripheral guidance (Zhang et al., 2026). Neither implements peripheral diminishment for attention on real passthrough. Further back, FocalSpace (Yao et al., 2013) demonstrated the 2D ancestor of this project's workstation scenario: synthetic background blur in video conferencing, improving memory for meeting content.

2.4 What the Periphery Tolerates

The perception literature describes peripheral vision not as blurry central vision but as a differently tuned system, one that notices contrast loss and motion change most, and chromatic detail loss least. One perceptual finding is directly responsible for a parameter in this system: Patney et al. (2016) established that peripheral blur is detected primarily through the *contrast reduction* it causes, and that restoring local contrast after blurring roughly doubles the tolerable blur radius. That chain also offers a mechanistic explanation for the irritation finding of Grogorick et al. (2018), since naive whole-periphery blur announces itself through contrast collapse. It is why the dark modes attenuate luminance, which the periphery tolerates better, rather than blurring structure.

A second literature approaches the periphery from the comfort side, and the manner of restriction matters more than its degree. Fernandes and Feiner (2016) showed a soft, dynamically applied field-of-view vignette reduces discomfort in VR locomotion without measurable presence loss. But Norouzi et al. (2018) found that vignetting *coupled to the user's own head motion* actually **increased** sickness relative to no vignette, a directly actionable negative result that justifies this system's inverted motion policy, where effects fade *out* during head motion and restore during stability. Cao et al. (2021) found granulated rest frames outperform opaque black restrictors on search performance while preserving peripheral awareness, the source for this system's world-locked static-grain variant (GranulatedPeriphery). Barhorst-Cates et al. (2016) contribute the sobering datum at the far end: spatial learning survives restriction down to a 10° field, but anxiety is elevated in *all* restricted conditions and rises monotonically as the field narrows. Waldner et al. (2014) map the comfort envelope for the flicker channel, giving the parameters this project adopts for its detection-highlighting pulse rather than anything in the photosensitivity band.

Collectively this draws the corridor the system must fly through. Peripheral attenuation is tolerable and even beneficial *if* it is soft-edged, decoupled from head motion, and mindful that total blackout carries a measurable anxiety cost. Those constraints are visible in the final system as the dark modes' graduated opacity ceilings, every mode's motion-triggered suppression, and the evaluation's insistence on measuring peripheral awareness alongside benefit. One honest gap remains, inherited rather than resolved: no study measures the comfort of peripheral *desaturation* specifically over sessions longer than ten minutes.

2.5 What Consumer Hardware Permits

Everything above presumes the ability to manipulate what the user sees of reality. On consumer hardware in 2026 that ability is narrowly rationed, and Chapter 3 develops the rationing in full as a three-tier design space. Two literature points attach to it here. The one published feasibility study of on-device camera compositing projects 720p30 with thermal throttling within five to ten minutes, a simulation-based estimate rather than a measured run, which functions as a warning about sustained camera processing (Laghari et al., 2025). And Varjo's research headsets grant full camera control, yet this review found no published study that shader-processes their passthrough region-selectively for attention. The question appears to be unexplored so far, although work not surfaced by the search may exist.

One further platform datum constrains not the system but its *evaluation*: psychophysical measurement shows that no current video see-through headset reaches normal human acuity at any light level, with Quest 3 significantly worse in low light (Wang et al., 2026). Any evaluation task requiring fine real-world detail read through passthrough therefore confounds the manipulation under test with the medium's acuity ceiling. This drives two decisions in Chapter 5: the focal task uses large look-alike shapes whose legibility through passthrough is verified by a pilot gate, rather than anything requiring fine detail, and room lighting is bright and constant across sessions.

2.5.1 Prior systems, and a documented search

Every constituent mechanic of this architecture has a documented precedent, and none of the precedents combine them. QuestCameraKit demonstrates camera-fed materials with GPU shaders (Coviello, 2025), but always as content *within* the scene rather than a full-surround treated periphery. Meta's Passthrough Transitioning motif uses a head-centred sphere whose transparency reveals system passthrough (Meta, 2024b), but the sphere carries virtual content for scene fades rather than a processed camera feed. Alpha punch-through to the passthrough underlay is standard technique (Meta, 2024d), shipped commercially by Immersed's Passthrough Portals (Immersed, 2022), but the hole is always surrounded by *virtual* workspace rather than by re-rendered reality. A peripheral dimming vignette shipped in Quest OS v67, but explicitly not over passthrough (Meta, 2024a). Two patents describe adjacent ideas with no evidence either was implemented (Patents US 12548271 B2 and US 12524072 B2, 2026).

A structured prior-art search conducted for this dissertation on 5 July 2026, comprising 26 web queries and 4 GitHub API queries across ISMAR, CHI, UIST, IEEE VR, DIS and arXiv from 2022 to 2026, Meta developer documentation, all 136 public forks of the PCA samples repository, XR trade press, and the visionOS and Varjo API surfaces, found **no publication, product, or open-source project that composites system passthrough with a live shader-processed camera feed in a single view, for any purpose**, and none that implements peripheral passthrough manipulation for attention guidance on a standalone consumer headset. The full query protocol is archived in the project record. The nearest systems are the five named above, each contributing one ingredient. Chapter 3, Section 3.5 states the resulting novelty claim, scoped precisely to the composite rather than to DR on passthrough in general.

2.6 Evaluating Attention Guidance in XR

The evaluation norms of this field are within-subject designs with modest samples and many repeated measures. Caine (2016)'s census of CHI found the modal sample size to be twelve, with in-person studies drawing their power from paired comparisons and trial-level repetition rather than headcount. For non-normal aggregates the HCI norm has been the aligned rank transform (Wobbrock et al., 2011), though recent critique of its error properties (Tsandilas and Casiez, 2024) motivates this dissertation's preference for mixed models with Wilcoxon fall-backs.

Quest 3 carries no eye tracker, so the measures must come from behaviour and head telemetry, and the literature validates both channels. Reaction-time probe detection is standardised as the Detection Response Task (ISO, 2016), and the Peripheral Detection Task places its probes at 11–23° eccentricity (Jahn et al., 2005), just outside the roughly 8° road-centre region where driver gaze concentrates (Victor et al., 2005). That lineage fixes the eccentricity bands of Chapter 5. Head orientation is a validated proxy for gaze at the eccentricities that matter here: Higgins et al. (2022) validated head pose as a gaze surrogate for object-level attribution in VR, and Sitzmann et al. (2018) showed with concurrent head and eye tracking that fixations concentrate at low head velocity. Simulator sickness is measured with the nine-item VRSQ (Kim et al., 2018), administered before *and* after exposure because post-only scores are uninterpretable without a baseline (Brown et al., 2022). The study administers no per-condition workload instrument, and Chapter 6 records what that costs. Finally, equivalence claims of the "no meaningful loss of peripheral awareness" kind require dedicated machinery: the two-one-sided-tests procedure with a pre-registered margin (Lakens, 2017), which converts a would-be null result into a falsifiable claim.

2.7 Gaps and Position

Read together, the literature above leaves five identifiable gaps, three of which this dissertation addresses in whole or part. Table 2.1 states each gap and what this work does about it.

Table 2.1: Research gaps identified by the review, and the standing of each in this dissertation.

#	Gap	Status in this dissertation
1	**No unified multi-mode DR system.** Guidance techniques and diminishment effects have been studied singly. No prior work combines multiple DR guidance modes in one configurable system for comparison within one design.	Addressed: ten modes built in one configurable system spanning the design-space tiers of Chapter 3, of which two are carried to confirmatory comparison within one study.
2	**No hybrid of system passthrough and app-processed camera feed in one view.** Colour-only global restyling of system passthrough exists. Full pixel control of an inferior camera feed exists. No prior work composites the two, and none implements peripheral passthrough manipulation for attention on a standalone consumer HMD (documented search, Section 2.5.1).	Addressed: the window-as-absence hybrid architecture of Chapters 3 and 4, the dissertation's strongest and most precisely scoped novelty claim.
3	**No temporal transition mechanics.** Gradual formation, easing, and motion-contingent suppression of DR effects have not been formally parameterised or studied, despite adjacent evidence that ramps evade noticeability (Hata et al., 2016) and that head-coupled strengthening harms comfort (Norouzi et al., 2018).	Partially addressed: formation ramps and inverted motion-coupling are implemented and parameterised in Chapter 4 and grounded in the cited evidence. Their isolated evaluation remains future work.

#	Gap	Status in this dissertation
4	**No user-controlled DR intensity.** User-initiated, graded release and re-application of diminishment is unstudied.	Not addressed, groundwork only. The trigger-painted window gives users spatial but not intensity control.
5	**No dual-channel DR technique.** Desaturation and blur have been studied separately, never combined and parameterised as a compound diminishment channel in AR attention guidance.	Not addressed as a confirmatory question. The compound channel exists inside ColorPop, but the study tests the unconfounded dark modes confirmatorily and leaves channel decomposition to future work.

Two published studies stand closest to this dissertation, and the delta from each defines its contribution. **Cheng et al. (2022)** established what users want from diminishment, but inside VR simulations, because deployable DR did not exist. Theirs is a perception-and-preference study of imagined systems. **McLaughlin et al. (2025)** established that distractor attenuation improves task performance, but likewise inside a virtual task environment, with the diminishment simulated by the VR scene itself. Both are simulation studies *of necessity.* This dissertation contributes the piece both were missing: a working diminishment system on real consumer passthrough hardware, evaluated with a design that pairs a real-hardware block against a simulation block within the same participants. If the two blocks agree in direction, the simulation methodology of Cheng and McLaughlin gains a real-hardware anchor. If they diverge, the divergence localises what simulation misses. Either way the evaluation assembles the field's strongest methodological norms, for the first time, around a deployable diminished-reality attention system.

One further applied result matters to that evaluation, because it uses driving *footage* as a stimulus context. AR cueing of roadside hazards improves response time *without* degrading detection of non-target objects (Rusch et al., 2013, N = 27), an early demonstration that guidance benefit and situational-awareness cost are separable, which is the separation this dissertation's secondary hypotheses formalise. That literature motivates the scenario aesthetics and supplies the salience-engineering precedent for ColorPop's red, orange and green hierarchy. It licenses nothing further: per Chapter 1, no claim about driving, drivers, or road safety is made anywhere in this dissertation.

Chapter 3

A Design Space for Diminished Reality under Compositing Constraints

3.1 The Compositing Constraint

Chapter 2 established that diminished reality (DR), the selective removal or attenuation of real-world visual information (Mori et al., 2017), has a substantial conceptual literature and a thin implementation record on consumer hardware. This chapter asks the technical half of the dissertation's question: what degree of diminishment control is actually achievable on a consumer video-see-through (VST) headset, without operating-system-level access to the passthrough layer?

The answer is not a single number but a structured design space, and this taxonomy is Contribution C1. The central claim is that on a platform such as the Meta Quest 3, the feasibility of any DR effect is decided almost entirely by **which layer of the compositing stack an application is permitted to touch**, and that these permissions stratify into three tiers with sharply different capabilities and costs.

On Quest 3 the reconstructed view of the real world that the user experiences, the system passthrough, is produced and owned by the operating system. An application **cannot read, modify, or subtract from the passthrough layer. It can only composite content on top of it.** The passthrough image is never exposed to application shaders as a texture. It is composited *behind* application content by the OS compositor, at a stage of the pipeline applications do not reach. The naive formulation of subtractive DR, sample the passthrough pixel, darken it, write it back, is therefore not merely difficult on this platform. It is architecturally impossible.

This is a deliberate privacy and safety boundary rather than an incidental limitation, and it is shared in stronger form by the other major consumer platform. On Apple Vision Pro camera pixels are unavailable to consumer applications entirely, since raw camera access requires an enterprise-only entitlement incompatible with App Store distribution, and the only world-appearance control offered is a global, non-spatial surroundings dimming effect (Apple, 2024). On true optical see-through (OST) glasses the constraint is harsher and physical rather than administrative. An additive display can only *add* light to the optical path. It can superimpose brightness but cannot remove photons arriving from the world, so darkening is unavailable without auxiliary hardware such as per-pixel dimming layers (Itoh et al., 2021).

The motivation for the boundary predicts its persistence. The passthrough image is a continuous camera view of the user's home, workplace, and bystanders. Granting arbitrary applications read access to it is a privacy exposure, and granting write access is a safety exposure, because an application could imperceptibly alter the user's view of the physical world they are walking through. Both concerns seem likely to strengthen rather than weaken over time, so research that assumes future read-modify-write access to system passthrough is betting against the platform vendors' incentives. This dissertation makes the opposite bet. It treats the constraint as permanent and asks what can be built inside it, which is what gives the resulting design space its claim to durability.

Subtractive attention guidance must therefore be *synthesised* on consumer VST hardware through whatever compositing rights the platform does grant. Figure 3.1 shows the three layers of the stack and which of them each tier of access is permitted to touch.

3.2 The Three-Tier Access Taxonomy

The Quest 3 grants applications three distinct kinds of access relevant to DR, summarised in Table 3.1.

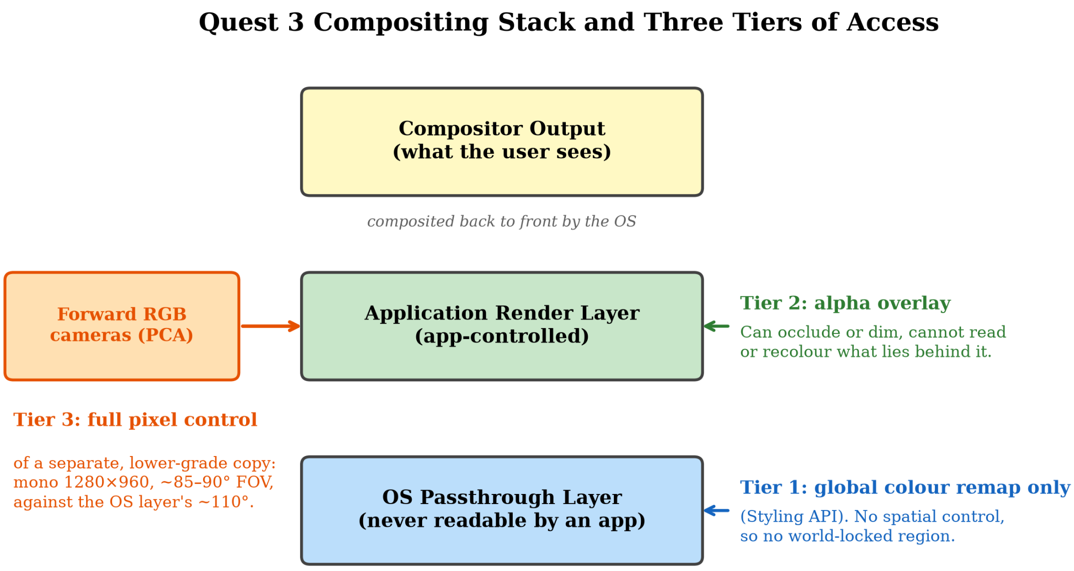

Figure 3.1: The Quest 3 compositing stack and the three tiers of access.

Table 3.1: Diminished-reality capability by compositing-access tier on Quest 3.

Tier	**Access mechanism**	**Achievable DR**	**Ceiling / cost**	**Modes located here**
1: OS passthrough styling	`OVRPassthroughLayer` Styling API: brightness / contrast / saturation scalars, 3D colour look-up tables, edge tint	**Global colour remapping only.** Whole-layer desaturation, tinting, posterisation	No blur. No per-pixel or spatial masks. No world-locked regions. Layer can never be read. Surface-projected passthrough deprecated at SDK v83	Not applicable (bounding case)
2: Overlay compositing	Application draws alpha-blended geometry in front of the passthrough layer	Dimming, blackout, occlusion. A shaped *absence* of overlay passes OS-native passthrough through untouched	Can only attenuate or occlude reality. Cannot recolour, blur, or filter it	Soft Dark, Hard Dark, Granulated-Periphery, SpotLift. The focus window itself

Tier	Access mechanism	Achievable DR	Ceiling / cost	Modes located here
3: Camera re-render (PCA)	Passthrough Camera Access: the raw forward camera feed as a GPU texture, with intrinsics and pose	Arbitrary per-pixel manipulation: blur, desaturation, salience re-grading, glare compression	Mono 1280×960 at ∼85–90° horizontal FOV vs the ∼110° OS view. Added latency. Binocular-fusion burden (Chapter 4). Sustained on-device processing is thermally bounded	SignPop (evaluated, built on the ColorPop pipeline), ColorPop, Blur (superseded), Chromatic-Cool, Conspicuity-Squeeze, Outlined-Dark

3.2.1 Tier 1: the styling ceiling

Tier 1 needs precise description, because some adjustment of the system passthrough layer is possible and overlooking it would overstate the constraint. Meta's Passthrough Styling API allows an application to apply brightness, contrast and saturation adjustments, full 3D colour look-up tables, and edge tinting to the layer as a whole (Meta, 2024c). A global desaturation of the entire visual field, one conceivable DR primitive, is achievable at Tier 1 without touching a camera pixel.

What Tier 1 categorically lacks is **spatial control**. Every styling operation applies to the whole layer uniformly. There is no mechanism for per-pixel masks, gradients, world-locked regions, or any effect that treats one part of the visual field differently from another. The one historical exception, surface-projected passthrough, allowed passthrough to be projected onto application-defined geometry with per-surface styling. It was deprecated at SDK v83 and is unavailable going forward. Attention guidance is *definitionally* spatial, since its entire purpose is to treat the focus region differently from the periphery, so Tier 1 alone cannot implement any mode in this dissertation. **The absence of spatial control, not the absence of control, is the finding.**

3.2.2 Tier 2: diminishment by occlusion

Tier 2 is ordinary alpha compositing. The application draws translucent or opaque geometry in front of the passthrough layer. Because the VST display is an opaque screen whose every pixel the compositor ultimately controls, an overlay can darken the apparent world without limit, up to and including full blackout, which is physically impossible on additive OST optics. The constraint runs the other way: a Tier-2 overlay knows nothing about the pixels behind it, so it can attenuate or occlude reality but cannot recolour, sharpen, blur, or respond to its content.

Two modes live entirely at Tier 2. **Soft Dark** composites a uniform black overlay at up to 75% opacity across the periphery. **Hard Dark** takes the same overlay to full opacity. Both leave a shaped hole, the world-locked focus window, through which untouched OS passthrough remains visible.

That hole is the load-bearing architectural insight of the whole system, and it is worth stating in its general form. **At Tier 2, the highest-fidelity rendering of reality is an absence of rendering.** Inside the focus window nothing is drawn at all, so the user receives OS passthrough at its full native quality, full $\sim 110°$ field of view, and zero added latency, precisely where they are being asked to direct their attention. The manipulation is confined to the periphery, where visual acuity is lowest and quality loss matters least. This inverts the naive design of re-rendering the whole view from the camera feed and accepting camera-grade quality everywhere, and it is why the system escapes the passthrough-acuity ceiling exactly where task performance depends on seeing well. Alpha punch-through as a mechanism is documented platform practice (Meta, 2024d) and shipped commercially in Immersed's Passthrough Portals (Immersed, 2022), but in all found prior uses the hole reveals passthrough *surrounded by virtual content*. Using the hole as the focus region of a diminished periphery is, per Section 3.5, unprecedented.

3.2.3 Tier 3: full pixel control at a price

Tier 3 is the Passthrough Camera Access (PCA) API (Meta, 2025), which since early 2025 exposes the headset's forward RGB camera feed to applications as a GPU texture together with camera intrinsics and pose. At Tier 3 the application is no longer manipulating a proxy for reality. It holds camera pixels and can apply arbitrary per-pixel computation: Gaussian blur, luminance-weighted desaturation, hue-selective salience re-grading, glare compression.

The price is paid in four currencies. **Resolution and field of view**: the accessible feed is a single mono stream at 1280×960 covering roughly 85–90° horizontally, against the OS passthrough's $\sim 110°$ stereo view, so a Tier-3 re-render is categorically lower-fidelity than what the user would otherwise see. **Latency**: the application-side pipeline of camera capture, texture upload, shader, and display adds delay the OS passthrough path, with its dedicated reprojection, does not exhibit. **Binocular fusion**: the feed is monocular, and presenting it convincingly to two eyes is non-trivial, since the naive approach measurably fails (Chapter 4). **Thermal budget**: sustained on-device processing of the camera stream is bounded. The only published work to quantify this class of hardware for PCA-style live compositing is a simulation-based feasibility study projecting 720p30 operation with thermal throttling after five to ten minutes (Laghari et al., 2025). That figure is projected rather than measured, since no benchmark ran on physical hardware, but it means Tier-3 modes are on today's hardware modes for sessions rather than workdays.

3.2.4 The hybrid: composing Tier 2 and Tier 3

The tiers are not mutually exclusive, and the most capable configuration this work demonstrates is a composite. The PCA camera feed, shader-processed, is rendered onto a head-centred sphere as the periphery at Tier 3, with the world-locked focus window left transparent so native OS passthrough shows through the hole by Tier 2's absence trick. The user experiences full-fidelity reality inside the window, seamlessly surrounded by a re-graded interpretation of reality outside it. This exploits the quality and field-of-view asymmetry between the two layers rather than suffering it. Figure 3.2 separates the composite into its layers and shows the single view they fuse into. The implementation is Chapter 4's subject.

The composite itself was never placed in front of a participant. Hard Dark was evaluated at the workstation and SignPop in the video scene, so no session put a shader-processed camera periphery around a native passthrough window. The composite is validated on the device rather than on people (Chapter 4, Section 4.11), and no claim here rests on how it is experienced.

3.3 What Is Impossible, and Why It Matters

A design space is defined as much by its walls as by its rooms. Three impossibilities structure everything above.

- **No read access, at any tier, to the pixels the user actually sees.** Even Tier 3 supplies a different, lower-grade copy of reality rather than the passthrough image itself. Any effect that must respond to what the user sees, such as content-adaptive dimming or saliency-weighted attenuation (Lu et al., 2012), can respond only to the camera proxy.

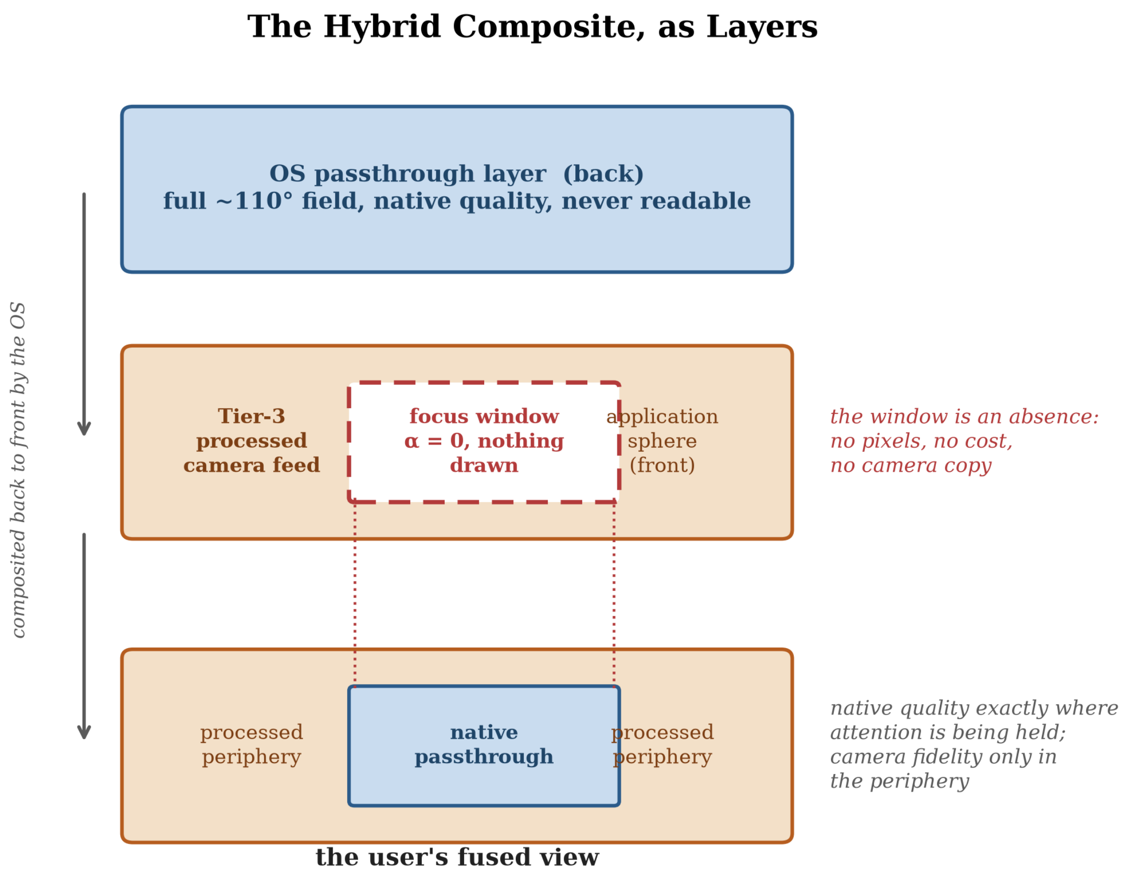

Figure 3.2: The hybrid composite, shown as exploded layers.

- **No spatial modulation of the native layer.** The full-quality view can be revealed or hidden at Tier 2 but never partially processed. A mode that blurs the real passthrough periphery, arguably the perceptually gentlest DR primitive per the foveated-rendering literature (Patney et al., 2016), cannot exist on this platform except via the Tier-3 copy.
- **No subtraction on OST.** The entire Tier-2 family is VST-specific, for the physical reason given in Section 3.1. What does and does not transfer is taken up in the conclusion.

These walls explain the system's shape. The dark modes exist because attenuation-by-overlay is the only manipulation available at native quality. ColorPop exists because re-grading requires pixels, and pixels are only available as the Tier-3 copy. The window-as-absence exists because it is the sole mechanism delivering native fidelity *and* spatial selectivity simultaneously. Had the platform offered read-modify-write access to the passthrough layer, none of this architecture would be necessary. That counterfactual is what makes the taxonomy transferable: any future platform can be located in it by asking which of the three walls it removes.

3.4 Design Axes and the Placement of the Modes

Three trade-off axes fall out of the taxonomy, and the modes carried into the study were chosen to span them.

Fidelity versus control. Tier 2 offers native fidelity with a single degree of freedom, attenuation. Tier 3 offers unlimited control at camera-copy fidelity. Hard Dark and Soft Dark sit at the fidelity pole, since the periphery they show, when they show it, is real passthrough merely darkened. SignPop sits at the control pole, since its periphery is a synthetic re-grading of the camera copy.

Suppression versus awareness. Hard Dark removes the peripheral signal entirely: maximal distractor suppression, zero residual awareness. Soft Dark attenuates to $\sim$75% opacity, so peripheral events remain detectable through the veil. SignPop suppresses selectively, dimming the red, orange and green signal band less than other colours by default and lifting that band to full visibility once a signal lamp or sign is confirmed (Chapter 4, Section 4.5). The study's safety hypothesis lives on exactly this axis. The axis matters because the literature puts a real cost at its far end: users asked what they want from diminishment prefer partial attenuation over complete removal (Cheng et al., 2022), and studies of DR in working contexts find the focus gain paired with a situational-awareness risk (Murph et al., 2021; McLaughlin et al., 2025). That is why the family keeps a graded point between untouched vision and full blackout, and why the evaluation measures peripheral awareness rather than assuming it survives.

Deployability versus capability. Soft Dark and Hard Dark run without the camera pipeline at all, since their shader path samples no texture. They draw negligible GPU power, measured on device at parity with an empty passthrough scene (Chapter 4, Section 4.11), require no camera permission, and face no thermal ceiling, so they are deployable for hour-long sessions today. SignPop inherits Tier 3's full cost structure.

Together, off → Soft Dark → Hard Dark forms a single-variable intensity ladder (overlay alpha 0 → 0.75 → 1.0), while SignPop extends the space along the orthogonal control axis. Figure 3.3 places the three on both axes, with deployability shown alongside.

One of the three was never placed in front of a participant. Soft Dark occupies the midpoint of the attenuation ladder, and its 75% ceiling is Cheng et al.'s partial-attenuation preference implemented as a number, but the session budget did not allow a third exposure and it was cut (Chapter 5). This dissertation therefore reports no participant data on Soft Dark of any kind. Its place here rests on design-space grounds and on its shader path, never on evidence about how it is experienced. Every subsequent claim about it is a claim about the design space, not about people.

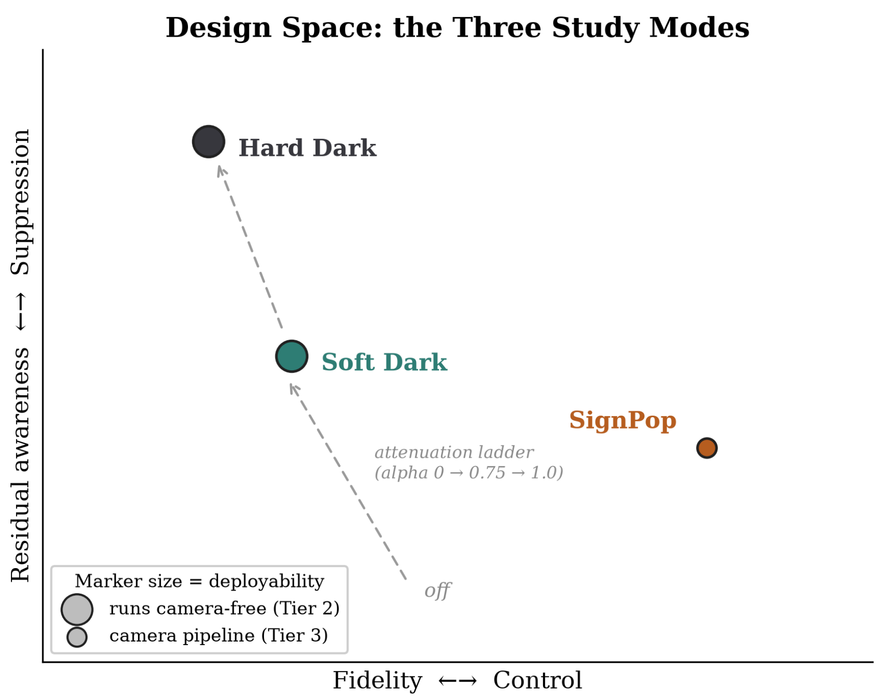

Figure 3.3: Two-axis plot (fidelity/control × suppression/awareness) with the three modes placed, deployability annotated.

These three are not the whole family. Ten peripheral treatments were built, and the remaining seven fill in the same two axes at points the study had no budget to test (Chapter 4, Section 4.5.3). They also divide cleanly along the tier boundary, which is the best evidence the boundary is real rather than imposed. Four modes never read a camera pixel and are therefore Tier 2, including SpotLift, whose focus-window brightness lift exists only in the video path because OS passthrough cannot be brightened. The other six read the camera feed and are therefore Tier 3. The tiers were not fitted to three convenient examples. They were derived from building the whole family and finding that every member's ceiling was set by the tier it had to run at.

A fourth dimension deserves naming even though this dissertation treats it as design rather than as an experimental factor: **time**. Every mode has temporal behaviour, a formation transition when the window locks, a fade-out when the head moves fast, a fade-in when it settles. The literature review found these transition mechanics essentially unstudied in DR, since prior work evaluates diminishment as a static state rather than as something that arrives, yields, and returns. The system therefore exposes every temporal parameter as tunable, and a controlled study of DR transition dynamics is identified in the conclusion as future work seeded by this design space rather than resolved by it.

3.5 Novelty Position

The claims of this chapter are calibrated against the structured prior-art search reported in Chapter 2, Section 2.5.1. Three verdicts matter here.

1. **The hybrid composite**, system passthrough revealed through a window in a live shader-processed camera-feed sphere: *nothing found.* No paper, product, or open-source project, for any purpose.

2. **Peripheral passthrough manipulation for attention guidance on a standalone consumer HMD**: *nothing found as an implemented system.* The nearest work is concept-level or differently situated. McLaughlin et al. (2025) evaluate DR distraction-attenuation entirely inside VR simulation. The DIS 2026 ADHD co-design study is formative, with no prototype. DiminishAR and the CHI 2026 Vision Pro study target specific objects or persons with occlusion rather than the periphery. Sutton et al. (2022) modulate real-world saliency but on a bench-mounted optical see-through rig that can only add light.

3. **A dark peripheral vignette over passthrough**: *near precedent, no direct precedent.* Quest OS v67's Theatre View ships an adjustable peripheral dimming vignette, but explicitly not over passthrough. Vision Pro's surroundings dimming is

global rather than windowed.

Each constituent mechanic, taken alone, has documented precedent that this dissertation cites rather than claims: camera-feed geometry with GPU effects (Coviello, 2025), a head-centred fade sphere revealing passthrough (Meta, 2024b), alpha punch-through windows (Meta, 2024d; Immersed, 2022), OS peripheral dimming in immersive mode (Meta, 2024a), and saliency modulation of reality on OST hardware (Sutton et al., 2022). The defensible novelty claim is therefore precisely bounded: **the composite**, native passthrough through a world-locked focus window in an effect-processed camera-feed periphery, exploiting the quality and field-of-view asymmetry between layers, **applied to peripheral attention guidance on a consumer standalone headset**, together with the world-direction fusion sampling that makes the mono feed binocularly viewable. An alpha-blended dark overlay, by itself, is just compositing, and is not claimed as novel. The claim is worded around the design space rather than the single technique. If a concurrent system were to surface, the contribution would remain a characterised design space with an independently built implementation.

Two adjacent patents (Patents US 12548271 B2 and US 12524072 B2, 2026) describe claim-level ideas in this territory: attenuating external stimuli while preserving spatial awareness, and blurred external video feeds composited into VR. Neither shows evidence of implementation. They are noted for completeness.

3.6 Summary

On consumer VST hardware, diminished reality is not one problem but three, stratified by compositing access: a colour-only, spatially blind styling tier, an attenuation-and-occlusion overlay tier whose most powerful move is a shaped absence, and a full-control camera-copy tier that pays in fidelity, latency, fusion and heat. The modes carried into the user study span the space this stratification creates, the hybrid composite exploits the asymmetry between its tiers, and the walls of the space, no reads, no spatial modulation of the native layer, no subtraction on OST, are findings in their own right. Chapter 4 descends from this map into the machine.

Chapter 4

System Design and Implementation

4.1 Overview

This chapter presents the reference implementation, Contribution C2: a multi-mode diminished-reality attention-guidance system running on a consumer Meta Quest 3, built in Unity 6 on Meta's Passthrough Camera Access sample framework (Meta, 2025). The system realises the design space of Chapter 3 in a deliberately economical architecture, **one shader and one manager component on a single piece of geometry**, and the economy is itself a result: every capability tier of Chapter 3's taxonomy is reachable from one head-centred sphere and one fragment program.

Throughout, parameter names in `code font` are the actual serialized fields and shader properties in the repository (`CameraSphereVignette.shader`, `CameraSphereVignetteManager.cs`, `VideoTestSceneManager.cs`), so every quantitative claim is checkable against the artifact. The full repository is publicly available at `https://github.com/batunii/Arjuna`, including the shader and manager sources, the study harness, the analysis scripts that regenerate every statistic in Chapter 5, and the raw capture records behind Section 4.11.

Figure 4.1 shows the components and the per-frame flow between them.

4.2 The Sphere and the Window

4.2.1 Geometry

The entire visual effect is carried by a single inverted sphere of approximately 10 m radius, centred on the user's head and re-centred every frame. The translation is applied without rotation, which is what keeps the sphere's surface coordinates world-stable. The mesh is rendered inside-out with `Cull Front`, on the transparent queue, with depth writes disabled and conventional alpha blending. Every fragment the user can see, in

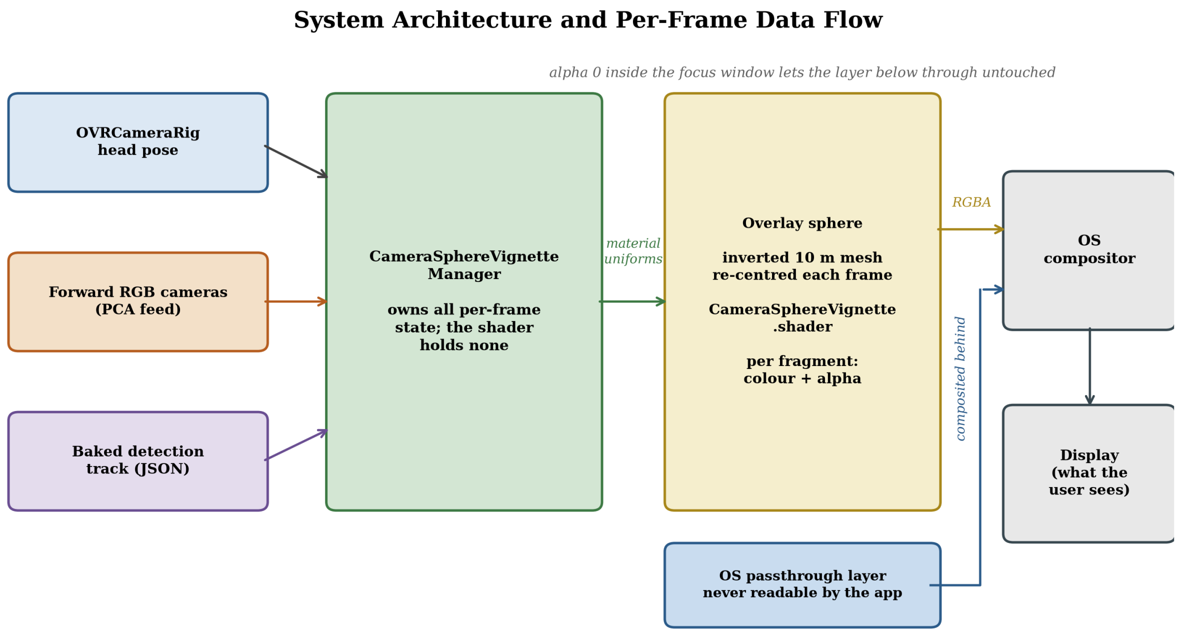

Figure 4.1: System architecture and per-frame data flow.

any direction, is a fragment of this sphere. The shader's output alpha at that fragment decides, per pixel, whether the user sees the diminishment effect or whatever lies behind the sphere, which in the passthrough scene is the operating system's native passthrough layer.

This choice of a fragment-alpha-controlled full-surround surface is what makes one object span all of Chapter 3's tiers. Alpha 0 is Tier 2's absence, so native passthrough passes through untouched. Alpha 1 with a black colour is Tier 2's blackout. Alpha 1 with a camera-fed colour is Tier 3's re-render. The mode logic reduces to deciding, per fragment, a colour and an alpha.

4.2.2 The world-locked focus window

Two different things in this system can look similar, and the chapter keeps them apart by name. The **focus window** is the region the user places and locks onto a spot in the room. It stays fixed to that spot, the application draws nothing inside it, and it does not move again until the user resets it. The **detection aperture** is a smaller opening the system creates over an object a detector has just recognised. It appears and disappears as objects are recognised and lost, and the user has no control over it. Section 4.5.5 describes the aperture.

The focus window is stored not in screen space or head space but in **world spherical coordinates**: a rectangle `_FocusRect = (azMin, azMax, elMin, elMax)` in radians of azimuth and elevation about the sphere's centre. Each fragment reconstructs its own direction from first principles:

```
dir = normalize(worldPos - _SphereCenter)
az  = atan2(dir.x, dir.z)
el  = asin(dir.y)
```

and computes a signed distance outside the rectangle, `max(dAz, dEl)`, which a smoothstep across the soft edge converts into the per-fragment vignette coordinate `t`: 0 inside the window, rising smoothly through the soft edge, and 1 in the far periphery. Every mode consumes this single scalar. The dark modes use it directly as overlay alpha. ColorPop uses it to interpolate each pixel between its inside and outside treatments.

The soft edge defaults to `m_softEdgeDeg` $= 20°$, widened to 32° for ColorPop (Section 4.5.2). These are the component defaults; the scene the user study ran serialises 18° for the dark path and 24° for the colour-pop path, against an $8° \times 6°$ window, which is the geometry Chapter 5, Section 5.7.3 reads its eccentricity bands against. The 20° default is chosen against the useful field of view, the region within roughly 30° of fixation from which people extract task-relevant information without moving their eyes or head (Ball and Owsley, 1993). Placing the soft edge inside that radius means the transition band falls where the eye is still resolving detail, so the effect reaches full strength just as the useful field runs out rather than cutting sharply across it. It also sits above the $\approx 15°$ eccentricity beyond which head yaw becomes a reliable pointer to attention (Higgins et al., 2022), which matters because the window is placed by head-directed painting and must be larger than the noise floor of the pointing method that places it.

Because azimuth and elevation are computed from world direction, the window is **world-locked**. It stays fixed to the room, to the desk, the doorway, the whiteboard, as the head rotates, rather than travelling with the gaze. Direction is computed per-fragment from `worldPos` rather than interpolated from vertices, which avoids interpolation error at the sphere's poles. Figure 4.2 shows the rectangle on the sphere, the soft-edge band, and the resulting profile of `t` along one axis.

4.2.3 The window as absence

In the passthrough scene the shader's output inside the window is `alpha = t = 0`, so nothing is drawn. The consequence is the system's central design decision. Inside the window the user sees the operating system's own passthrough at full native quality, full $\sim$110° field of view, and the OS pipeline's own latency compensation, none of which any application-rendered reconstruction can match. The effect budget is spent entirely in the

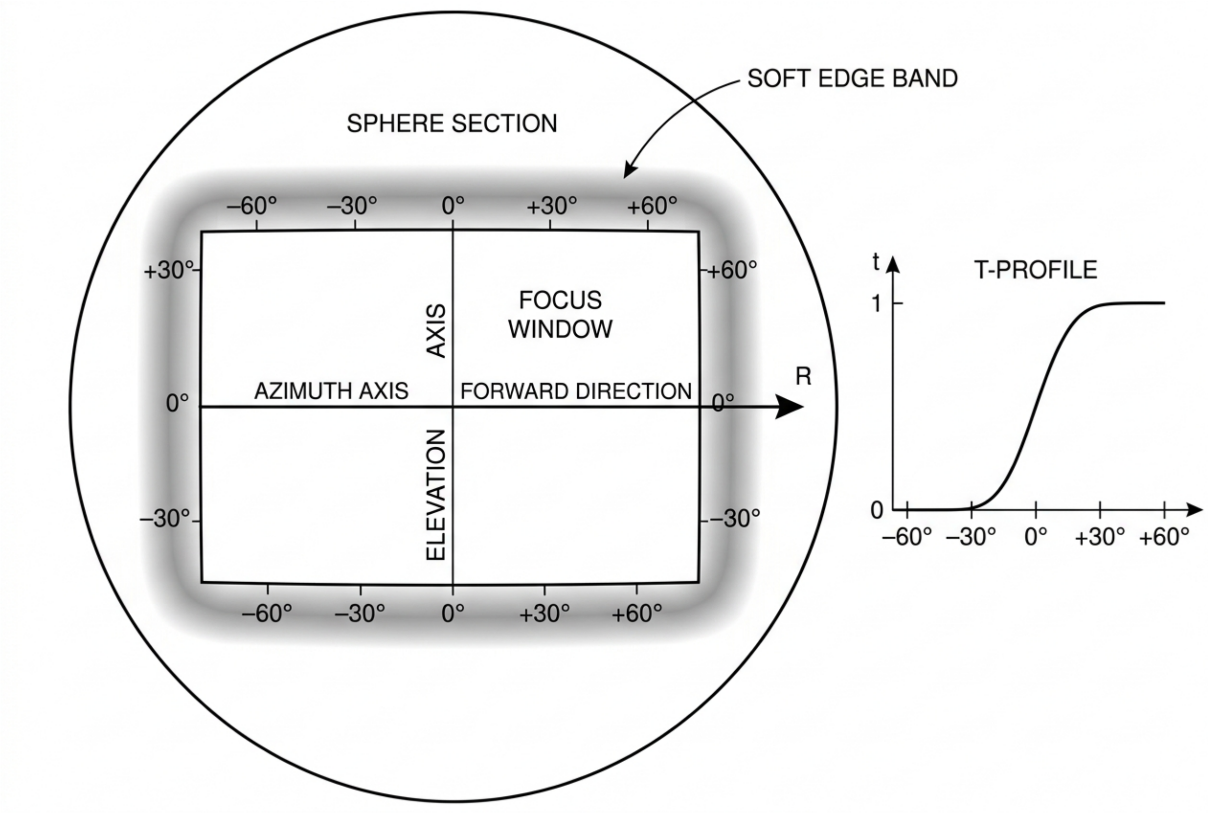

Figure 4.2: The az/el window geometry.

periphery, where human acuity is lowest. One line of shader arithmetic simultaneously solves the field-of-view problem, since the accessible camera feed covers only ∼85–90°, and the fidelity problem, since that feed is 1280×960 mono against a native stereo view, by ensuring the reconstruction is only ever shown where neither limitation matters much.

4.2.4 Per-frame data flow

The manager owns all per-frame state and feeds the shader exclusively through material uniforms, so the shader holds no state of its own. Each frame it re-centres the sphere on the head anchor and uploads `_SphereCenter`, the one value the whole angular coordinate system depends on, the focus rectangle and soft edge, the combined formation and motion-suppression strength, and the active mode's parameter block. For camera-fed modes it also uploads each camera's pose basis and half-FOV tangents, derived from the intrinsics the PCA API reports rather than an assumed field of view, which is what keeps the camera periphery in scale with the native window across the seam. The shader is written for single-pass instanced stereo and, by the world-direction design of Section 4.4, contains *no* eye-dependent sampling anywhere in the camera path, so the renderer needs no per-eye scripting.

4.3 The Hybrid Composite

The system's flagship configuration composes the tiers: a **Tier-3 processed camera feed as the periphery, with the Tier-2 transparent window revealing native passthrough as the focus region**. These are two different renderings of the same reality, fused into one continuous view.

Mechanically, the PCA feed is bound to the sphere material (`_MainTexL`, and `_MainTexR` where the right camera is available), the fragment shader computes each periphery fragment's colour from the feed with whatever manipulation the active mode specifies, and outputs it at `alpha = t*strength`. Inside the window, `t = 0` yields transparency and the OS layer shows through. The seam between the two renderings is the window's soft edge, where the processed feed fades out over `_SoftEdge` radians. Because both renderings depict the same world from nearly the same viewpoint, the transition tends to read as an effect boundary rather than as two images.

The construction has one honest seam-level caveat. The two renderings do not share a latency budget: the OS passthrough inside the window is served by the platform's dedicated low-latency reprojection path, while the camera-fed periphery carries the application pipeline's additional delay. In static or slowly moving views the mismatch is not noticeable. Under fast head rotation the periphery can momentarily lag the window at the seam. The motion-based suppression of Section 4.6 incidentally masks the worst case, since the effect fades out precisely when the head moves fast, but the residual mismatch is a real property of the hybrid rather than something designed away.

The claim this construction supports is deliberately precise (Chapter 3, Section 3.5): the *composite* is the novelty, exploiting the quality and field-of-view asymmetry between a native layer that cannot be touched and a camera copy that can, not any of its ingredients. Its costs are Tier 3's costs, paid only in the periphery. Figure 4.3 shows the composite as recorded on device.

4.4 Binocular Fusion by World-Direction Sampling

The accessible camera feed is monocular, but the display is binocular, and how the mono image is presented to two eyes determines whether the system is usable at all. The governing requirement is that the sampling domain be **eye-independent**. Under stereo rendering the same world point projects to different screen positions in each eye, so any scheme keyed to screen position feeds each eye a different camera pixel for the same world point, and the two images do not fuse (Section 4.10, failure 1).

The shader therefore samples by **world direction**. Each fragment's direction from

Tier-3 processed periphery (Blur), native window on the monitor   Tier-2 extreme: Hard Dark, the window as the only rendered absence

Figure 4.3: The window as absence, from on-device recordings of the same room. Left: a Tier-3 processed camera periphery (the Blur variant, whose contrast restoration reads as an edge drawing) surrounds the world-locked window, through which the monitor is served by native OS passthrough in full colour. Right: the Tier-2 extreme, Hard Dark, where the window is the only region rendered at all.

the sphere centre, already computed for the window test, is projected through a head-centred pinhole model of the camera. With the camera's forward, right and up basis and half-FOV tangents derived from the camera intrinsics rather than an assumed field of view, the shader computes

```
uv = ( (dir.rt, dir.up) / max(dir.fwd, eps) ) / tanHalfFov * 0.5 + 0.5
```

Because world direction is eye-independent, both eyes sample the *same* camera pixel for the same world point. The mono image behaves like a texture painted onto the world shell, which the visual system can fuse, as verified on device (Section 4.10). The residual and acceptable artefact is that the painted shell has no true stereo disparity of its own, which is consistent with the project's standing comfort constraint that a mono feed must never be presented as if it were geometry-correct stereo across the full field of view.

Where both cameras are available, the shader samples both and selects per fragment by in-view weight, a **hard split** rather than an alpha crossfade, because a crossfade of two horizontally offset viewpoints produces exactly the ghosting the world-direction scheme exists to avoid. The two forward cameras share an orientation and differ only by a small horizontal baseline, and no stitch line was visible in normal use. Figure 4.4 contrasts the two sampling schemes on the same world point.

4.5 The Modes

The mode enumeration in the codebase (`VignetteMode`) contains eleven entries. Two are carried into the user study, SignPop and Hard Dark, both to confirmatory testing. A third, Soft Dark, is specified here but was never tested with participants (Chapter 3,

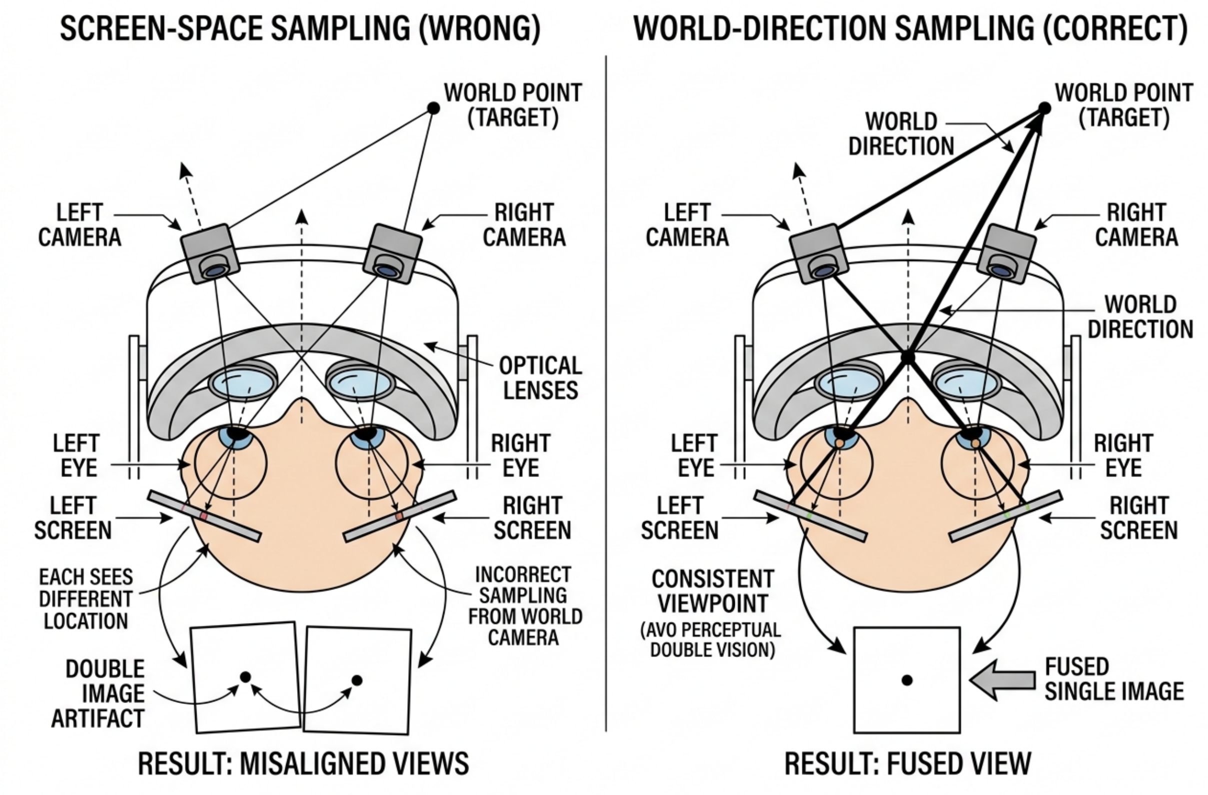

Figure 4.4: Screen-space versus world-direction sampling of a mono camera feed.

Section 3.4). ColorPop is the base grading pipeline SignPop extends with a detection gate. Blur is their superseded ancestor, and five further entries are retained design-exploration variants. All share the geometry, window, and t-coordinate of Section 4.2. A mode is, in implementation terms, a per-fragment colour and alpha policy. Mode names are the code identifiers, kept so every claim stays checkable against the repository, and where a name suggests a standard image-processing operation the description given with it is authoritative: Blur, for instance, names a progressive two-ring Gaussian of the camera periphery with contrast restoration, not a whole-image convolution.

4.5.1 Peripheral occlusion (Hard Dark and Soft Dark)

The simplest and strongest manipulation is a black overlay whose alpha is the vignette coordinate scaled by the current strength:

```
periColor = black
periAlpha = t * _VignetteStrength * _MaxVignetteAlpha
```

with `_MaxVignetteAlpha` = 1.0. At full formation the periphery is completely occluded and the user's visual world *is* the focus window. The shader path (`_SimpleMode = 1`) samples no textures and does no camera mathematics. An early return delivers what is effectively a fixed-function overlay, with correspondingly negligible GPU cost and no camera permission or thermal exposure, which is Chapter 3's deployability argument in code. Hard Dark is the confirmatory mode of the workstation block: the strongest available manipulation, chosen to maximise the chance of detecting the distraction-suppression mechanism if it exists.

The same shader path run with `_MaxVignetteAlpha` set from `m_mode2MaxAlpha` (default **0.75**) gives Soft Dark, where the periphery darkens substantially but never fully, so peripheral motion and events stay visible through the veil. The ceiling is set below 1.0 rather than at it because Cheng et al. (2022) found users asked what they want from diminishment prefer partial opacity reduction to complete removal, and want enough residual structure to know something is still there. Soft Dark is that preference implemented as a number.

Both dark modes **form gradually**. On window lock, a coroutine animates `_VignetteStrength` from 0 to 1 along a SmoothStep over `m_vignetteFormTime` (default **3 s**), so the periphery dims as a transition rather than a cut. The ramp is there because Hata et al. (2016) showed a visual change introduced gradually enough stays below the awareness threshold while still biasing gaze, so the guidance survives the ramp even though the announcement does not. Formation runs independently of the motion suppression of Section 4.6: if the head moves during formation, suppression multiplies the strength down while the formation clock keeps counting, and the effect resumes at its accumulated level when the head settles.

4.5.2 Colour- and saliency-gated peripheral re-grading (ColorPop)

ColorPop is the Tier-3 mode. It re-grades the camera feed's colour and brightness rather than dimming it, using the literature's strongest guidance channel, luminance and saturation contrast, with no hue shifts (Bailey et al., 2009; Grogorick et al., 2017; Sutton et al., 2022). The re-grading is a standing property of the whole visual field rather than a transient cue that appears and disappears.

Its grading is a **four-level hierarchy** on two tracks, gated by colour class and interpolated by the window coordinate t:

$$\text{outside-other} < \text{outside-ROG} < \text{inside-other} < \text{inside-ROG}$$

Colour classification. A shader helper computes membership in the kept red, orange and green (ROG) set, the colour classes of signal lamps and signage. Membership is defined by hue bands, a warm band and a wrap-around red band above 0.85, plus a green band from 0.22 to 0.48, each gated by a saturation floor. The warm band uses a deliberately higher floor (`_PopWarmSatMin` = 0.35) because warm-*white* light, headlights at hue ≈ 0.1 and saturation 0.1 to 0.25, must not ride into the orange class, while genuine lamps and signs sit above saturation 0.5. The floor is therefore placed in the gap the two populations leave.

The other track, meaning non-ROG pixels, stays near-natural inside the window, with a slight desaturation that keeps the window from feeling artificially vivid against its own periphery, and is graded down to dim grey outside (`_PopGreyDim` = **0.4**). The floor sits at 0.4 rather than lower because a periphery approaching black would carry almost no peripheral events, and a monitoring mode whose own dimming floors peripheral detection would fail its safety hypothesis by construction rather than by finding. This track covers the great majority of peripheral pixels, so its floor, rather than the signal track's, is what decides whether the periphery stays usable.

The ROG track boosts saturation about the pixel's own luminance inside the window, applies a sigmoidal midtone-contrast transfer per Sutton et al. (2022)'s published recipe ($\alpha = 10$, $\beta = 0.5$, normalised so 0→0 and 1→1, scaled by $(1 - t)$ so it is full inside the window and absent outside), and lifts brightness slightly. Outside the window, ROG pixels keep their **natural colour, dimmed** to `_PopBrightOut` = 0.65. This is deliberate: signal-relevant colours remain detectable *everywhere*, merely privileged inside the window, which fits a monitoring context in which a peripheral traffic light must never become invisible.

Glare compression. Pixels that are bright, unsaturated, and not ROG, the signature of headlight glare, are compressed toward a knee at `_PopGlareKnee` = 0.6 luma, a level above which a light source tends to veil the detail around it rather than merely being bright. They are compressed toward that knee and never to black, so the oncoming vehicle stays legible while its veiling glare is attenuated. The compression is mild inside the window and strong outside it, because inside the window the user is looking directly at the source and needs it to remain readable, whereas in the periphery an uncompressed glare bloom washes out precisely the region the mode is trying to keep usable. A **blown-core guard** protects the one case the mask would otherwise destroy, the clipped white centre of a bright signal lamp: before dimming, the shader samples a small Gaussian surround, and if the surround is a kept colour, meaning the lamp's saturated fringe, the pixel is exempted.

Global periphery dim. Finally the whole periphery, ROG included, is multiplied down by `_PopPeriphDim` = 0.85, so the focus window also wins on plain luminance, the strongest single guidance channel in the attention literature (Bailey et al., 2009; Grogorick et al., 2017). Net outside-ROG brightness at defaults is therefore $\approx 0.65 \times 0.85 \approx 0.55$ of natural. That figure is a deliberate midpoint rather than a maximum. Cheng et al. (2022) found partial attenuation preferred over removal, and Barhorst-Cates et al. (2016) found anxiety rising monotonically as the visible field narrows, so a monitoring mode that must keep a peripheral traffic light findable has reason to stop at roughly half natural brightness rather than push further.

Because a colour and grey boundary tends to read harsher than a blur or darkness boundary, ColorPop uses a widened window edge, `m_popSoftEdgeDeg` $= \mathbf{32}°$ against the global default of 20°. Figure 4.5 traces a fragment through the classification, the two grading tracks, and the periphery dim that merges them.

In the passthrough scene ColorPop grades the periphery only. The window must remain transparent, since the OS layer cannot be graded, and repainting the full field from the mono feed would violate the comfort constraint. In the video test scene the same shader grades the entire sphere, window included, which is the configuration the driving block evaluates.

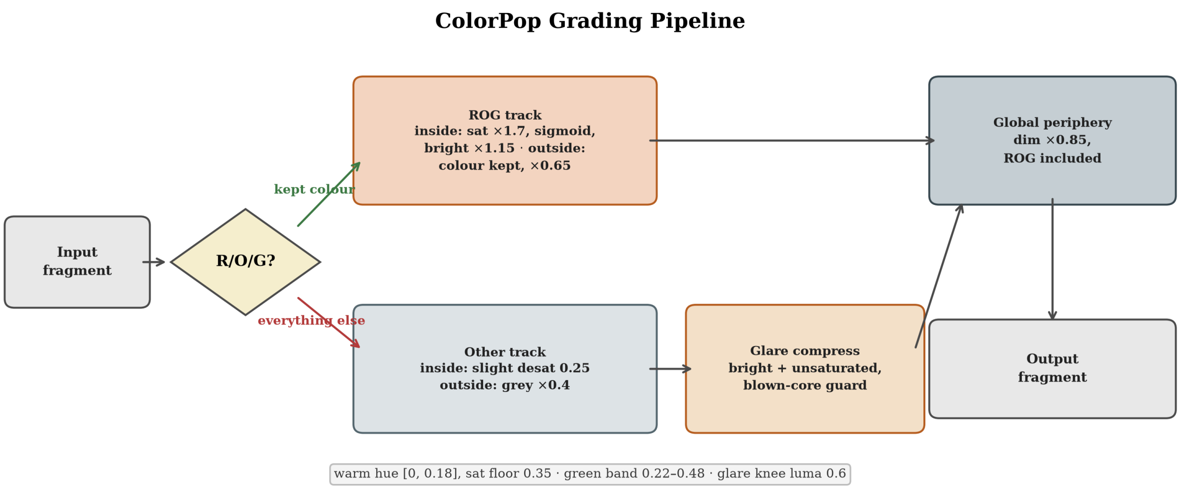

Figure 4.5: ColorPop grading pipeline flowchart.

4.5.3 The superseded and exploratory family

Six further modes are retained in shader and enumeration but excluded from evaluation. Table 4.1 gives each with the source it derives from and the reason it was not carried forward. Each is reported as an explored design point carrying no evaluation claim. Figure 4.6 shows nine of the ten peripheral treatments as captured on device, the study modes over live passthrough and the exploratory family over the driving scene.

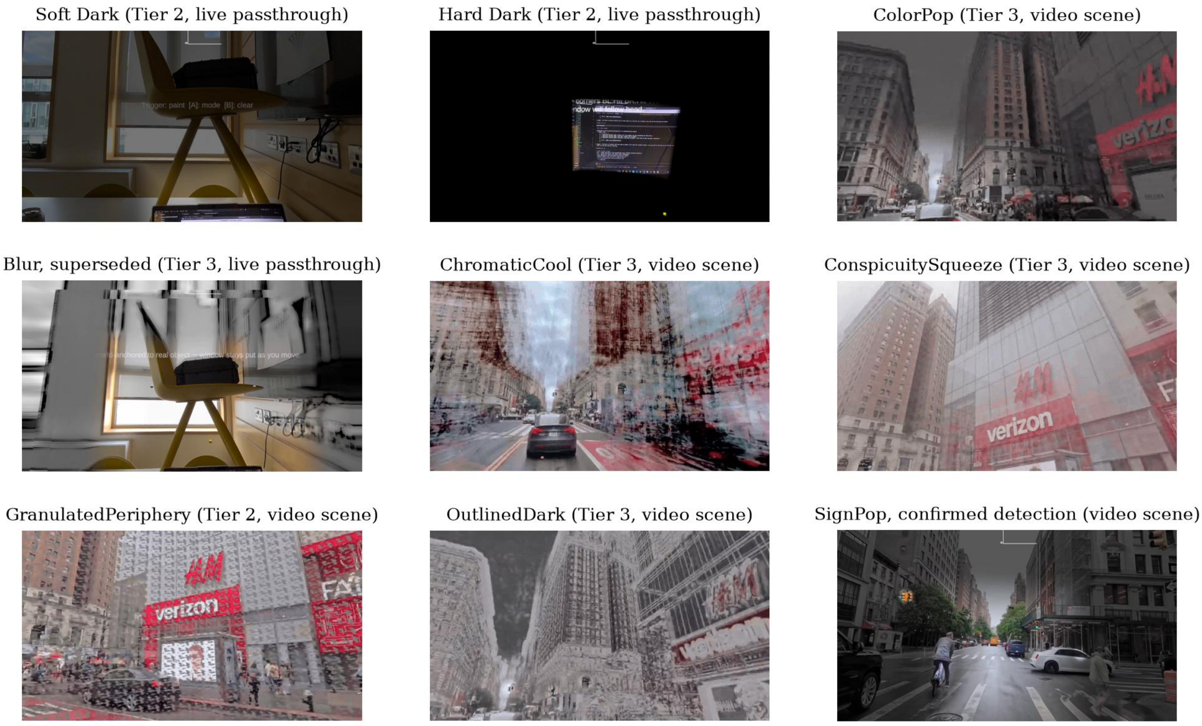

Figure 4.6: Nine of the ten peripheral treatments, from on-device recordings. The passthrough captures show the world-locked window revealing untouched passthrough; the video-scene captures show full-field grading over the driving clip. Blur's contrast-restoration pass reads as an edge drawing rather than a smooth defocus, one reason it was superseded. SpotLift is not shown: it exists only on the video path and no recording of it was retained. TintedDark is dead code (Table 4.1).

Table 4.1: Modes built but not evaluated, with the source each derives from and the reason it was not carried forward.

Mode	What it does	Why not evaluated
Blur (superseded)	Progressive Gaussian blur of the camera periphery with delayed desaturation, a perceptual noise term after Tariq et al. (2022), and contrast restoration after Patney et al. (2016)	Whole-periphery blur was the least subtle of five methods and irritated 59 of 102 participants in its own source study (Grogorick et al., 2018). Also the heaviest shader path in the family
ChromaticCool	Warm window against a cool periphery, chromatic opposition	Luminance modulation outperforms warm–cool chromatic modulation in its own source study (Bailey et al., 2009)
Conspicuity-Squeeze	Centre–surround contrast flattened toward the local mean, after Veas et al. (2011)	Brightened the periphery instead of squeezing its contrast, the opposite of the intent, confirmed by shader inspection
Granulated-Periphery	World-locked static noise grains with eccentricity-ramped density, after Cao et al. (2021)	Grains occlude fine detail, the wrong failure mode for a driving scene where the detail is the target
OutlinedDark	Near-blackout with luminance edges restored, following the finding of Cheng et al. (2022) that context-preserving outlines reduce anxiety	Surviving edges read as a comic-book style that early testers found more interesting than the task

Mode	What it does	Why not evaluated
SpotLift	Focus-region brightness lift over a soft peripheral dim	Cannot exist on the passthrough path at all, since OS passthrough cannot be brightened. Video path only

A seventh candidate, **TintedDark**, a configurable-colour overlay, is dead code. It has no proposing paper and no mechanism the dark family does not already have. Its shader branch and enum entry remain only because removing the entry would break a serialized scene value (Section 4.10).

One sampling detail from the Blur path generalises to any equirectangular source: taps must wrap horizontally for a seamless 360° join and fold back rather than clamp vertically, since clamping smears the top and bottom texture rows across the sky and floor, and the tap radius must fade toward the poles, where any UV-space offset corresponds to a huge angular distance.

4.5.4 Cost structure of the shader paths

The mode family spans roughly two orders of magnitude in per-fragment cost, and the gradient is structural rather than incidental. It is Chapter 3's deployability axis realised in instructions. The dark modes take an early return after the window test: no texture reads, a handful of arithmetic operations, one blended write. ColorPop reads the source texture once per fragment on its main path, plus a 17-tap Gaussian surround only in the small fraction of fragments where the glare mask fires with the core guard enabled, and its cost is dominated by per-pixel colour arithmetic. The Blur path is the heaviest: two Gaussian rings of eight taps each around a shared centre per camera, doubled when both cameras are live near the stitch, plus the noise texture read and contrast-restoration arithmetic.

These per-fragment counts are structural, and Section 4.11 confirms the gradient they predict with measured frame times: the dark modes cost nothing measurable over an empty passthrough scene, and the heaviest camera path costs roughly two and a half times the dark modes' frame time while still holding the display's frame rate. The design consequence stands in both the structure and the measurement: the modes the study asks participants to imagine using for long sessions, the dark family, are the modes that cost almost nothing, and the mode with the richest effect is the one whose Tier-3 pipeline carries the platform's thermal ceiling.

Figure 4.7: SignPop over the driving scene, from an on-device recording. Non-signal content is graded to dim grey, kept red–orange–green pixels stay visible everywhere, and a confirmed detection (right, enlarged) fires the full bundle: colour pop, detection aperture, saliency lift, and darkened surround.

4.5.5 Detection-gated salience re-grading (SignPop)

SignPop is the most capable of the ten peripheral treatments, live only in the video test scene, and it is the mode placed in front of participants in the driving block (Figure 4.7). It shares ColorPop's four-level ROG grading pipeline, but a confirmed detection from the offline oracle track (Section 4.8) gates a bundle of effects rather than colour alone.

The colour gate. ColorPop already keeps ROG hues partly visible in the periphery without any detection. SignPop restricts the *full* vivid treatment to pixels inside a confirmed detected box. An undetected ROG pixel, whether neon signage, brake lights, advertising, or a genuine bake miss, falls back to a partial keep (default 0.35) rather than full grey, a graceful miss so a missed real signal is dimmed rather than hidden. A short hold of 0.5 s keeps a detection alive after it drops out of the bake, so pops do not blink between samples taken every half second.

An independent detection-zone system. Beyond the colour gate, every confirmed detection drives three further, colour-independent effects computed from soft elliptical zones around each detection box. A **detection aperture** carves the vignette away directly over the object. This is not the focus window of Section 4.2: it is a separate, transient opening appearing wherever a detection is confirmed, inside or outside the focus window, and disappearing when the detection is lost. A **saliency lift** boosts saturation, brightness, and local contrast around the object, scaled down when the object already sits inside the focus window, since the standing window already reveals it there, and left at full strength in the periphery where the lift is doing the work. A **darkened surround** dims a soft annulus just outside the object's boundary, raising the object's centre–surround contrast from both sides, after the dual modulation of Veas et al. (2011). The saliency lift additionally oscillates at approximately 1 Hz, following Waldner et al. (2014) on low-frequency, low-amplitude temporal modulation attracting attention with low annoyance, far below the 15–25 Hz photosensitivity band.

This entire detection-zone system is suppressed for Hard Dark. Once the user-placed window is locked, no detected object anywhere gets a carve-out, however salient. This is a deliberate choice rather than an omission: Hard Dark's role is to test the strongest possible manipulation, full occlusion with no exceptions, against a workstation task with no peripheral object the participant needs to see. Adding detection apertures back into Hard Dark would weaken exactly the property the confirmatory test relies on. A detection-aware variant is a reasonable extension for contexts where a peripheral object genuinely matters, and the conclusion returns to it as future work.

Two gates on what counts as a detection at all. The offline bake considers only the forward view by default, azimuth ±85° and elevation ±50° around video-centre, so a detection behind the participant never enters the track. At runtime, a detection's window and boost stay suppressed until its apparent size clears a minimum of ∼1.2°. In the current bake, 33 of 246 post-debounce traffic-light lifetimes never clear it, so a distant, still small light does not open a window over what is still, perceptually, a speck.

What a confirmed detection actually switches on. Six effects fire together on the same gate: the colour-pop upgrade, the window carve-out, the saturation lift, the brightness lift, the contrast expansion, and the darkened surround, layered on the ∼1 Hz pulse. **None of the six is isolable in the current build.** This is the single most important limitation on what any measured effect can be attributed to, and it is stated here, where the mechanism is, rather than repeated wherever a result appears. The conclusion proposes the three-arm design that would separate them.

4.6 Motion-Based Suppression

Head-coupled restriction of the visual field is a known comfort hazard: Norouzi et al. (2018) found that vignetting yoked to head rotation *increased* rather than decreased simulator sickness. The system therefore decouples in the opposite direction. The effect **fades out while the head moves fast, and returns when it settles.**

Each frame the manager computes head angular speed between successive head rotations. When speed exceeds a per-mode threshold, a suppression scalar animates toward 1 over 0.20 s, and once speed has stayed below threshold for the mode's hold time, back toward 0 over 0.70 s. Thresholds and holds scale with the mode's intrusiveness: Blur 30°/s with a 0.6 s hold, ColorPop 35°/s with 0.7 s, Soft Dark 50°/s with 1.0 s, Hard

Dark 70°/s with 1.5 s. The blackout mode tolerates the most motion before yielding and waits longest before returning, because wrongly suppressing a strong effect flashes the whole periphery back into view, while wrongly suppressing a gentle one costs little. The fade asymmetry follows the same logic: yielding should feel immediate to be trusted, and returning can be slower because little is at stake.

The shader receives `effectiveStrength = m_vignetteStrength * (1 - suppression)`, so suppression multiplies the formation state rather than resetting it, and a glance over the shoulder during the three-second formation does not restart the transition. State the user created, the locked window and the accumulated formation, is never destroyed by a comfort mechanism, only veiled.

One refinement matters for the interaction loop. If the head's direction enters the focus window, with a +15° margin, the hold timer is zeroed immediately, so arriving back at the focus region restores the effect at once. The asymmetry is deliberate. Looking *away* is treated as a possible need for situational awareness, so the effect yields generously, while looking *back at the task* is treated as re-engagement, so it returns promptly. Figure 4.8 shows the threshold, the hold timer, the two fade ramps, and the focus-arrival shortcut as one state diagram.

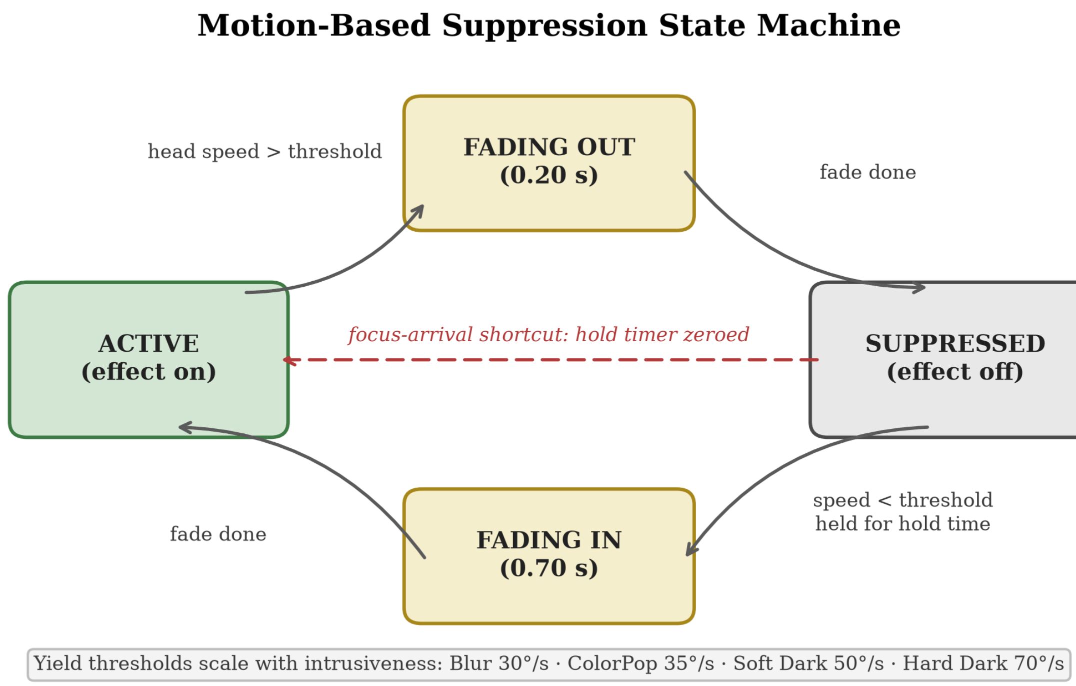

Figure 4.8: Motion-based suppression: speed threshold, hold timer, and fade ramps.

4.7 Interaction Design

The user defines the focus window by **painting** it. Holding the right controller trigger sweeps the controller's aim across the scene, and the window rectangle grows to enclose the swept region, seeded at the aim point with a small brush pad. Releasing the trigger **locks** the window and, in the dark modes, starts the three-second formation. The rectangle is computed and stored directly in the world azimuth and elevation coordinates of Section 4.2.2, so what the user paints is what stays locked to the room. The A button cycles modes and B clears the window.

Feedback is deliberately minimal. Five dot markers, four corners plus an aim cursor, show the window during painting, dim on lock, and auto-hide after three seconds. A world-space toast announces mode changes and fades away. Hand tracking is disabled and input is controller-only, a scope decision that removes a class of input ambiguity from the study sessions.

Three design rationales are worth making explicit. *Painting rather than pointing*: a focus window is a region rather than a point, and sweeping it makes the region's extent a deliberate act, where auto-sizing a window around a pointed target would hide a consequential parameter inside a heuristic. *World-locking rather than head-locking*: the window belongs to the task surface rather than the gaze, and a head-locked window would chase attention rather than anchor it, which is also the comfort hazard Section 4.6 exists to avoid. *Formation as transition*: the ramp exists because an instantaneous peripheral blackout would be startling, and it doubles as the user's confirmation that the lock succeeded. Figure 4.9 follows one placement from painting to locked formation.

4.8 Perception: Detection Islands and the Offline Oracle

4.8.1 Live detection support

The live passthrough manager supports up to eight simultaneous **detection islands**, azimuth and elevation rectangles fed by an optional `YoloRunner` component, inside which the vignette is cleared so the detected object shows through undiminished. Each island renders as a soft-edged ellipse with the dual modulation after Veas et al. (2011) described in Section 4.5.5. Detections expire after 0.6 s without renewal. The class filter is deliberately narrow: COCO classes 9 (traffic light) and 0 (person) only, the two object classes the driving block's marker task uses as targets.

Two properties of the aperture system are stated here because they bound what it can

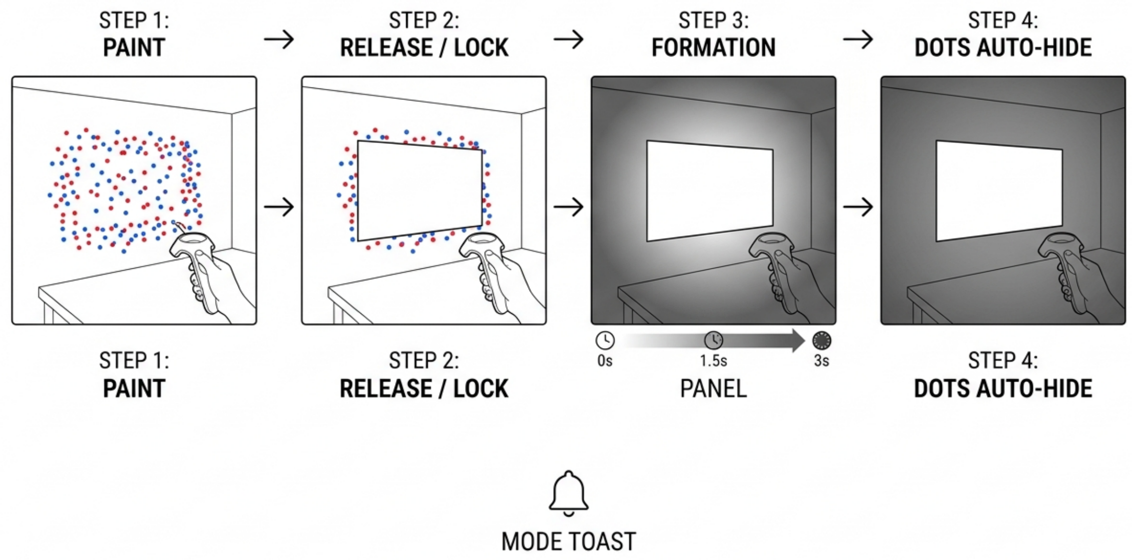

Figure 4.9: Interaction sequence for placing and locking the focus window.

do in a crowded scene. First, the caps above are a hard budget: the live path holds eight simultaneous apertures and the video path sixteen, slots fill in the order detections arrive, and any detection beyond the cap is dropped. There is a budget but no prioritisation, so a crowded scene degrades by dropping the newest detection rather than the least important one. Neither cap was reached in any reported session, but that is a property of the study footage rather than a design guarantee. Second, a detection is characterised only by its angular position and apparent size, evaluated per frame with the minimum-lifetime gate and smoothing fades described below. There is no velocity or trajectory term anywhere in either path, so a pedestrian stepping off a kerb and one standing still receive identical treatment if they subtend the same angle. Both limits are carried into Chapter 6.

4.8.2 The offline oracle pipeline

For evaluation, live inference is replaced by a **pre-baked detection track**. This is the oracle-perception commitment named in Chapter 1: the study asks whether an effect built on excellent detection helps attention, without first building excellent detection on-device. A PC-side script (`Tools/bake_detections.py`) runs YOLO11 (Jocher and Qiu, 2024) over the full-resolution 3840×2160 study video as 3×2 overlapping tiles plus a full-frame pass merged by per-class non-maximum suppression, sampling at 0.5 s intervals, and writes a JSON track keyed by video time. The runtime loads it and plays detections back deterministically.

Full resolution is the point rather than an indulgence. Detections that serve as ground truth or as pre-scripted salience input must be computed at a resolution the live pipeline cannot afford, which is exactly what makes them an *oracle*. Distant traffic lights are the binding case: they are fully visible to a human viewer but subtend few pixels, so they fall below detectable size in a downsampled frame.

The bake over the 8-minute-39-second study clip produced 13,939 detections across 1,040 sampled frames, with 99.5% of sampled frames containing at least one detection (median 14 per sample; 2,952 traffic lights and 10,987 people). Two properties matter for the rest of the dissertation. First, it confirms the footage is genuinely event-dense, a precondition for its role as the driving-block stimulus. Second, it fixes a versioned, reproducible detection set, so the study's stimuli are artifacts rather than descriptions.

What the bake does *not* establish is equally important. It says nothing about on-device feasibility, and nothing about detection quality in absolute terms, since full-resolution YOLO11 is a *reference* rather than truth verified against human annotation. The gap between this oracle and a live on-device detector was never measured. Every statement in this dissertation about detector-dependent guidance is therefore conditioned on oracle-quality detection, and the conclusion carries closing that gap as the first item of future work. Unlike the live manager, the video scene's manager draws on the shader's full sixteen-slot detection array, which absorbs the bake's density, so no ordering or survival policy is needed for the great majority of samples. Figure 4.10 shows the pipeline from source video to deterministic playback.

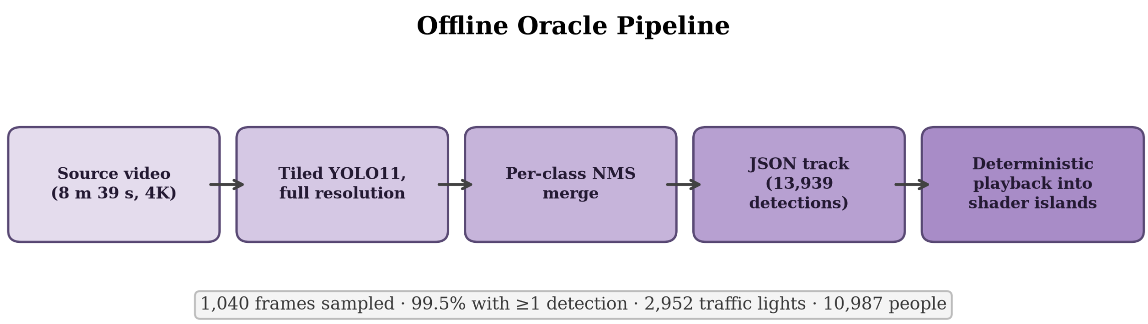

Figure 4.10: The offline oracle pipeline, from source video to deterministic playback.

4.9 The Video Test Scene

All dynamic-scenario evaluation runs in a dedicated harness scene in which the passthrough layers are replaced by an 8-minute-39-second equirectangular driving clip decoded onto the sphere. The scene preserves everything else, the same shader, window logic, modes, interaction, and detection islands, so a mode behaves identically over video and over the live camera feed, while gaining what live passthrough can never provide: **identical, repeatable, event-dense stimuli for every participant**, plus a deterministic ground-truth detection track. The harness is thus the simulation-based evaluation instrument of Chapter 5's driving block.

The equirectangular mapping exposes UV offsets and a vertical flip so any source clip's forward direction and horizon can be registered without re-encoding the video, and a live-inference path maintains a dedicated downsampled render target for the detector via Unity Sentis (Unity Technologies, 2024) so it never reads the 4K texture directly. For evaluation that path is bypassed in favour of the baked track.

Two harness-specific notes have study consequences. In the video scene SignPop's grading covers the full sphere including the window, which is the intended driving-block configuration. And the video manager's motion-suppression master switch defaults to **false** while the driving block specifies suppression active, so the session sequencer asserts the required configuration at start-up rather than leaving it to the scene's saved state.

4.10 The Failure Museum

This section reports the failures that shaped the system, on the view that documented dead ends are reusable knowledge: each with what was tried, why it failed, and what replaced it.

1. Screen-space camera sampling → double vision. The first shader sampled the camera texture by screen position. Per-eye the image was correct. Binocularly it could not fuse, because each eye sampled a different camera pixel for the same world point. The on-device symptom was frank double vision. Replaced by world-direction sampling through a head-centred pinhole (Section 4.4), verified fused on device. Lesson: in stereo rendering, eye-independence of the sampling domain is a correctness criterion, not an optimisation.

2. Surface-projected passthrough → deprecated API. The obvious platform mechanism for putting real passthrough on application geometry was evaluated and rejected because it is deprecated as of SDK v83. Any architecture built on it would be built on removed ground. Replaced by the window-as-absence composite, which uses only stable compositor behaviour. Lesson: on young platforms, prefer primitives the OS must keep, such as alpha compositing, over conveniences it may withdraw.

3. `Shader.Find` on device → null. Runtime materials created by looking a built-in shader up by name worked in the editor and silently returned null on Quest, because Android builds strip shaders that no serialized asset references, and the failure surfaced as a crashed startup coroutine rather than an error. Replaced by a serialized template material that runtime dots clone, since the asset reference guarantees build inclusion. Lesson: never construct device materials from name lookups. Ship a serialized template.

4. Downsampled detection baking → missing traffic lights. The first oracle bake ran YOLO on a 640×360 downsample of the 4K source. Distant traffic lights fell below detectable pixel size and the track silently omitted most of them, an oracle that was not an oracle. Replaced by the full-resolution tiled baker of Section 4.8.2, raising coverage to 99.5% of samples. Lesson: validate ground-truth pipelines against the *smallest* objects they must contain, and never let the convenience path overwrite the reference track.

5. Two operational traps, recorded for the protocol rather than the architecture. The platform's camera-based screenshot service returns black frames whenever the application holds the passthrough camera, because the two contend for the same resource, so the study's redundant session recording must use framebuffer capture instead. And the headset sometimes boots with controllers in `CONNECTED_INACTIVE`, leaving the application with no input at all and no way to distinguish that from a broken build, so the study build boots straight into the experiment scene, hand tracking is disabled, and waking the controllers is a line on the pre-session checklist. Lesson: input liveness is an environmental precondition, so it belongs in the protocol rather than the debugger.

4.11 Technical Validation: Frame Cost and Legibility

Two claims have carried this chapter so far on structural argument alone: that the camera-free dark modes are cheap enough to deploy for long sessions, and that the native passthrough served through the window is visibly better than the Tier-3 camera re-render it spares the user from. Both were measured on the study hardware on 7 August 2026, against criteria fixed before the data was seen.

4.11.1 Per-mode frame cost

Frame cost was captured with the platform's OVR Metrics Tool, which logs one row per second of the application's GPU frame time, stale-frame count, and per-core CPU and GPU utilisation. One continuous run in the passthrough scene stepped through effects off, Soft Dark, Hard Dark and Blur, and a second run in the video scene covered SignPop and effects off, with every switch logged against the wall clock. Five seconds are trimmed from both ends of each segment so formation transitions do not contaminate a segment's statistics, leaving 79 to 245 samples per configuration. The pre-stated pass criterion for the study configurations is a 95th-percentile GPU frame time within the 13.88 ms budget of the 72 Hz display and a stale-frame rate below 1%. Blur was pre-stated as an interpretation band rather than pass/fail, since its cost relative to Soft Dark is the price of Tier-3 camera re-rendering either way. Table 4.2 gives the results and Figure 4.11 plots them against the budget; the analysis script and raw captures are in the project record.

Table 4.2: Per-mode frame cost from the OVR Metrics captures. GPU times are the application's own frame time in milliseconds against the 13.88 ms budget; stale rate is stale frames as a share of display frames; CPU is the worst core's mean utilisation.

Configuration	**GPU p50**	**GPU p95**	**Stale**	**CPU worst**	**GPU util**
Effects off (passthrough)	3.23	3.37	0.30%	68%	41%
Soft Dark	3.25	3.34	0.00%	62%	41%
Hard Dark	3.17	3.32	0.02%	61%	41%
Blur	8.01	8.15	0.09%	63%	79%
SignPop (video scene)	8.62	11.42	7.29%	48%	71%
Effects off (video scene)	10.07	12.36	7.89%	49%	79%

Three findings. First, **the dark modes cost nothing measurable**: their 95th-percentile frame times sit within 0.05 ms of the empty passthrough scene, at a quarter of the budget, with stale rates at or near zero. The deployability argument of Chapter 3 is now empirical rather than structural. Second, **Blur is the price of Tier-3 re-rendering made visible**: two and a half times the dark modes' frame time and nearly double the GPU utilisation, yet still comfortably inside the budget, so the Tier-3 ceiling on this hardware is thermal and sustained-load, not per-frame. Third, **the video scene misses the stale-frame criterion, in both arms equally**: 7.3% with SignPop on and 7.9% with it off, against the 1% threshold, while 95th-percentile GPU time stays inside the budget. The miss is a property of the shared 4K video-decode path, not of the treatment, and SignPop's own grading adds no stall on top of it. The paired Block B comparison is therefore undisturbed, since both conditions ride the same pipeline, but the configuration misses its pre-stated criterion and is reported as missing it.

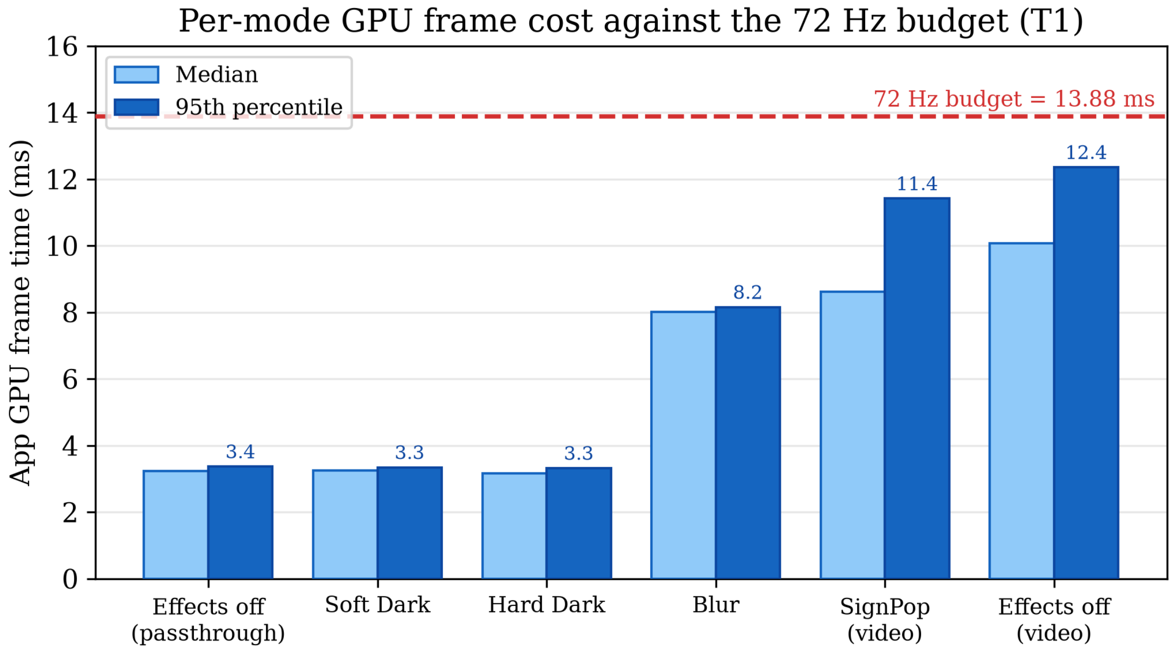

Figure 4.11: Per-mode GPU frame time against the 72 Hz budget.

4.11.2 Legibility across the two rendering paths

The window-as-absence architecture rests on the claim that native passthrough through the window is visibly better than the camera re-render that Tier 3 could put in its place. To measure it, a Landolt C acuity chart covering logMAR 1.3 to 0.0 in 0.1 steps, sized for a 600 mm viewing distance, was printed at 100% scale and fixed at 600 mm from the headset, which rested on a stable support and did not move. A debug toggle switched the whole view between the two paths: native passthrough, and the full-field camera re-render with the grading neutralised at runtime, so the comparison isolates the rendering path rather than any mode's styling. Each segment identifies itself in the recording through an on-screen debug label. The pre-stated criterion: the architecture is vindicated if the native path resolves at least one logMAR step finer than the camera path.

A cast recording is a valid instrument here because both paths share everything downstream of the compositor, so any resolution difference originates upstream and survives into the capture. The capture (1280×685 per eye) undersamples both paths, the native path far more severely than the camera path, and the camera path additionally renders the chart larger in frame because the physical camera sits closer to it than the eye does. Both effects compress or favour the camera path's side of the difference, so the measured gap is a lower bound on the true one, which is the right direction for a one-sided criterion. Figure 4.12 shows the same chart through both paths, from the same recording, seconds apart.

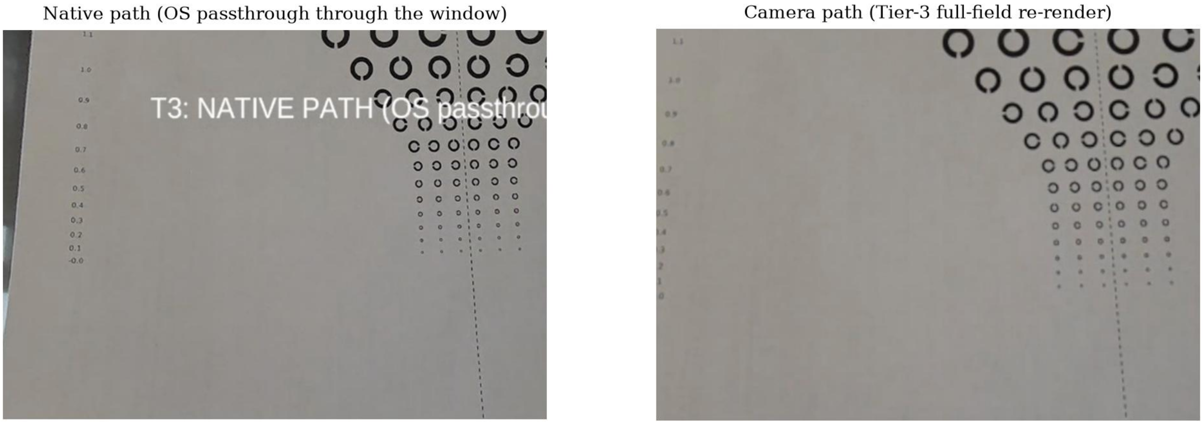

Figure 4.12: The Landolt C chart at 600 mm through the two rendering paths, from one continuous cast recording with the headset stationary. Row labels are logMAR values. The native path holds ring-gap orientation about two 0.1-logMAR rows further down the chart than the camera path.

The criterion is met with margin. In the capture, the native path holds ring-gap orientation clearly through the 0.6 row and marginally to 0.5, while the camera path's gaps blur closed below the 0.8 row. That is a native advantage of at least two 0.1-logMAR steps, roughly a factor of 1.6 in resolvable detail, despite every capture bias running the other way. The application's frame-rate overlay held 72 fps in both segments, so neither path bought its appearance with dropped frames. The limits of the method are stated plainly: one chart, one distance, one lighting condition, and a legibility read taken from the recording by the author rather than a psychophysical procedure with naive observers. Within those limits, the measurement confirms what the architecture bet on, and what Wang et al. (2026) measured for passthrough acuity generally: the place where full pixel control exists is also the place where the image is worst, and the window spares the focus region exactly that penalty.

4.12 Summary

One sphere, one shader, one manager: the implementation realises Chapter 3's design space with a per-fragment colour and alpha policy over a world-locked spherical window. Its load-bearing techniques, the window as absence, the hybrid composite of native passthrough and processed camera feed, and eye-independent world-direction sampling, are each answers to a specific platform wall, and its failures are documented with the same care as its successes because both are findings about building DR on consumer hardware. Its two central trade-offs are measured on device: the dark modes cost nothing over an empty scene, and the native path the window preserves resolves at least two logMAR steps finer than the camera copy. Chapter 5 asks whether any of it helps a person.

Chapter 5

User Study: Design and Results

Status note. This study pre-registered a target of 20 analysed participants, run to a frozen protocol after a formal pass/fail pilot. *Pre-registered* is used throughout in the sense defined in Chapter 1, Section 1.4. Testing is complete: **twenty people were tested**, across an initial pilot day and eight further collection days, preceded by informal naive-pilot iteration used to tune stimulus parameters ahead of the frozen protocol, against a recruitment target of twenty. The pre-registered exclusions applied below remove one pair from the workstation block and three from the driving block. Nineteen usable pairs exist for the workstation block and seventeen for the driving block, at 79.8% and 89.0% achieved power for the primary contrast and the pre-registered eccentricity band, against the design's 80% target. All tests are two-tailed. The hypotheses are directional, and a one-tailed test would have been more sensitive to the improvement they predict, but a one-tailed test cannot register movement in the other direction at all. This study asks a safety question as well as a benefit question, so the possibility of harm was retained as a detectable outcome rather than defined away, and the two-tailed convention was fixed in the analysis plan before collection (Appendix A, Section A.6). Section 5.7 reports the final analysis of those pairs, read against the pre-committed decision table.

5.1 What the Study Must Be Able to Conclude

The purpose of this study is to determine whether the system of Chapters 3 and 4 *works*, and just as importantly to be able to determine that it *does not*. A study that could only confirm would be a demonstration rather than an experiment. Two design choices commonly produce that defect: an underpowered between-subjects comparison, and safety hypotheses phrased as bare nulls, since a non-significant difference in a small sample carries little evidential weight.

The design maps every outcome to a pre-committed conclusion by reframing the question around an effect already established in the literature. Peripheral visual distractors reliably impose a measurable performance cost on a focal sustained-attention task in a head-mounted display: in a virtual-classroom Go/No-go study, commission errors more than doubled in the presence of peripheral distractor events (Ai et al., 2025, $N = 66$). The primary question is therefore not the unconstrained "does the vignette make people better?" but the falsifiable:

> *Peripheral distractors impose a measurable cost on a focal task. Does the DR vignette reduce that cost?*

Under this framing every outcome is informative. The design relies on that established literature for confidence that the distractor reel imposes a real cost, rather than on an internal manipulation check, and that dependency is carried into the limitations.

Two methodological commitments govern the study's position. The driving block is a **simulation**, with participants viewing pre-recorded equirectangular footage rather than a live street, in the tradition established by Cheng et al. (2022), which gives every participant identical, repeatable, event-dense stimuli at a declared cost to ecological validity. And detection comes from the **offline oracle** of Chapter 4, Section 4.8.2 rather than a live model, so the study can ask whether an effect built on excellent detection helps attention without first engineering excellent detection on-device.

The two confirmatory blocks sit at opposite ends of that spectrum. Block A runs on **real passthrough with real physical distractors** and evaluates the artefact as built. Block B runs on the simulated video scene and evaluates the concept under the ideal-context assumption. Convergence strengthens both, and divergence is itself a reportable finding about what simulation-based XR evaluation misses.

The design is fully within-subjects with a high trial count per condition, following the two closest published DR evaluations (Cheng et al., 2022; McLaughlin et al., 2025) and the field norm Caine (2016) documents, where in-person studies draw their power from repeated measures rather than headcount. Each participant contributes up to 140 task events per Block A condition and 40 marker events per Block B condition, so the nineteen analysed workstation pairs contribute 4,681 file-backed scored Block A trials (task form and provenance: Section 5.7).

5.2 Research Questions and Pre-Registered Hypotheses

RQ1 (primary). Does peripheral DR dimming (Hard Dark, world-locked window) reduce the performance cost that peripheral visual distractors impose on a focal workstation task?

RQ2 (secondary). Does detection-gated salience re-grading (SignPop) improve detection specifically where the gate has room to act, without degrading detection pooled across the whole field of view beyond an acceptable margin?

RQ3 (exploratory). After experiencing the system, what do participants report about perceived focus benefit, perceived awareness cost, and visual comfort?

Every hypothesis is stated with the observation that would refute it, as Table 5.1 sets out. A hypothesis without a refuting observation is not admitted to the design.

Table 5.1: Pre-registered research hypotheses, their refuting observations, and confirmatory status.

ID	Hypothesis (directional, falsifiable)	Refuting observation	Status
H1	Error rate on the Block A task, with distractors present throughout, is lower with the vignette on than off, by at least the smallest effect size of interest (default $\geq$ 50% reduction from the vignette-off error rate, finalised from pilot data).	The paired difference's 90% CI excludes the SESOI, or the difference is absent or reversed.	Primary confirmatory
H2a	With SignPop on, marker-detection hit rate in the 20–30° eccentricity band is higher than with the vignette off. This is the band where the colour-pop gate has visible room to act but the target is still within useful glance range.	Band-level difference's CI includes 0 or favours off.	Secondary confirmatory

ID	Hypothesis (directional, falsifiable)	Refuting observation	Status
H2b	With SignPop on, hit rate pooled across the whole field of view is *equivalent* to off within a margin of 10 percentage points (two one-sided tests).	Equivalence not established **and** the point estimate shows a drop > 10 pp. A conclusive finding that the mode harms situational awareness.	Secondary confirmatory (safety)

H2a was pre-registered on the 20–30° band specifically, before any data existed, because the band follows from the system's own geometry. In the scene the user study ran, the colour-pop soft edge is 24°, so the gate reaches half strength near 20° and full strength near 32°, while the useful field of view runs out around 30° (Chapter 4, Section 4.2.2). That annulus is where the gate has room to act on a target the eye can still use. The four-band breakdown reported in Section 5.7.3 is a secondary analysis around that pre-registered band, not a search across bands for one that moved.

Margins. H1's smallest effect of interest is expressed as a proportion of the vignette-off error rate rather than a raw count, because the raw rate depends on stimulus tuning the pilot legitimately adjusts. The 50% default embodies a practical judgement: a vignette that removes less than half of the distraction-laden error rate does not justify wearing a headset to obtain it. H2b's 10-point margin reflects the safety framing inherited from the focus-versus-awareness trade-off documented by McLaughlin et al. (2025) and Murph et al. (2021). Losing more than one detection event in ten to the overlay was judged an unacceptable awareness cost for a monitoring context. Equivalence testing follows Lakens (2017), which gives the safety hypothesis a form that can fail rather than a bare null.

5.3 Participants, Ethics, and Apparatus

The study was reviewed and approved through the School of Computer Science and Statistics research ethics process (REAMS application *Enhance user focus in real world using XR*, principal investigator the author, supervisor John Dingliana; approved 27 May 2026, reference 6061) before any participant was recruited. The blank participant information leaflet (version 2.0, 13 May 2026) and informed consent form are reproduced in Appendix A, Section A.9; signed originals are retained by the supervisor under the School's retention policy rather than submitted, since they carry personal data. Participants received the information leaflet at least 24 hours before their session and signed consent at the start of the session itself.

Recruitment targeted 20 analysed adults with spares against dropouts and pre-registered exclusions; twenty people were tested in all, completing the full two-block protocol (Section 5.7). Inclusion criteria follow the approved ethics protocol: aged 18 or over, normal or corrected-to-normal vision by self-report, no self-reported vestibular disorders, and no history of epilepsy or photosensitive conditions. Recruitment is by direct approach from the postgraduate computer-science cohort, unpaid. No demographic instrument was administered, so age, gender and background distributions cannot be reported; the inclusion criteria and recruitment route above are what a reader has to judge the sampling frame by, and the absence of demographic data is carried as a limitation in Chapter 6. Prior VR experience is recorded on a four-level scale rather than used as a selection criterion, since the deployed system must serve VR-naive users; it is not analysed further in this dissertation.

Sickness risk is managed by design and by rule. All tasks are seated, there is no artificial locomotion, total headset time is approximately 35 minutes, and a headset-off break separates the two confirmatory blocks. A written **stop rule** is applied without discussion: the VR portion terminates immediately if the participant reports nausea or dizziness at a moderate or worse level, or if the experimenter observes distress. The Virtual Reality Sickness Questionnaire (Kim et al., 2018) is administered pre and post, because post-only scores are uninterpretable without a baseline (Brown et al., 2022). Across all twenty sessions the stop rule was never triggered and no session was terminated. One participant reported feeling unwell, rested during the mid-session break, and chose to complete their remaining runs. The session timeline, pilot gates, analysis plan, decision table, operational protocol, threats to validity and exclusion rules are set out in full in Appendix A.

All sessions use the same Meta Quest 3 with controllers, in the same room, with bright constant lighting at a marked setting. Quest 3 passthrough acuity degrades markedly in low light (Wang et al., 2026), so lighting is a controlled variable rather than a convenience. Every stimulus is deterministic and scripted from versioned schedule files, and no live object detection runs anywhere in the study.

Within each block the stimulus material, seating, lighting, controller and task are identical across conditions, and only the shader state differs. In vignette-off conditions the focus window is still painted and locked by the participant, so the motor and procedural experience is identical across conditions and the vignette is isolated as the sole independent variable.

5.4 Block A: Primary, Workstation Focus

The participant sits at a desk with the task display directly ahead, approximately 0.6 m from the eyes. The display carries three regions of a single task page: the shape stream in the centre, and two **distractor panels** at the display's left and right edges, centred at approximately **±35° azimuth**. Rendering the distractors on the task display itself, rather than on separate devices, fixes their eccentricity relative to the task instead of leaving it to furniture placement. That eccentricity places the distractors in the near-periphery zone of maximal involuntary attentional capture identified by the useful-field-of-view literature (Ball and Owsley, 1993). Note that Ball and Owsley define and validate the UFOV *test*; the ≈30° extent used throughout is the conventional figure from that literature rather than a value stated in their paper. The panels play a prepared distractor reel of fast-cut, high-motion, bright video, embedded in the task page, started with the run and dark outside it, identical across participants, and running continuously in every condition. Figure 5.1 gives the plan view.

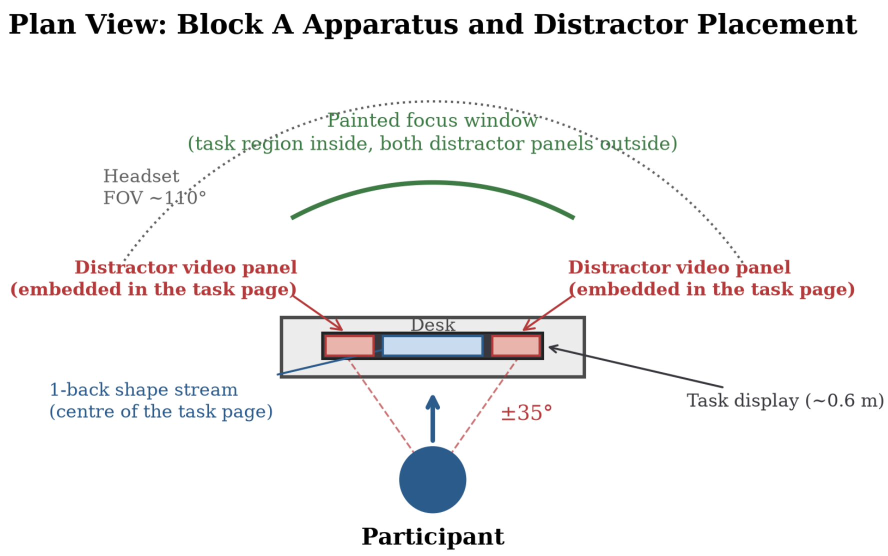

Figure 5.1: Plan view of the Block A apparatus and distractor placement.

The two conditions are **NoFilter**, where the window is painted but invisible, and **Filter**, Hard Dark on. In the 140-trial form each lasts 4 minutes 40 seconds, which is 140 trials at a 2.0 s stimulus onset asynchrony, run once each per participant in counterbalanced order. Earlier sessions ran the shorter form of the same task (Section 5.7). Before the first condition the participant paints the focus window around the central task region and releases to lock it, and the experimenter verifies that both distractor panels fall outside it and the entire task region inside. The same locked window persists across both conditions. Hard Dark's job here is to occlude the distractor panels from view entirely, not merely dim them at a distance.

The task is a forced-choice 1-back. The centre of the task display, inside the focus window, presents one look-alike shape at a time, drawn from a set of six at $\geq 3°$ visual angle, and the participant answers yes or no on every shape to "same as the previous shape?" via trigger press. Two choices in that sentence are load-bearing. The task is seen through the window's untouched native passthrough and its shapes are deliberately large, because Quest 3 passthrough does not reach normal visual acuity and a task requiring fine detail would floor out in every condition for reasons unrelated to the vignette; the shape stream's legibility through passthrough is a pilot gate (Appendix A, gate G1), not an assumption. And it is forced-choice rather than Go/No-go, because a Go/No-go stream ceilinged out at 96–100% accuracy in piloting regardless of condition, which would leave the block unable to detect any vignette effect at all. The 1-back adds a working-memory component on top of the reactive one, and its memoryless 50%-repeat sequence keeps performance off ceiling.

Error rate, the primary dependent variable, is misses plus commissions, logged to the millisecond. No per-trial correctness feedback is given during scored runs, only pacing feedback, so feedback cannot shift a participant's speed-accuracy strategy mid-run. The window uses room-geometry anchoring for this block specifically rather than the direction-lock used by default, so four corner rays captured on release are remembered as physical points and the window stays put as the participant leans toward or away from the desk. Head-turns toward the distractor panels beyond 30° yaw are recorded from 60 Hz telemetry as an exploratory measure; they are not analysed for this dissertation.

5.5 Block B: Secondary, Driving Marker Detection

The comparison is again Filter against NoFilter, and the task uses the driving footage's own content as the target. A ring marker is authored onto each real traffic-light or pedestrian object as it appears, and the participant presses the trigger on noticing it. Putting the target on a real object rather than on an abstract probe overlaid on the scene is what makes the measurement direct: the object under the marker is exactly the kind of object SignPop's detection gate is built to reveal.

Both conditions play the same clip in full, running 8 minutes 39 seconds each, so the two differ only in shader state and in which marker set is scored. The focus window is pre-set to the windscreen region and direction-locked for all participants, so the guided region stays constant across the sample. Condition order alternates by participant parity, and the two marker sets are alternated against condition so neither set is systematically tied to Filter or NoFilter.

Forty scored markers per condition, drawn from two disjoint sets of forty over the same clip, which is 1,360 marker presentations across the seventeen usable pairs. Using a different set in each condition stops a participant meeting the same target twice, which would turn the second condition into a memory test. Each marker fades in rather than switching on, because sudden onsets capture attention automatically and would swamp the effect being measured (Yantis and Jonides, 1984), and it modulates the pixels underneath rather than covering them, so it keeps its own internal contrast even where the filter has dimmed everything under it (Wallis et al., 2015). **The marker itself is dimmed by the same falloff as the rest of the scene regardless of condition, so a filtered marker is never easier to see as a marker.** Only the object beneath a detected marker is treated differently between conditions. Seventy of the eighty authored targets fall inside an active detection window while on screen. Figure 5.2 shows the marker's construction, rendered directly from the shader's arithmetic at the study's parameters.

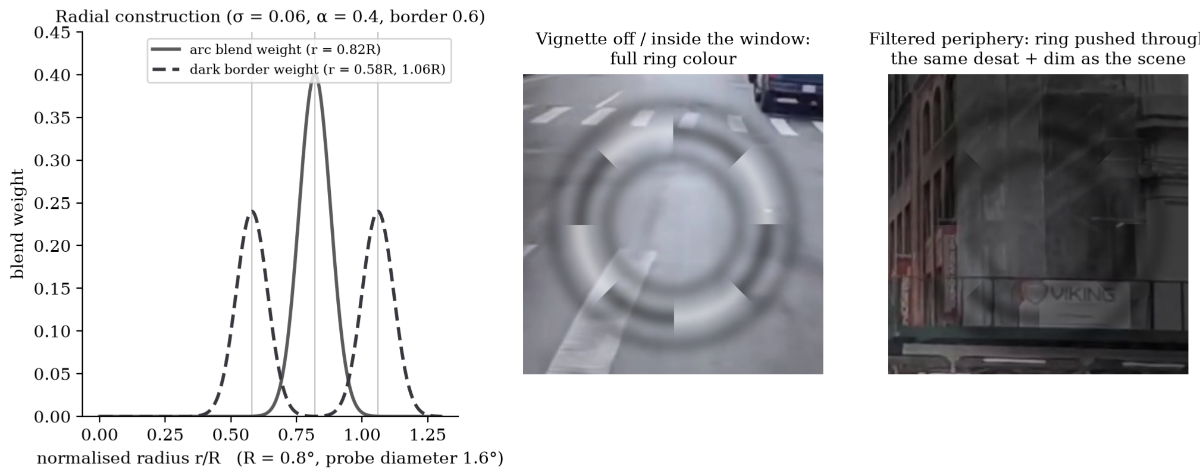

Figure 5.2: The ring marker, constructed from the shader's own arithmetic at the study configuration and shown enlarged (the probe subtends 1.6° in the headset). The marker is a translucent "rope": four alternating black-and-white arc pairs on a Gaussian annulus at 0.82 of the probe radius (40% opacity), flanked by two multiplicative dark borders. The pattern is achromatic, so the filter's desaturation cannot erase it, and its polarity alternates, so some arc contrasts with any background it lands on. The marker fades in over 0.5 s rather than onsetting abruptly (Yantis and Jonides, 1984). Centre and right: the same marker over an unfiltered and a filtered region of the driving scene. The borders darken multiplicatively and the white arcs are pushed through the same desaturation and dim the grading applies to the scene at that pixel, so the pattern's internal contrast survives while the marker as a whole dims like a real object rather than advertising itself (Wallis et al., 2015).

A **hit** is a trigger press while the marker is on screen, within 10° of its position at that instant. Presses landing on nothing are false alarms. Markers move while visible, by 17° of travel on average, so scoring against a marker's average position misassigns a meaningful share of presses to the wrong analysis band. The intended method is therefore to score every press against the marker's reconstructed position, from the recorded scene track, at the exact instant of the press. That reconstruction is not yet implemented in the analysis pipeline, so the band table reported in Section 5.7.3 assigns each target by its mid-lifetime position instead, and the size of the resulting misassignment is stated there as a limit on the band finding.

Eccentricity bands. Each scored marker is assigned to one of four bands, fixed from published gaze landmarks before any data was seen. Driver gaze concentrates in a road-centre region of roughly 8° radius (Victor et al., 2005), so the under-10° band covers what a viewer is looking at anyway. The Peripheral Detection Task, the standard instrument for measuring spare attentional capacity under load, places its probes just outside that region at 11–23° horizontal eccentricity (Jahn et al., 2005), and the Detection Response Task standardises the same logic with a fixed, background-independent stimulus (ISO, 2016). The 10–20° and 20–30° bands bracket that published range. The outer boundary is the conventional useful-field extent of roughly 30° (Ball and Owsley, 1993), past which a stronger peripheral signal has little left to buy. One point of fit is worth stating precisely, because the 20–30° band carries H2a: the published probe range stops at 23°, so it covers 10–20° completely but only the lower part of 20–30°. The boundaries bracket the published landmarks rather than reproducing any one study's geometry.

Figure 5.3 places the four bands against the published landmarks they were drawn to bracket.

Head pose serves as the attention proxy in place of eye tracking, which the Quest 3 does not provide. Head orientation is a validated gaze surrogate at the eccentricities used here (Higgins et al., 2022; Sitzmann et al., 2018).

5.6 Block C, and What It Cost to Cut

Block C answers RQ3 and carries no confirmatory claims. A three-mode sampler was specified here, 75 seconds each of SignPop, Soft Dark and Hard Dark on common footage, and **it was cut from the running protocol** because the two confirmatory blocks already account for roughly 35 minutes in the headset and the sampler would have added two further exposures at the point of maximum fatigue. Two consequences follow and are carried through this dissertation rather than absorbed quietly. Soft Dark was never shown to a participant, so no participant data on it exists (Chapter 3, Section 3.4). And

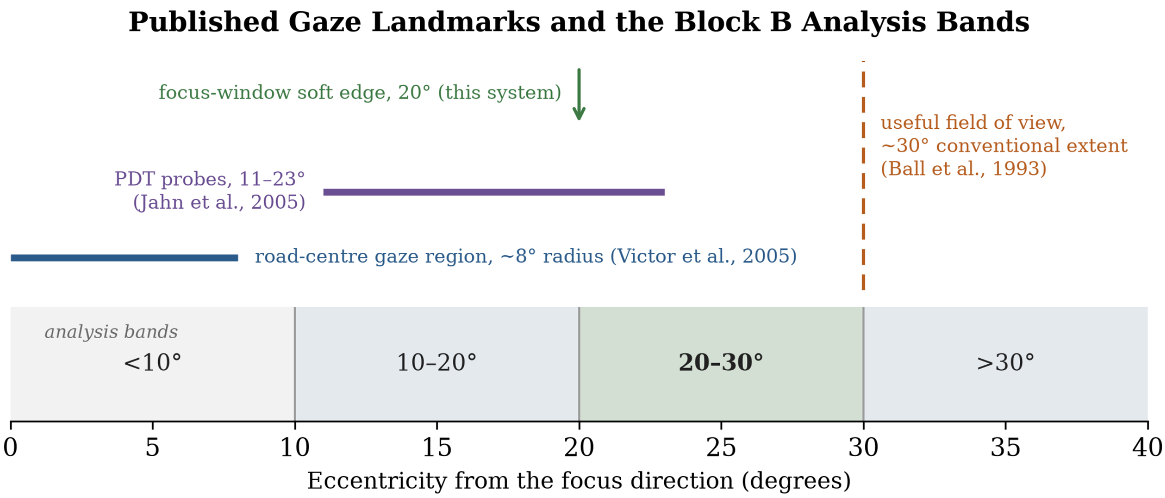

Figure 5.3: The four Block B analysis bands against the published gaze landmarks that fixed their boundaries: the road-centre gaze region (Victor et al., 2005), the Peripheral Detection Task probe range (Jahn et al., 2005), the conventional useful-field extent (Ball and Owsley, 1993), and the system's own soft edge, which in the study scene reaches half strength near 20° and full strength near 32°. The published probe range stops at 23°, so the 20–30° band is only partly covered by it.

no participant ever saw two modes on a common stimulus, so Hard Dark is tested only at the workstation and SignPop only in the driving scene. Every confirmatory claim here is therefore a *mode-in-context* claim, never a mode-generalised one, and nothing in the design softens that confound.

The instrument is a single end-of-session questionnaire, self-completed with the headset off after the post-session sickness questionnaire and before the debrief, so opinions are captured before the hypotheses are revealed. It carries 15 rated items across three sections, one section per scored construct: perceived focus benefit, perceived awareness cost and visual comfort, each mixing positively and negatively keyed items to counter acquiescence bias. The first construct is the subjective counterpart of the workstation block's focus question and the second of the driving block's awareness question, so each has a behavioural result of its own to be read against. Because it asks for one considered opinion of the whole session, it cannot separate Hard Dark's contribution from SignPop's. It is a system-level instrument, reported as exploratory.

5.7 Results

Twenty people were tested and testing is complete. The pre-registered twenty completed the full two-block protocol, so no supplementary participants were run. The completed sample delivers 79.8% achieved power for the primary contrast and 89.0% for the pre-registered eccentricity band, both two-tailed as registered, against the design's 80% target.

Block A yields **nineteen** usable paired participants and Block B **seventeen**, the two sets overlapping but not identical because some participants contributed a usable result to one block and not the other. One participant is excluded from Block A: P1's task rounds ran about fifteen minutes after the session's Unity blocks closed, so the arm assignment cannot be verified from the session ledger. That exclusion is specific to Block A, and P1's Block B pair is unaffected and is analysed. Three participants are excluded from Block B under documented rules, never inferred ones. Two fall under the pre-registered response-validity rule, which excludes a pair whose false alarms across both conditions together outnumber its eighty scheduled markers (Appendix A, Section A.4). P3's filter run alone logged 539 false alarms, with provably looser hits (Mann-Whitney $p = .0301$). P43's two arms logged 135 together. The highest retained pair total is 66, so the achieved data sit clear of the line on both sides. The third is P9, the one discretionary exclusion, made by experimenter decision on the day on task-comprehension grounds: the response rule was not yet understood during that participant's first video block, which was their first exposure to the task. Separately, P15 and P16 are two ledger IDs for one participant, reconciled into a single record under the relaunch rule of Appendix A, Section A.4. That participant's P15 Block B run is excluded for 206 false alarms, so the P16 pair is their Block B record and the P15 pair their Block A record. Block A is retained for P3, P9 and P43, since those exclusions concern the marker task alone. The arithmetic closes on the twenty tested: nineteen contribute a Block A pair, the twenty less P1, and seventeen contribute a Block B pair, the twenty less P3, P9 and P43.

Two provenance facts belong with the numbers rather than in a footnote. The first is the study's **one protocol deviation**, and it is reported here as one. Block A pools a short and a long form of the same task: five participants ran the short form before the trial count was raised, once, to 140, and the remaining fourteen ran the 140-trial form. Participant numbers are ledger-assigned rather than collection order, so the split is chronological and is not read off the identifiers. The change was made deliberately and for one stated reason, to move performance off the ceiling the short form sat on, and the achieved data show it partly worked. Across the five short-form participants the mean accuracy over their file-backed arms was 96.1% with a floor of 84.6%, leaving little room to detect a benefit; across the fourteen who ran the 140-trial form it was 92.5% with a floor of 74.8%, so participants after the change do miss, and the measure has room to move in both directions. Nothing else about the protocol changed after collection began: no hypothesis, no margin, no decision-table row and no exclusion rule. Pooling the two forms is justified by the robustness split reported below, where the two halves of the sample return the same effect at the same size. And one of the thirty-eight Block A arms, P4's filter-off arm, is the experimenter's contemporaneous record rather than an exported file, which is why the trial count in Section 5.1 is given as file-backed. Dropping that arm changes nothing ($p = .0085$ against .0104, d_z unchanged at 0.678).

5.7.1 H1: workstation focus

The pre-registered primary test. Table A.4 registers H1's primary test as a trial-level logistic model with a participant random effect, and the paired test as its robustness check. Across the 4,681 file-backed scored trials from nineteen participants the vignette term is significant: **odds ratio 0.600** (95% CI 0.431 to 0.836, $z = -3.020$, $p = .0025$), so Hard Dark **reduces trial-level error odds by 40.0%**. The estimator fitted is a logistic GEE with an exchangeable working correlation clustered by participant, because statsmodels offers no strict frequentist mixed model with p-values, a substitution already documented in `Tools/analysis/h1_lmm.py`. The literal random-intercept specification agrees, at odds ratio 0.594 with a 95% credible interval of 0.496 to 0.712. Trial-level models use 37 of the 38 Block A arms, because P4's filter-off arm has no exported file. The robustness check agrees in direction and significance, and is the weaker of the two because it averages within participants before testing rather than using the trials themselves.

The registered robustness check. Accuracy on the memory task rose from 92.01% (SD 6.57) with the vignette off to 95.08% (SD 4.42) with Hard Dark on. That is a gain of **+3.07 percentage points** (SD 4.52, median +2.16; 95% CI +0.89 to +5.25, paired $t(18) = 2.95$, $p = .009$; Wilcoxon $W = 23.0$, $p = .007$; $d_z = 0.678$). Fourteen of nineteen participants improved, four declined, one was unchanged. Achieved power at this sample and effect size is 79.8% two-tailed, against the design's 80% target.

H1's direction is confirmed; its pre-set size is not, and the decision table is applied as written. Both the parametric and the rank test agree, the rank test, the more conservative of the two here, returns the lower p-value, so the result does not depend on the normality assumption, and the refuting observation, a difference absent, reversed, or with its 90% interval excluding the smallest effect size of interest from below, did not occur. But the hypothesis asked for more than a direction. Its default smallest effect size of interest was half of the vignette-off error rate, which at the achieved data is 3.99 of 7.99 points, and the observed reduction is 3.07 points, 38% of the distractor-laden error rate, with a 90% interval from 1.27 to 4.87 that spans the 3.99-point mark. That is row 4 of the pre-committed decision table: *the benefit is real-signed but cannot be claimed at the pre-set size.* The bounded conclusion the table prescribes is therefore the claim: Hard Dark removes between roughly 16% and 61% of the error rate that distractor-laden work carries (90% CI), with 38% the best estimate, a real benefit that falls short of certainty at the half-the-errors bar the design set itself.

The estimate itself is stable. It barely moved as the sample grew, from +3.34 at fourteen pairs to +3.09 at seventeen and +3.07 at the final nineteen, which is what a real but modest effect looks like under accumulation. One robustness split deserves stating rather than burying: the five who ran the shorter form against the fourteen who ran the 140-trial form, where the two groups give the same effect at the same size, +3.30 and +2.98 points. What differs is spread (SD 2.62 against 5.12), because the early task sat near ceiling and compressed every delta, while the harder longer form moved performance off ceiling at the cost of individual variation. Pooling therefore combines two measurements of one effect rather than two different effects. Figure 5.4 shows every participant's pair.

5.7.2 H2b: the pooled safety check

Hit rate pooled across the whole field of view rose by **+7.35 percentage points** with SignPop on, from 50.15% (SD 15.60) to 57.50% (SD 9.14), with ten of seventeen participants improving ($t(16) = 2.19$, $p = .0434$; Wilcoxon $W = 29.0$, $p = .0435$; $d_z = 0.532$). The 90% confidence interval runs +1.50 to +13.20, entirely above zero. Achieved power for this pooled contrast is **54.0% two-tailed**, well below the 79.8% and 89.0% the primary contrast and the pre-registered band reach, which is the first of two reasons this chapter treats the pooled rise as a description rather than a finding. The second is its leave-one-out behaviour, in Section 5.7.4.

H2b's refuting observation did not occur, so the safety claim stands. Section 5.2 committed in advance to what would refute H2b: equivalence not established *and* a point estimate showing a drop greater than 10 points. The estimate is a significant rise of 7.35 points, so the second condition is not met and the mode is not shown to harm situational awareness. The worst case the data support, at 90% confidence, is a gain of about one and a half points, far inside the ten-point loss margin.

The pre-registered statistic is nonetheless worth reporting exactly. The two one-sided tests return $p < .0001$ on the lower bound and $p = .221$ on the upper, so formal equivalence within ±10 points is not established at this sample. The reason is entirely one-directional: the data cannot exclude a *benefit* larger than ten points. That is a specification issue rather than a result. H2b asks a one-sided question, whether re-grading costs peripheral awareness, and a two-sided procedure additionally requires ruling out a large gain, which the safety framing never cared about. The instrument matched to the question is a non-inferiority test against a −10-point margin, which is the lower one-sided test above, and it passes decisively. Both numbers are given so the reader can see the

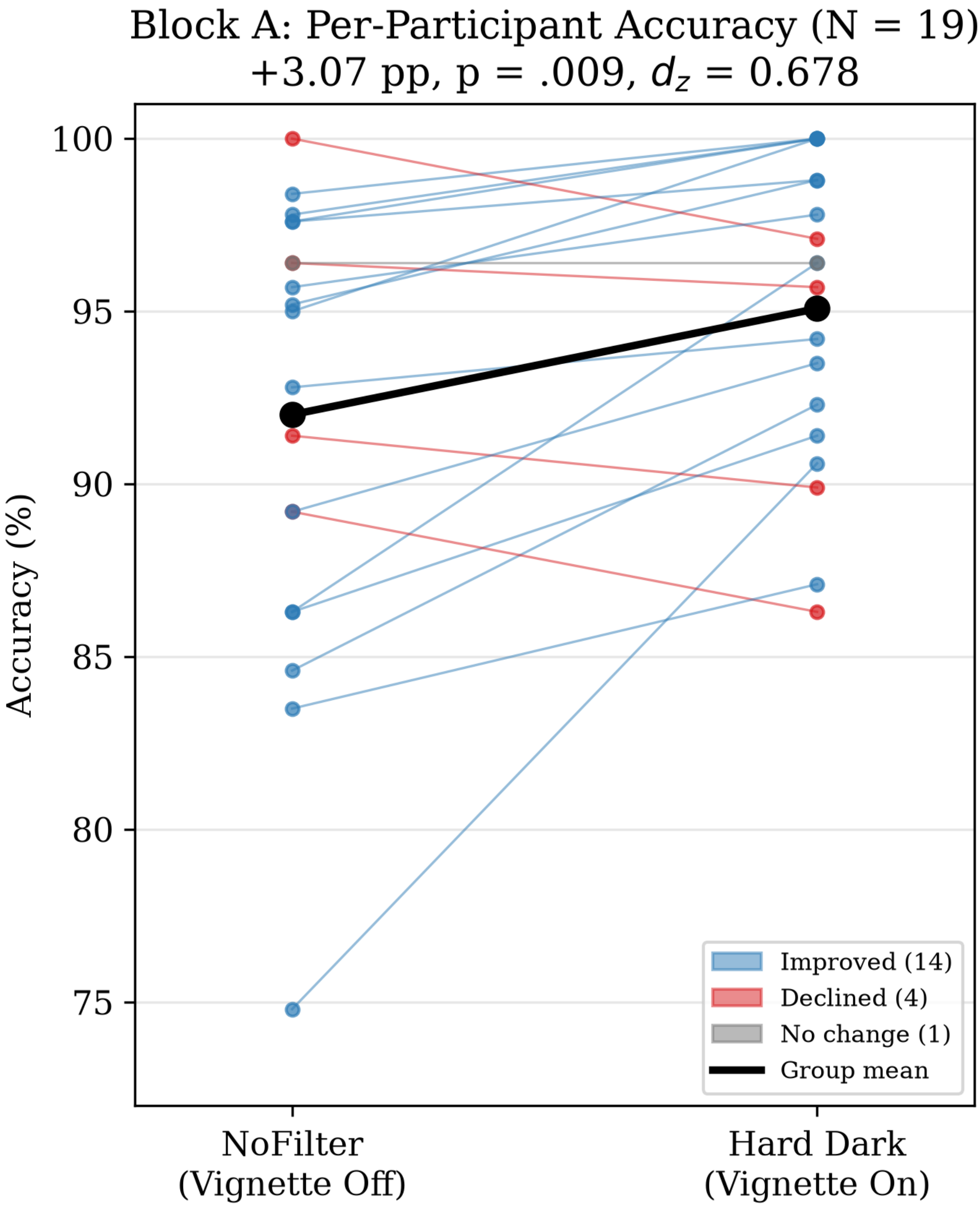

Figure 5.4: Block A per-participant accuracy, vignette off versus Hard Dark on (N = 19).

pre-registered procedure and the question it was meant to serve. What this dissertation does not do is quietly substitute the better-fitting reading for the pre-registered one after seeing the data. A study writing a safety bound should state the direction it cares about and test that direction alone.

A significant pooled rise was not a pre-registered hypothesis, and it is not promoted to one here. Raising pooled detection across the whole field was never SignPop's job, and the design bought power for the band contrast below, not for this pooled estimate. The rise is reported as an observed benefit with its interval, and one adjacent observation qualifies it: false alarms also rose under the filter, a mean of 11.6 presses per run against 6.9 with it off, with ten of seventeen participants pressing more. Hit rate does not correct for response criterion, so part of the pooled rise may be a more liberal response strategy rather than better detection. The exclusion rule bounds the worst of this, since the pairs excluded on false-alarm grounds held 557, 206 and 135 false alarms against a highest retained pair total of 66, and the band analysis below is unaffected in kind, but the pooled estimate carries this qualification. Figure 5.5 places the estimate against the margin.

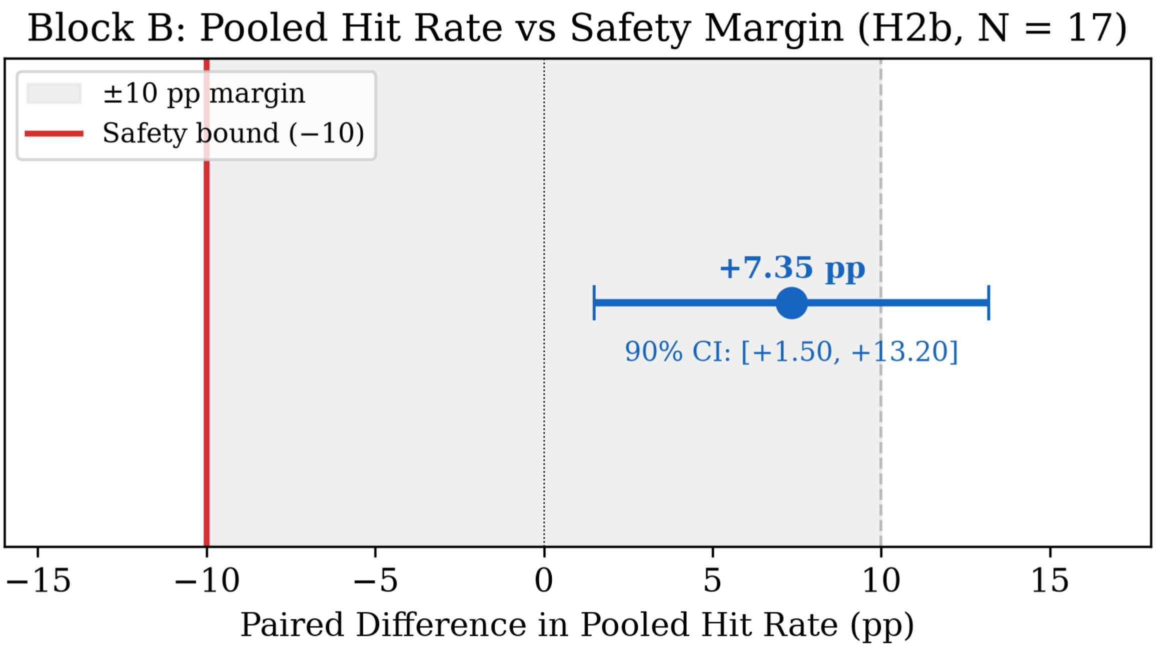

Figure 5.5: Block B pooled hit rate against the safety margin (H2b, N = 17).

5.7.3 H2a: the 20–30° band

Hit rate broken out by eccentricity band, in Table 5.2, shows a pattern rather than a uniform shift. Three of the four bands move in the predicted direction on the per-participant estimator and the innermost moves slightly against it, and only one moves reliably.

Two estimators are reported because they answer different questions and, in one band, disagree. The **pooled** rate weights every marker presentation equally and is what the bars of Figure 5.6 show. The **per-participant** mean weights every participant equally, and it is the estimator the paired tests and confidence intervals belong to. Participants contribute unequal numbers of markers to a band, so the two can differ, and in the under-10° band they differ in sign.

Table 5.2: Block B hit rate by eccentricity band. Pooled rates weight every marker presentation equally; the difference column and its interval are per-participant, which is what the tests use.

Band	**Off, pooled**	**On, pooled**	**Difference (per participant)**	**90% CI,** p
<10°	75.0%	78.3%	−3.68 pp	$[-12.0, +4.6]$, $p = .452$
10–20°	40.8%	50.2%	+9.74 pp	$[-0.2, +19.7]$, $p = .108$
20–30°	**38.9%**	**59.6%**	+20.82 **pp**	$[+10.1, +31.5]$**,** $p = .004$
>30°	54.4%	55.4%	+0.98 pp	$[-5.7, +7.7]$, $p = .802$

H2a is supported, in the band it was pre-registered on. The 20–30° band improves by 20.8 points, with fourteen of seventeen participants improving, and it is the only band significant on both the parametric and the rank test ($d_z = 0.82$, Wilcoxon $p = .008$, achieved power 89.0% two-tailed), so its refuting observation did not occur. It is twice the size of the next band. The neighbouring 10–20° band trends the same way at +9.7 points, but its 90% interval now spans zero and its tests do not reach significance ($p = .108$, 36% power), so it is reported as a trend rather than a finding.

The pre-registered multiplicity correction leaves both secondary verdicts unchanged. Appendix A, Section A.6 registers H2a and H2b as a two-member secondary family, Holm-corrected across the two. Applied at $\alpha = .05$, the matched non-inferiority test's $p < .0001$ is the smaller of the two and clears the .025 threshold, and the band's $p = .004$ clears .05, so both survive; had the two-sided equivalence statistic been carried into the family in place of the non-inferiority test, its $p = .221$ would have failed the correction exactly as it fails uncorrected. No secondary conclusion in this chapter depends on the uncorrected values.

What makes it more than a lucky slice is that the geometry was fixed before any data existed, and it was fixed against the driving-attention literature rather than against these results. The clear window is 8° by 6°, sized to the road-centre gaze region drivers actually watch (Victor et al., 2005), and the filter grades outward from its edge over 24°, reaching half strength near 20° and full strength near 32°. That places the graded zone across the 11–23° band the peripheral detection task uses as the region where events remain task-relevant while awareness is already falling (Jahn et al., 2005), and completes the ramp where the useful field of view runs out (Ball and Owsley, 1993). The measured benefit follows that curve: absent inside 10°, where there is almost no filter to work against, partial through 10–20° as the ramp opens, largest in the 20–30° annulus where the gate is near full and the eye can still use what it reveals, and flat beyond 30°, where the filter is at its strongest but the eccentricity is past the field the eye can act on. Two qualifications apply to reading these bands as retinal: the window is video-locked, so eccentricity here is scene eccentricity, and a participant who looks directly at a peripheral marker sees it through the filter.

Two limits belong beside the finding. Band membership in this table uses each target's mid-lifetime position, which is a weak proxy: targets move, the median lifetime swing is 17.2°, and roughly 31% of catches land in a different band from the one the mid position assigns. And with the pooled rate rising in every band while the inner band is near ceiling in both conditions, the honest reading is a general positive trend with the 20–30° band carrying most of the signal and its inner neighbour trending the same way, rather than three null bands and one live one.

5.7.4 Leave-one-out robustness

A small sample invites the question of whether any single participant is carrying a result, so the same test is applied to all three headline estimates rather than to whichever ones survive it. Each result was refitted with every participant removed in turn (`Tools/analysis/loo_sensitivity.py`).

Both pre-registered results hold throughout. H1 stays significant in all nineteen leave-one-out samples, with the worst case at $p = .0175$; dropping the participant who gains most (P15, +15.83 points) leaves it at +2.36 points, $p = .0091$. The 20–30° band stays significant in all seventeen, worst case $p = .0081$, and this holds even when the participant contributing its largest single gain is removed (+60.32 points; the band is then +18.35 points, $p = .0077$). The registered secondary therefore does not rest on any one

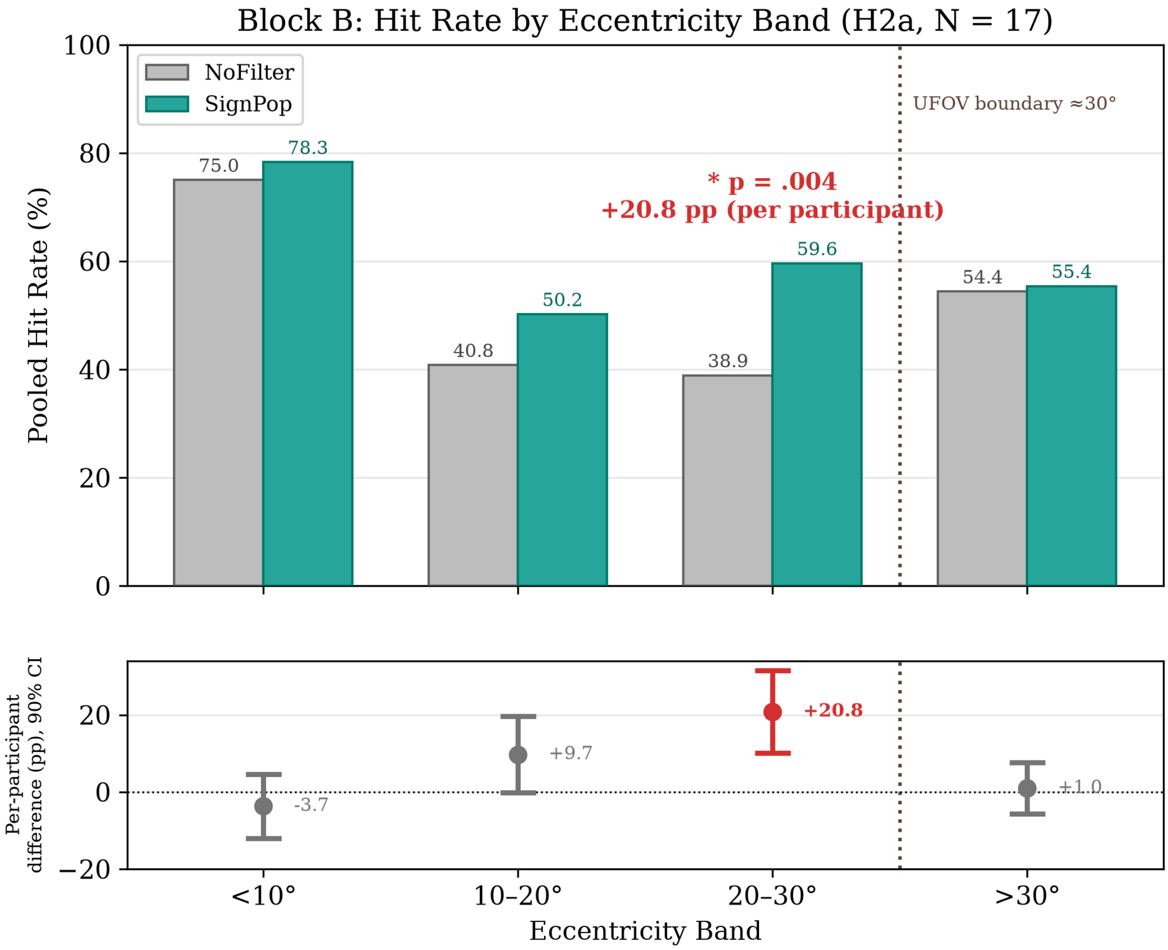

Figure 5.6: Block B hit rate by eccentricity band, with the useful-field boundary overlaid (H2a, N = 17). The upper panel pools every marker presentation in a band. The lower panel gives the paired per-participant difference with its 90% interval, which is the estimator the tests use, and which for the <10° band differs in sign from the pooled figure because participants contribute unequal numbers of markers to a band. Red marks the 20–30° band, the only one significant on both the parametric and the rank test; the starred annotation gives its per-participant difference.

participant, which is the specific thing a reader of a seventeen-pair result should want checked.

The pooled Block B rise does not survive the same test, and that is reported rather than smoothed: removing either of its two largest contributors returns $p \approx .086$, and it is significant in seven of seventeen leave-one-out samples. This is what a borderline result at this sample size looks like, and it is one more reason the pooled rise is not treated as a finding in this chapter. It does not disturb H2b's verdict. Every leave-one-out sample remains a rise, so the refuting observation Section 5.2 committed to, a drop greater than ten points with equivalence unestablished, does not occur in any of them.

5.7.5 What participants said, and where it disagrees

Twenty end-of-session questionnaires show a divergence worth flagging rather than resolving. Perceived focus benefit is strongly positive, 89 of 100 responses favouring the effects across all twenty respondents. Perceived awareness cost splits close to evenly at 58 of 100, so roughly four in ten responses report feeling cut off from the surroundings or noticing side events late. The split is item-shaped rather than person-shaped: "I still felt aware of my surroundings" runs 17 of 20 in the filter's favour, while "I noticed side events later than I would have" runs 9 of 20 and "I was worried I would miss something important" 10 of 20. Visual comfort sits at 55 of 80 across its four answered items. The focus-benefit items on holding attention, looking away and knowing where to look were answered favourably by 19, 18 and 19 of the twenty respondents. At the earlier ten-participant analysis these three were unanimous and they no longer are, which is a useful reminder of how little weight a unanimous result at that sample size could carry. One respondent dissents strongly across the focus-benefit block, answering one of its five items favourably, so the construct is not uniformly endorsed even where it is strongest. Figure 5.7 shows the three constructs side by side.

The subjective impression of lost awareness is not supported by the behavioural hit rate, which shows no measured loss and, in the 20–30° band, an improvement. Two explanations are worth taking seriously rather than explaining away. First, noticing that the periphery has been changed is a different experience from missing something in it, and the DR manipulations studied to date have all been reported as noticeable (Sutton et al., 2022). "I can tell my vision has been altered" and "I failed to detect something" are not the same report, even when only the first is true. Second, narrowed-field anxiety is documented independently of task performance, rising in every restricted-field condition tested in the literature, including conditions where spatial learning is unaffected (Barhorst-Cates et al., 2016). Reassuring people directly, for instance with a visible cue

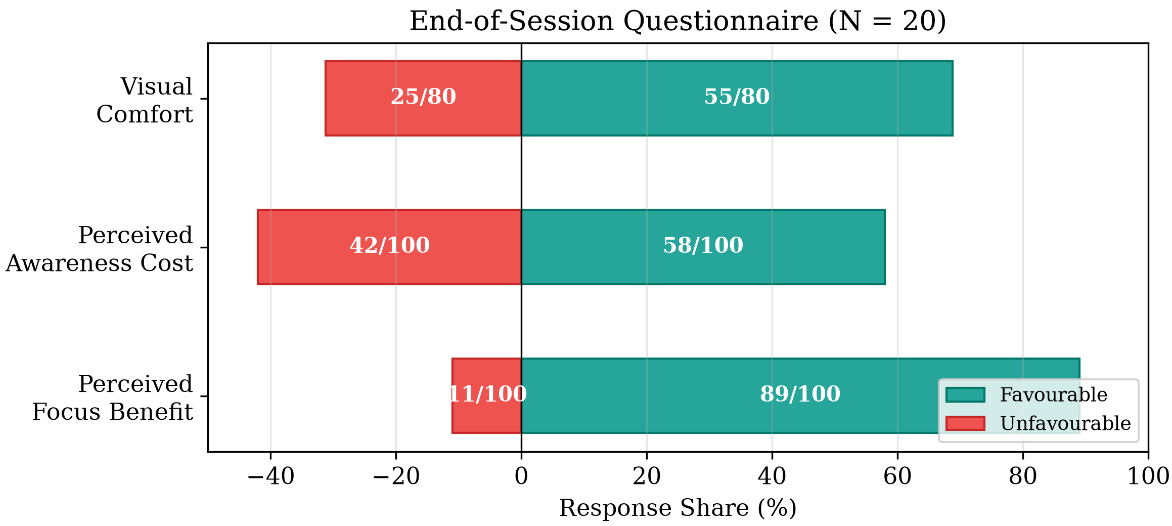

Figure 5.7: End-of-session questionnaire responses by construct (N = 20). The visual-comfort total covers its four answered items rather than five, one item having been skipped by every respondent.

when something happens outside the window, would reintroduce exactly the salience the filter exists to remove. That is the trade-off in its sharpest form, and it is left unresolved here.

5.7.6 What this result does and does not show

Read against the decision table, the study lands on the working-system end of the range the literature left open, with its verdicts stated at their honest strength. The primary contrast is a real 3.07-point accuracy gain, at close to the sensitivity the design planned for, on a real-hardware task with real physical distractors, surviving both a parametric and an assumption-free test, claimed as a bounded benefit because its interval spans the pre-set half-the-errors bar (row 4). Peripheral awareness showed no measured loss: the pooled measure rose rather than fell, and the worst case the data support at 90% confidence is a gain of about one and a half points against a ten-point loss margin. Raising pooled detection across the whole field was never SignPop's job, so the pooled rise stands as a secondary description with its response-criterion qualification, not a claim the design depends on. And detection improved specifically where the mechanism predicted it would, in the pre-registered 20–30° band, at 89.0% achieved power.

Two qualifications bound all of it. The pooled Block B estimate carries the response-criterion qualification of Section 5.7.2. And the bundle limitation of Chapter 4, Section 4.5.5 applies in full, so nothing above attributes the 20–30° gain to any one of the six effects a detection fires. Chapter 6 takes up what follows.

Chapter 6

Discussion and Conclusion

6.1 The question, revisited

This dissertation asked one question with two halves: *can subtractive attention guidance be delivered on consumer XR hardware, and when it is delivered, does it actually help people focus?*

The first half is answered in the affirmative, with precision about what delivered means. On the Meta Quest 3 the question reduces to a compositing-rights problem, and the access an application has stratifies into three tiers: colour-only styling of the OS-owned passthrough layer, overlay compositing on top of it, and full pixel control over a lower-quality camera feed. That narrow space turns out to be sufficient for a working multi-mode system, built around an architectural inversion in which the focus region is rendered as an *absence* in an overlay, so native passthrough at full quality serves the region that matters while the treatments occupy the periphery around it. The hybrid composite at the heart of the most capable mode survived a structured prior-art search with no published or shipped precedent found. The architecture's two central trade-offs were then measured on the device itself: the camera-free dark modes cost nothing measurable over an empty passthrough scene, and the native path the window preserves resolves at least two logMAR steps finer than the camera re-render it spares the user from (Chapter 4, Section 4.11).

The second half is answered by the study, completed at twenty people tested: the pre-registered twenty through the full two-block protocol, giving nineteen workstation pairs and seventeen driving pairs, with the decision table applied as committed. The answers: a 3.07-point accuracy gain on a real-hardware task, real, significant, and at close to the planned sensitivity, but smaller than the pre-set half-the-errors bar, so claimed as a bounded benefit under the table's fourth row rather than at the pre-set size; peripheral awareness with no measured loss, the safety hypothesis's refuting observation never occurring and the pooled measure instead rising, though the two-sided equivalence statistic itself was not established, for the one-directional reason set out in Chapter 5, Section 5.7.2; and a detection benefit appearing in the 20–30° band the system's own detection-gated geometry predicts, supported at 89.0% power and bounded by the band-assignment proxy of Chapter 5, Section 5.7.3. Every outcome was mapped to a pre-committed conclusion before the data existed, and the outcome that did occur is a specific, mechanistically coherent one: the system helps, by a real amount whose exact size a larger sample would pin more tightly.

RQ3's answer does not simply echo the behavioural one. Perceived focus benefit is strongly positive, but perceived awareness cost splits close to evenly, with roughly four in ten responses reporting feeling cut off from the surroundings or noticing side events late (Chapter 5, Section 5.7.5). The behavioural hit rate shows no such loss, so the subjective impression and the measured outcome disagree, and that disagreement is reported rather than resolved.

6.2 What the design space contributes, independently of the data

The central finding of Part T does not await any result. It is the shape of the design space. On a consumer VST headset, diminished reality is not one problem but three, indexed by compositing access. At Tier 1 the platform grants global colour remapping and nothing spatial: no per-pixel masks, no world-locked regions, no read access, and, since the deprecation of surface-projected passthrough at SDK v83, no per-surface control either. The significant fact is the *absence*: a developer who wants to dim the periphery and spare a window cannot do it at Tier 1 at all. That absence is a platform-policy finding rather than an engineering one, and it arguably shapes what DR research is possible on consumer hardware more than any algorithm does.

The system's response is an existence proof that windowed subtractive DR is achievable at Tier 2 without OS cooperation, and that hybrid Tier-2/Tier-3 composites are achievable with Passthrough Camera Access. The composite, precisely scoped, is the novelty claim, and the documented search protocol is what entitles the dissertation to make it.

Four engineering lessons generalise beyond this project, and Chapter 4, Section 4.10 records each with the failure that produced it. **Sample by world direction, not screen space**, when re-rendering a mono camera feed, because only an eye-independent sampling domain fuses binocularly. **Exploit the quality asymmetry**: on any platform where the accessible feed is worse than the native view, put the manipulation where the eyes are not, so the task-relevant region never pays the quality tax. **Decouple effects from head motion**, inverting the common default of effects that follow the head into one that yields to it, because head-coupled restriction increases discomfort (Norouzi et al., 2018). And **consumer-platform DR research inherits platform-policy risk**: surface-projected passthrough, a capability this project might have built on, was deprecated mid-project. Tier boundaries on consumer hardware are set by vendor policy and move without notice, so designs that keep a working path at every tier are more robust to that movement than designs premised on a single capability.

6.3 Limitations

The limitations below are collected from Chapters 3 to 5 so they appear once, together, rather than diluted across the text.

1. **Mode × scenario confound.** Hard Dark is tested only in the workstation context and SignPop only in the monitoring context, because the three-mode sampler was cut from the running protocol. Every empirical claim is mode-in-context. The design cannot separate mode effects from scenario effects, and nothing in it mitigates this.

2. **SignPop's manipulation is a bundle.** A confirmed detection switches on six effects at once: colour pop, window carve-out, saturation lift, brightness lift, contrast expansion and darkened surround (Chapter 4, Section 4.5.5). None is isolable in the current design, so no measured benefit can be attributed to any one of them.

3. **Soft Dark carries no participant data at all.** It was never placed in front of a participant. Its place in the dissertation rests on design-space grounds and on its shader path, never on evidence about how it is experienced.

4. **The oracle gap was never measured.** SignPop is detector-dependent, and the detection driving it is an offline full-resolution bake rather than a live on-device pipeline. How far a live detector would fall short of that oracle, in recall, throughput and latency, is unquantified. Every detector-conditioned statement inherits that qualifier.

5. **The technical measurements cover two properties, not five.** Frame cost and legibility are measured (Chapter 4, Section 4.11); end-to-end latency, world-locking stability, and thermal endurance remain argued from structure rather than measured. Within what was measured, one criterion was missed: the video scene's

stale-frame rate exceeds the pre-stated 1% bound in both study arms equally, a property of the shared 4K playback path rather than of any treatment. The legibility read is the author's, from a cast recording, not a psychophysical procedure.

6. **No eye tracking.** Quest 3 has none. Head pose is a validated proxy at the eccentricities used (Higgins et al., 2022; Sitzmann et al., 2018), but covert attention shifts, distraction without head movement, are invisible to telemetry. The dependent variables were chosen so that *performance outcomes* rather than attention allocation carry the hypotheses, so the mechanism remains partially inferred.

7. **Sample, power, and Block B's response-criterion qualification.** The pre-registered twenty completed the full two-block protocol, and the primary hypothesis and the 20–30° band sit at 79.8% and 89.0% achieved power, two-tailed as registered against the design's 80% target. Block B lost three pairs to its response-validity and comprehension exclusions, all reported in Chapter 5 and including one discretionary case, made by experimenter decision on the day and disclosed as such. The pooled Block B estimate carries a response-criterion qualification and sits at 54.0% achieved power (Chapter 5, Section 5.7.2), and it is the one headline estimate that does not survive leave-one-out, holding significance in seven of seventeen such samples where the primary contrast and the pre-registered band survive every single-participant removal (Chapter 5, Section 5.7.4).

8. **Band membership is assigned by a proxy.** The eccentricity bands carrying H2a are assigned from each target's mid-lifetime position rather than its position at the instant of the press, because the press-time reconstruction is not implemented in the analysis pipeline. Targets move while visible, the median lifetime swing is 17.2°, and roughly 31% of catches land in a different band from the one the mid position assigns (Chapter 5, Sections 5.5 and 5.7.3). The band result is the dissertation's largest single effect, and this is the measurement caveat that bounds it.

9. **Workload is not among the measured constructs.** Subjective response was collected with a purpose-built end-of-session questionnaire covering perceived focus benefit, perceived awareness cost and visual comfort, the first two written as the subjective counterparts of the two blocks' own questions (Chapter 5, Sections 5.6 and 5.7.5). Task demand in the NASA-TLX sense is not one of those constructs and was not collected separately per condition, so the study reports what participants thought of the effects but makes no claim about whether the vignette made the task feel less effortful as well as more accurate. Separately, the distractor reel's cost is taken from the published literature (Ai et al., 2025) rather than verified internally.

10. **Single experimenter, single site, single hardware generation, first exposure.** Scripts and checklists bound experimenter variance but cannot eliminate it, and every number is specific to Quest 3 and the frozen OS and SDK versions. Practice-to-criterion and counterbalancing distribute first-exposure effects but do not abolish the fact that every participant meets peripheral diminishment for the first time that hour. The workstation distractors are video panels on the task display and the monitoring stimulus is a video: the study measures attention mechanisms under controlled distraction, deliberately, and its conclusions end where those controls end.

11. **No demographic data.** The post-session questionnaire carries no age, gender, vision or background items, so participant demographics cannot be reported. What is reported instead is the sampling frame: adults recruited by direct approach from the postgraduate computer-science cohort, unpaid, under the inclusion criteria of Chapter 5, Section 5.3. The frame is narrow, and findings generalise from it accordingly.

12. **The detection system reads position and size, never motion.** A detection aperture responds to an object's angular position and apparent size per frame, with no velocity or trajectory term, so an approaching pedestrian and a stationary one subtending the same angle are treated identically. Approach speed is exactly the cue a driving-relevant deployment would need. Relatedly, the aperture budget (eight live, sixteen in the video path) fills in arrival order with no prioritisation, so a scene dense enough to exhaust it would degrade by dropping the newest detection rather than the least important one. Neither cap was reached in any session, a property of the footage rather than a guarantee.

6.4 Transfer: what leaves this dissertation, and under what conditions

To optical see-through hardware, the dark overlay does not transfer. The Tier-2 modes work because a VST display is an opaque screen under full render control, and additive OST optics cannot darken the world. On OST, "draw black" means "draw nothing". What transfers is the taxonomy and the question. On OST hardware the taxonomy collapses toward display-side attenuation, of which segmented dimming hardware such as the Magic Leap 2's dedicated low-resolution dimming panel (Magic Leap, 2024) is the nearest existing analogue of a Tier-2 windowed dim, and toward the additive salience modulation demonstrated by Sutton et al. (2022). Critically, the **human-factors findings are expected to transfer**, because what peripheral diminishment does to attention should depend on the human visual system more than on the display that implements it. The engineering is platform-bound. The psychology should be far less so, though display properties such as field of view and latency could still moderate the effect. A replication of the Chapter 5 paradigm on a dimming-capable OST device would be the cleanest test of that expectation.

From the simulation, transfer is conditional, and the conditions are enumerable. Monoscopic video removes motion-in-depth, so peripheral capture by looming stimuli is untested. The passenger viewpoint removes task coupling, which is why no statement here concerns driving performance. Video dynamic range compresses glare, so SignPop's glare-suppression track is exercised only weakly. And identical stimuli for all participants trade ecological breadth for internal validity, a trade that is intended. The workstation block, running on the real system against physical distractors, is the one component that transfers without these caveats, which is why it carries the primary hypothesis.

The two blocks also carry an incidental methodological payload. They run the same guidance principle at opposite ends of the feasibility spectrum, so if they agree in direction the simulation methodology that the nearest published work relies on entirely (Cheng et al., 2022; McLaughlin et al., 2025, both VR-simulated) gains external credibility from this project's real-hardware anchor, and if they diverge the divergence localises what simulation misses. Agreement is read at the level of sign rather than magnitude, since the blocks use different tasks, dependent variables and modes, so any cross-block statement is observational rather than powered.

6.5 Future work

Six directions follow directly from the material, in the order their absence limits the present claims.

Separating the bundle. The 20–30° finding is the clearest reason this item leads. A three-arm design, no filter against filter with the detection gate off against filter with it on, would need no new code, since the gate already has a toggle. It would directly separate "dimming helped" from "passing the signal through helped".

Closing the oracle gap, and finishing the measurement campaign. SignPop is detector-dependent on offline oracle-quality detection. The natural next iteration replaces the offline gate with a live detector and re-runs the driving block to test whether the measured benefit survives the accuracy and coverage gap. That programme also needs the measurements this dissertation did not take: end-to-end latency, world-locking stability, and above all thermal endurance, since the frame-cost results of Chapter 4, Section 4.11 show the Tier-3 path holding frame rate comfortably while saying nothing about holding it for an hour. A prioritised aperture budget belongs to the same iteration: ranking detection candidates by urgency, for example eccentricity, apparent size and time-to-contact, and spending the fixed slots on the top few, so a crowded scene degrades by dropping the least critical detection rather than the newest. There may be no clean solution at high density, since a filter that opens twenty holes has stopped filtering, and at some density the honest behaviour is to disable the effect rather than dilute it.

Dose–response. The study tests only the extreme of the attenuation axis. Soft Dark, the midpoint, was built but never tested, so the middle of the axis has no data behind it. A discrete-levels design, off against partial against full attenuation, counterbalanced, would establish the shape of the benefit curve and answer the deployment question this dissertation cannot: how much diminishment buys how much focus, at what comfort cost.

Three further directions follow less directly but are seeded by the existing architecture. **Adaptive introduction**: rather than switching the effect on, a deployed system might introduce it gradually as focus wanes, a question the formation animation and motion-coupled fade mechanics already make tractable, though as a validation instrument a ramp confounds time with intensity and was excluded here. **Hour-long deployment**: the camera-free dark modes are the only plausible basis for session-length use today, and a longitudinal study with workload and comfort instruments administered across days is the bridge between controlled findings and any claim about practice. **Saliency-weighted diminishment**: the review identified dimming weighted by computed scene saliency as unstudied in AR attention guidance, and since the shader already computes per-fragment treatment, driving its intensity from a saliency model of the periphery is a natural composition of the Tier-3 path with the saliency-modulation literature.

6.6 Closing

The premise of this work was that a consumer headset can be an instrument for *removing* the world as well as adding to it, and that both halves of that proposition deserve rigour: the systems half, where an OS-owned display layer had to be subtracted from by purely additive means, and the human half, where "it helps you focus" had to be converted from a hope into a hypothesis that could fail.

The evaluation apparatus built to do that is part of the contribution. A single primary endpoint with a smallest effect size of interest, equivalence margins for safety claims, a gated pilot whose failure aborts rather than degrades the main study, and a decision table committed before data collection, is not yet standard practice at masters-project scale. The apparatus costs little, since its artefacts are documents and thresholds rather than equipment, and it converts every outcome into a claim with stated bounds. One of those bounds turned out to bind on the study itself: the safety hypothesis was written as a two-sided equivalence test when the question it encoded had only one side, and Chapter 5 reports both the pre-registered statistic and the mismatch rather than quietly substituting the better-fitting one.

The system exists and runs on target hardware. The hypothesis was tested, on nineteen workstation pairs and seventeen driving pairs, and answered under the rules committed to before the data existed: a real focus benefit, claimed at its bounded size rather than its hoped-for one; peripheral awareness with no measured loss; the benefit appearing exactly where the geometry said to look. The question was asked on the hardware people actually own, and that is the standard the answer inherits.

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Appendix A

Study Protocol Artefacts

This appendix holds the operational material of the Chapter 5 protocol: the verbatim scripts read to participants, the pass/fail gates the pilot had to clear before the main study launched, the session timeline, the analysis plan and decision table, the operational protocol, the threats to validity, and the pre-registered exclusion rules. It is separated from Chapter 5 so the chapter can argue the design while the artefacts that make it reproducible remain available in full.

A.1 Block A participant instructions (verbatim)

> "A panel in front of you will show one shape at a time. **After the first shape, answer whether each new shape is the same as the one before it**, one controller trigger for yes, the other for no. Respond as fast as you can without guessing. Ignore everything else in the room, only the panel matters. Each round lasts about four and a half minutes. There are two rounds. The rounds may look different from each other. Just do the same thing every round."

Participants are never told which conditions belong to "our system" or what the vignette is expected to do. The session framing throughout is that they are "comparing display modes", a defence against demand characteristics (Section A.8).

A.2 Block B participant instructions (verbatim)

> "You'll watch a driving scene. Small **ring markers will appear briefly** on things in the scene, sometimes in front of you, sometimes off to the side. **Press the trigger as soon as you notice one. That's the only thing**

you need to do. Don't press for anything else. Keep watching the road ahead as if you were the driver, and stay seated."

A.3 Pilot study with pass/fail gates

A pilot with n = 3–5 participants (excluded from the main sample) runs in the first week. Its purpose is structural: to settle the task and stimulus parameters so that every main-study outcome maps to a pre-committed conclusion. The gates govern tuning rather than launch: where a gate is not cleared, the minimal viable adjustment is made and collection continues, with the shortfall recorded and carried into interpretation. Pilot data never enters the main analysis. Stimulus parameters may be tuned between pilot participants, but hypotheses, margins, and analysis choices may not.

Table A.1: Pilot study pass/fail gates, tests, criteria, and actions on failure.

Gate	Test	Pass criterion	On fail
G1: task form	The forced-choice 1-back tried through passthrough	Panel fully legible; accuracy off both ceiling and floor across pilot participants	Retune timing/difficulty (SOA, answer window, lure ratio) against pilot data, then re-test. Redesign the task only if retuning cannot bring it off ceiling or floor
G2: marker calibration	Block B with candidate marker fade timing and size	Baseline hit rate 70–90% across the sample; unimodal RT distribution	Adjust marker timing/size, then re-test
G3: protocol dry run ×2	Two full end-to-end sessions with the final configuration	Timeline within ±5 minutes; zero log gaps (validator clean); battery ≥ 20% at end; no sickness terminations; questionnaire flow smooth	Fix the specific failure, then re-run one dry run

Gate outcomes. G2 passed at pilot; across the full sample the NoFilter baseline converged lower, settling at 50.15%. G1 was not cleared. Block A accuracy sat on the ceiling the gate guards against; the trial count was raised once, to 140, in response, and the achieved data show the adjustment partly worked, moving the mean from 96.1% to 92.5% and the floor from 84.6% to 74.8%, so the measure has room to move in both directions but remains above the range the gate intends. Collection continued on the tuned form under schedule constraint rather than re-piloting, under the rule in row 5 of the decision table. Both shortfalls are recorded here rather than left to inference, and both run in the direction that preserves headroom: the lower Block B baseline removes any ceiling explanation for the 20–30° band result, and the residual Block A ceiling is among the reasons H1's effect is reported as small and bounded. G3 was cleared: both dry runs completed end to end on the final configuration. They were rehearsals rather than data collection, so the outcome is recorded observationally.

The pilot concludes with a one-page **lock-in memo** to the supervisor recording the chosen task form, the tuned stimulus parameters, and the finalised absolute SESOI for H1. After this memo, nothing changes until data collection ends, with one exception that is recorded as a deviation and reported in Chapter 5, Section 5.7: the Block A trial count was raised once, after five participants had run, to move task accuracy off the ceiling the short form sat on.

A.4 Pre-registered exclusion and data-quality rules

Table A.2: Pre-registered exclusion and data-quality rules.

Rule	**Action**
VR-sickness termination	Session ends under the stop rule; partial data retained but participant excluded from confirmatory analysis; replaced from spares
App crash / restart during a condition	That condition void; re-run once at the end of its block if time allows; otherwise the participant contributes remaining conditions to the LMM (which tolerates missingness) and is excluded from the paired robustness test

Rule	**Action**
App crash / restart spanning a **new session launch**, since the ledger assigns a fresh participant ID on relaunch rather than resuming	The two IDs are reconciled into one canonical participant record before any analysis is run. Cleanly completed blocks are joined under the canonical ID, and a block attempted under both keeps only the later complete attempt. The join, both IDs, and the ledger timestamps justifying it are recorded in the incident log and not revisited afterwards
Task accuracy < 60% in NoFilter	Task not performed; exclude and replace
Block B: false alarms across both conditions together outnumber the pair's eighty scheduled markers (see the note below this table)	Response strategy invalid (indiscriminate pressing); exclude Block B for that participant
Log integrity failure (gaps > 1 s or missing condition markers)	Affected condition void; incident log entry; same re-run rule as crashes
Participant recognises the driving footage	Noted; excluded only on reported strong familiarity affecting behaviour

Note on the response-validity threshold. The Block B screen is absolute and is assessed on the pair as a whole rather than on either arm alone: a participant's Block B record is excluded when its false alarms across both conditions together outnumber the eighty markers the pair schedules, forty per condition (Section 5.5). The combined basis is deliberate. Which arm draws the heavier pressing is not predictable before the data exist, so a per-arm screen would amount to a threshold applied to whichever side happened to run hotter; taking the pair as the unit removes that choice. The level follows from the task's structure rather than from the data: a participant cannot legitimately register more false presses than the task asked them to make responses for, counting both arms. The achieved data sit clear of the line on both sides, the highest retained pair holding 66 false alarms against 135 for the lowest pair excluded under it, and any threshold between those two values selects exactly the same set, so no verdict here turns on the precise figure. The one discretionary exclusion, P9, is made on task-comprehension grounds and not on false-alarm count, and is not covered by that separation.

A.5 Session timeline

The full session runs ~48.5 minutes, inside the 40–60-minute tolerance window participants typically accept. Headset-on time is ~35 minutes of that. The information sheet is sent in advance, and the closing questionnaires are done with the headset off. The block order shown below is nominal: the two confirmatory blocks are counterbalanced across participants, so for about half the sample the driving block ran first and the two Block B rows precede the Block A rows.

Table A.3: Session timeline: clock time, duration, and activity for the full ~48.5-minute session.

Clock	Duration	Activity
00:00	4 min	Welcome; written consent (information sheet pre-read); VR-experience form; VRSQ (pre)
04:00	4.5 min	Headset fitting and comfort check; tutorial: paint/lock a practice window; practice to criterion ($\geq$ 90% accuracy on a 1-minute practice stream; marker practice: 6 markers, $\geq$ 4 hits). One repeat allowed.
08:30	1 min	**Block A**: paint and lock focus window; experimenter verifies geometry
09:30	4 min 40 s	Condition i (Filter or NoFilter, order counterbalanced)
14:10	1 min	Reset (distractor reel re-cued, window geometry re-verified)
15:10	4 min 40 s	Condition ii
19:50	0.5 min	Noticing question
20:20	1.5 min	Headset-off break; verbal comfort check (stop rule applies throughout); water offered
21:50	8 min 39 s	**Block B**: condition i (Filter or NoFilter, order counterbalanced), the study clip in full
30:30	0.5 min	Reset
31:00	8 min 39 s	Condition ii, the same clip in full with the other marker set
39:39	0.5 min	Fatigue and comfort check; headset off
40:09	1 min	VRSQ (post)
41:09	5 min	**Block C**: end-of-session questionnaire (ESQ), self-completed
46:09	2.5 min	Study-aims debrief; thanks

Clock	Duration	Activity
48:39	n/a	Experimenter runs the session validator **before the participant leaves**. Incident log entry if anything deviated

Arithmetic: opening 8.5; Block A 11.8 (1 + 2×4.67 + 1 + 0.5); break 1.5; Block B 18.3 (2×8.66 + 2×0.5); close 8.5 (1 VRSQ + 5 ESQ + 2.5 debrief). Total 48.6 minutes.

The practice-to-criterion tutorial serves a specific validity function beyond familiarisation: the practice includes the vignette itself, so its first appearance is never during a measured trial, neutralising first-exposure novelty effects at the point where they would otherwise contaminate data.

A.6 Analysis plan and decision table

All analysis will be performed in Python (pandas, pingouin, statsmodels) or R, with scripts written and dry-run on pilot data before main collection begins. $\alpha = .05$ throughout. Effect sizes and confidence intervals are reported for every test: 95% two-sided by default, 90% where one-sided or equivalence logic applies (the H1 SESOI bound and the H2b TOST).

A.6.1 Confirmatory analyses

Table A.4: Confirmatory analysis plan: primary test, robustness check, and effect size per hypothesis.

ID	Primary test	Robustness check	Effect size
H1	Linear mixed model on trial-level errors (logistic LMM: error $\sim$ vignette + (1 | participant)); the vignette term is the test	Paired t-test on per-participant error rate, NoFilter vs Filter; Wilcoxon signed-rank if Shapiro–Wilk rejects normality	dz on the paired difference; odds ratio from the LMM
H2a	Paired t-test (Wilcoxon fallback) on per-participant hit rate in the 20–30° eccentricity band, Filter vs NoFilter	LMM on trial-level hit/miss within the band (logistic)	dz

ID	Primary test	Robustness check	Effect size
H2b	TOST on hit rate pooled across the whole field of view (paired), margin $\Delta = 10$ pp (Lakens, 2017)	Bayesian paired test as an optional sensitivity check	Mean difference with 90% CI against the margin

Mixed-effects modelling at trial level follows the analysis approach of McLaughlin et al. (2025). The aligned-rank-transform alternative for nonparametric factorial data (Wobbrock et al., 2011) is noted but not preferred, in light of recent critiques of ART's error control (Tsandilas and Casiez, 2024).

Multiplicity policy. H1 is the single primary endpoint and receives no correction. H2a and H2b form the secondary family, Holm-corrected across the two; the correction is applied and reported in Chapter 5, Section 5.7.3. Everything else, the end-of-session questionnaire and the head-turn telemetry, is labelled exploratory and is never quoted as a confirmatory finding.

Power. For the primary contrast, computed per participant as a paired difference in error rate between NoFilter and Filter, a paired t-test at $\alpha = .05$ (two-tailed) with N = 20 achieves 80% power for dz $\approx$ 0.65. The two-tailed convention is deliberate and applies to every confirmatory test. The hypotheses are directional and a one-tailed test would be more sensitive to the improvement they predict, but it would also be unable to register a decrement of any size. This study asks whether the treatment costs peripheral awareness as well as whether it helps focus, so an adverse outcome is kept detectable rather than defined out of the test. Directional logic is confined to the places it was registered for: the one-sided SESOI bound on H1 and the equivalence and non-inferiority readings of H2b. The distractor-driven error rate underlying this comparison is large in the source literature (commission errors more than doubling at N = 66), so a vignette recovering half of it is plausibly detectable at this N. The pilot verifies the locally achievable error rate before the main study commits, and if it proves small, the SESOI conversation with the supervisor happens *before* data collection rather than after. Trial-level mixed models over $\sim$5,600 Block A trials provide additional sensitivity beyond the aggregate-level power figure.

A.6.2 Exploratory analyses

This subsection states the exploratory plan as pre-registered; the note at the end of it records which parts were carried out. End-of-session questionnaire items are to be reported per construct as medians with interquartile ranges, with the open answers quoted rather than coded, since a single end-of-session opinion per participant supports description and not testing. Head-turn counts and dwell toward the distractor panels (Block A) and head angular speed (Block B) are to be analysed descriptively with paired tests, flagged exploratory. Interview responses are to be thematically coded into a compact table (mode $\rightarrow$ recurring likes and concerns), with quotes used to illustrate the quantitative findings. H1 is to be re-estimated within VR-experienced and VR-naïve subgroups as a robustness description, not a test.

Of these, the questionnaire analysis was carried out and is reported in Chapter 5, Section 5.7.5. The head-turn telemetry, the interview coding, and the VR-experience subgroup re-estimation were not carried out for this dissertation, so no result is reported for them.

A.6.3 Decision table

The following table was pre-committed before data collection. Collection completed at twenty people tested, the pre-registered twenty through the full two-block protocol, giving nineteen workstation and seventeen driving pairs, and the table is applied to that final sample. **The row that occurred is row 4** for the primary hypothesis: the estimate is positive, significant, and powered close to the design target (79.8% two-tailed against an 80% target, on the convention this power calculation was set in), but its 90% interval spans both the smallest effect size of interest and smaller values, so the benefit is reported as a bounded estimate rather than claimed at the pre-set size. On the safety side, H2b's refuting observation did not occur and the non-inferiority reading passes decisively; the formal two-sided equivalence statistic is not established, for the one-directional reason set out in Chapter 5, Section 5.7.2. H2a is supported in its pre-registered band.

Table A.5: Pre-committed decision table mapping outcome patterns to dissertation conclusions.

#	Outcome pattern	Dissertation conclusion
1	H1 supported (paired difference in predicted direction, $\geq$ SESOI) **and** H2b passes equivalence	**The system works**: peripheral DR dimming reduces distraction cost, and the salience mode does not measurably harm peripheral awareness.

#	Outcome pattern	Dissertation conclusion
2	H1 supported, H2b **fails** (drop > 10 pp)	The system works **for focus, at a quantified situational-awareness cost**, a conclusive, qualified result with a design implication (mode choice by context).
3	H1 **refuted** (90% CI on the paired error-rate difference excludes the SESOI from below)	**The system does not deliver a meaningful benefit** in its core use case, a conclusive negative. The dissertation reports the bounded estimate ("the vignette reduces the error rate by at most X%").
4	H1 estimate positive but CI spans both SESOI and smaller values	The benefit is real-signed but **cannot be claimed at the pre-set size**. Report the estimate with its CI as a bounded conclusion. (If it occurs, it is reported as such, not spun.)
5	Pilot gate not cleared after tuning	**The minimal viable adjustment is made and collection continues.** The adjustment is recorded as a deviation, and the gate's residual effect is carried into interpretation as a bound on what the affected measure can show rather than left implicit.

Interpretation of whichever row occurs, including the transfer limits of a positive result and the design implications of a negative one, is developed in Chapter 6.

A.7 Operational protocol

Three properties of the running protocol bear on whether the data can be trusted.

The session is deterministic. Every distractor clip, marker onset, and task trial derives from versioned schedule files in the repository, and condition order derives from the participant ID, so a session is reproducible from those two facts alone. No experimenter arithmetic happens mid-session and no live machine learning runs anywhere. The sequencer asserts the required scene configuration at start and refuses to run on mismatch, which matters because two settings that Block B depends on are off in the stock scene:

motion suppression, and baked-detection playback, without which SignPop's colour pop, window, and saliency boost do nothing at all.

Logging is redundant. Telemetry writes one CSV row per frame at roughly 60 Hz, flushed every frame, so an unbroken timestamp chain is itself evidence of integrity and a crash costs under a second of data. Every event is a labelled row. A `scrcpy` screen recording gives a second channel against which any disputed trial can be re-checked.

Data are validated before the participant leaves. A validator checks timestamp continuity, condition counts and durations, and per-condition event counts, for example exactly 40 scored markers per Block B condition, returning PASS/FAIL while the participant is still in the room, so a voided condition can be re-run under the pre-registered rule rather than discovered a week later. Every deviation goes to an incident log with the rule that applied, and those are reported rather than hidden. One deviation is recorded against this study: the Block A trial count was raised once, to 140, after five participants had run, to move task performance off the ceiling the short form sat on. It is reported with its effect on the achieved accuracies, and the pooling it requires is justified, in Chapter 5, Section 5.7. No other change was made after collection began.

A.8 Threats to validity

Table A.6: Threats to validity and their mitigations.

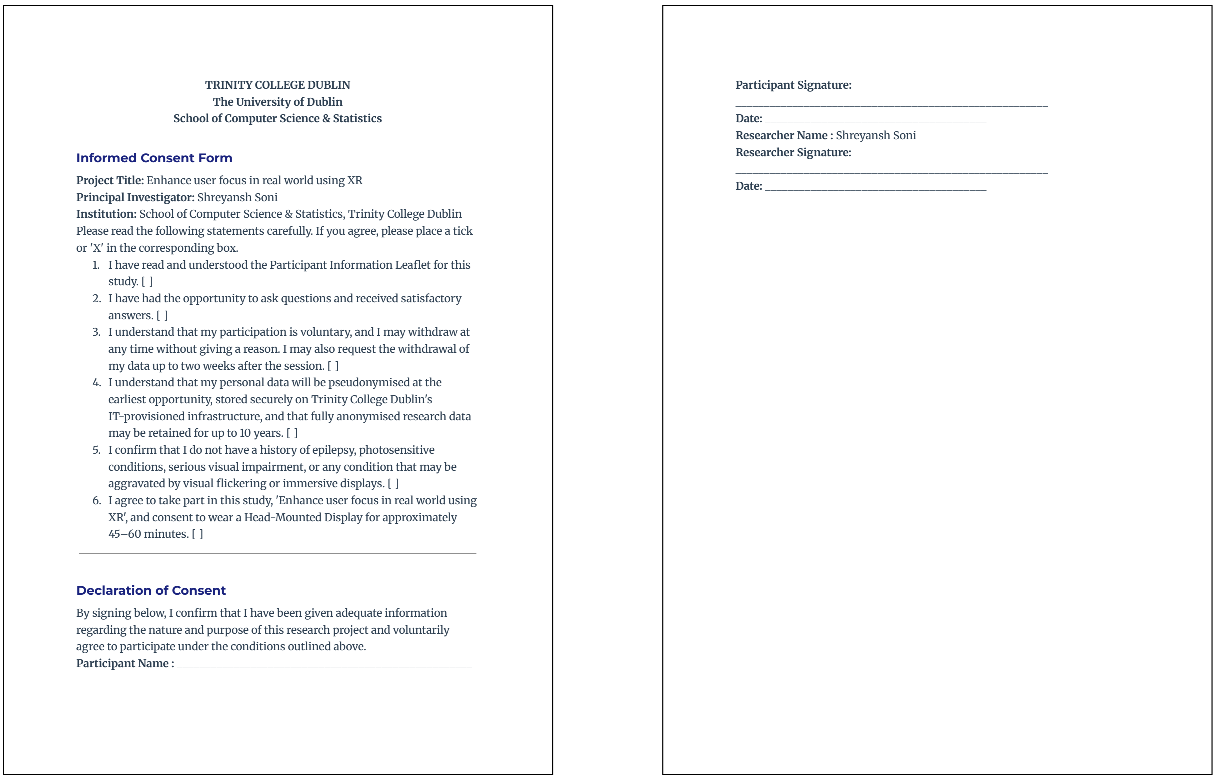
Threat	**Mitigation**
Passthrough acuity confound (Quest 3 VST below normal acuity, worse in low light)	No fine-text reading through passthrough anywhere; the task uses large look-alike shapes ($\geq 3°$) seen through the window's untouched native passthrough; legibility confirmed at G1; bright constant lighting on the checklist
Novelty effect	Practice-to-criterion tutorial including the vignette before any measured trial; counterbalancing distributes residual novelty

Threat	**Mitigation**
Order and learning effects	Parity counterbalance in both Block A and Block B; matched sheets/segments; equal trial counts. Block order is counterbalanced across participants as well as within each block, so neither block is systematically second and cross-block fatigue cannot load onto one of them; about half the sample ran the driving block first
Coverage limits from the session budget	The three-mode sampler was cut, so Soft Dark was never shown to a participant and no participant saw two modes on a common stimulus (Chapter 5, Section 5.6); the mode $\times$ scenario confound is therefore unmitigated, and subjective data is one end-of-session opinion per participant rather than a per-mode comparison
Workload not measured	Subjective response is collected through the purpose-built end-of-session questionnaire, covering perceived focus benefit, perceived awareness cost and visual comfort, with the first two written as the subjective counterparts of the workstation and driving blocks' own questions (Sections 5.6 and 5.7.5). Task demand in the NASA-TLX sense is not among those constructs and is not collected separately per condition, so no claim is made about whether the vignette made the task feel less effortful as well as more accurate
Demand characteristics	Objective primary DVs (errors, millisecond RT, telemetry); neutral "comparing display modes" framing; the noticing question follows Block A data collection
Distractor floor (distractors fail to distract)	Relies on the established distractor-cost literature cited in Chapter 5, Section 5.1 (Ai et al., 2025, N = 66) rather than an internal manipulation check; not independently verified within this study
Simulator sickness and dropout	Seated, no locomotion, ~35 min headset time, mid-session break, VRSQ pre/post, stop rule, N + 4 recruitment

Threat	Mitigation
Small-N insensitivity	Within-subjects design, high trial counts, mixed models; single primary endpoint; SESOI and equivalence margins make null results informative; effect sizes and CIs always reported
Experimenter variability	One experimenter; verbatim scripts; printed per-session checklist
Ecological validity (driving is a video; workstation distractors are on-screen video panels)	The claim tested concerns attention mechanisms under controlled distraction, with driving footage as stimulus context, explicitly not a road-safety claim (Chapter 1 scoping rules); developed further in Chapter 6
Stereo comfort of camera-fed effects	The dark modes are overlay-only (no camera repaint); SignPop runs in the video scene (no live camera fusion, and its detection gate is driven by the offline oracle track, never a live model); head motion is moderate by task design
SignPop's manipulation is a bundle	A confirmed detection switches on six effects at once: colour pop, window carve-out, saturation lift, brightness lift, contrast expansion, and darkened surround (Chapter 4, Section 4.5.5). So H2a and H2b cannot attribute any measured effect to one component. This is not mitigated in the current design. Chapter 6 proposes a three-arm follow-up (no filter, filter with detection off, filter with detection on) to separate them

A.9 Ethics artefacts

The study was approved through the School of Computer Science and Statistics research ethics process (REAMS) on 27 May 2026, under the application title *Enhance user focus in real world using XR*, reference number 6061, with the author as principal investigator and John Dingliana as primary supervisor. The blank participant information leaflet (version 2.0, dated 13 May 2026) and the blank informed consent form are reproduced in Figures A.1 and A.2 exactly as given to participants. Signed originals carry personal data and are retained by the supervisor under the School's retention policy rather than submitted with this dissertation. The leaflet lists a per-condition workload questionnaire among the measures. That instrument was not administered, so the study carries no per-condition workload data; participants were asked for their assessment of the effects they had just used, once, at the end of the session, through the end-of-session questionnaire reported in Section 5.7.5.



TRINITY COLLEGE DUBLIN
The University of Dublin
School of Computer Science & Statistics

Informed Consent Form

Project Title: Enhance user focus in real world using XR
Principal Investigator: Shreyansh Soni
Institution: School of Computer Science & Statistics, Trinity College Dublin
Please read the following statements carefully. If you agree, please place a tick or 'X' in the corresponding box.

1. I have read and understood the Participant Information Leaflet for this study. [ ]
2. I have had the opportunity to ask questions and received satisfactory answers. [ ]
3. I understand that my participation is voluntary, and I may withdraw at any time without giving a reason. I may also request the withdrawal of my data up to two weeks after the session. [ ]
4. I understand that my personal data will be pseudonymised at the earliest opportunity, stored securely on Trinity College Dublin's IT-provisioned infrastructure, and that fully anonymised research data may be retained for up to 10 years. [ ]
5. I confirm that I do not have a history of epilepsy, photosensitive conditions, serious visual impairment, or any condition that may be aggravated by visual flickering or immersive displays. [ ]
6. I agree to take part in this study, 'Enhance user focus in real world using XR', and consent to wear a Head-Mounted Display for approximately 45–60 minutes. [ ]

Declaration of Consent

By signing below, I confirm that I have been given adequate information regarding the nature and purpose of this research project and voluntarily agree to participate under the conditions outlined above.
Participant Name : ____________________

Participant Signature:
____
Date: ____________________
Researcher Name : Shreyansh Soni
Researcher Signature:
____
Date: ____________________

Figure A.1: The blank informed consent form.

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Information Leaflet: Enhance user focus in real world using XR

The study title is self-explanatory: Enhance user focus in real world using XR.

Invitation Paragraph

You are being invited to take part in a research study about how visual modifications in an Augmented Reality (AR) headset can help people stay focused in real-world situations.

Participation is voluntary.

Before you decide if you want to take part, it is important for you to understand why the research is being done and what taking part involves.

Please take time to read the following information carefully and discuss it with others if you wish. Please ask us if there is anything that is not clear or if you would like more information.

What is the purpose of the study?

We plan to study whether using shader-driven visual modifications (specifically desaturation, blur, and boundary highlighting) applied via AR passthrough on a Meta Quest headset can effectively guide and sustain user attention in real-world task contexts. The study aims to develop a Diminished Reality system that can improve task-relevant attention focus and reduce distractor salience.

The study is scheduled to run from the start of data collection on 25/05/2026 until the project end date of 14/08/2026.

For the purpose of this study, we will collect the following personal data:

- *Contact Information (Names and email addresses) for recruitment, scheduling, and obtaining informed consent.*
- *Performance Metrics (Task completion times and error rates).*

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- *Attention & Tracking Data (Head direction/tracking data, time-to-first-fixation on distractors, and fixation distribution).*
- *Survey/Questionnaire Responses (Cognitive load scores via NASA-TLX, situational awareness responses, and user preference ratings via Likert scales).*

Why have I been chosen to participate?

We are interested in understanding the experiences of adults in using head-mounted Augmented Reality systems for attention guidance across various tasks (driving, classroom, and workstation scenarios).

You are being invited to participate because you are an adult (18 or over), have normal or corrected-to-normal vision, and are comfortable wearing a head-mounted display for approximately 45–60 minutes. You must not have any known history of epilepsy, photosensitive conditions, or serious visual impairment.

We are hoping to have 20 participants in the study.

Do I have to take part?

It is up to you to decide whether or not to take part. If you do decide to take part you will be given this information leaflet to keep and asked to sign a consent form.

If you decide to take part you are still free to change your mind at any time and without giving a reason. There will be no penalty or loss of benefits should you decide to withdraw.

What will happen to me if I take part?

This study involves attending a single session, lasting approximately 45–60 minutes, which will take place in a quiet, controlled laboratory environment on the Trinity College Dublin campus.

The session will include a 5-minute briefing and consent process, followed by three task scenarios (each lasting approximately 10 minutes) with and without the AR system.

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You will complete post-condition questionnaires (NASA-TLX and Likert scales) to assess cognitive load and user preferences.

Data Handling:

- *All research data will be pseudonymised upon collection using a participant ID code, with the linking key stored separately and securely.*
- *Participants may request withdrawal of their data for up to two weeks after participation. During this period, the linking key will be retained securely to allow identification of participant data if withdrawal is requested.*
- *After the two-week withdrawal period, the linking key will be permanently deleted and the research dataset fully anonymised. After anonymisation, it will no longer be possible to identify or remove individual participant data.*
- *Contact information and signed consent forms will be stored securely on a password-protected university-managed drive accessible only to the researcher and supervisor, and retained in line with Trinity College Dublin postgraduate research retention guidelines.*
- *Fully anonymised research data may be retained for academic publication and research purposes.*

What are the benefits of taking part?

You will gain knowledge about your own attention dynamics and how peripheral distractions may affect your focus in different environments. The results of this study will help us to understand how customised Augmented Reality systems can be developed to act as attention coaches in real-world scenarios.

What are the possible disadvantages and risks of taking part?

We have taken all measures to minimise any risk to you. The primary risk is the potential for mild discomfort or cybersickness associated with wearing a Head-Mounted Display (HMD) for the 45–60 minute session.

In the event of you experiencing significant discomfort, motion sickness, or visual strain, the researcher will immediately stop the session at your request or if they observe signs of

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distress. The researcher will advise you to rest. All information collected will be treated confidentially.

Will my participation in this study be kept confidential?

Your privacy is important to us. All digital data, including performance metrics and survey responses, will be pseudonymised immediately upon collection using a unique participant ID code. The data will be stored securely on Trinity College Dublin's official IT-provisioned cloud infrastructure (Microsoft OneDrive/SharePoint).

The code-key linking participant identities to participant ID codes will be retained securely for up to two weeks after participation to allow data withdrawal requests to be processed. After this period, the code-key will be permanently deleted and the dataset fully anonymised.

Any information that leaves Trinity, for example in a publication, will be fully anonymised so that your identity remains confidential. Anonymised, aggregated data may be shared with the scientific community and industry.

We will never share personal audio or video recordings with any third parties, as no such files are created during this study.

*It is **not** foreseeable that participants could potentially reveal information that the research team has a legal obligation to disclose (e.g., child protection policy, malpractice, or criminal activity). However, please be aware that researchers are legally obliged to report any such inadvertent discovery as required under relevant policy, legislation, or code of ethics.*

What will happen to the results of this research?

The research is being conducted as part of a Taught Masters program and the results will be included in the final dissertation. Fully anonymised data may also be published in future scientific papers or presented at conferences. If this is the case, your identity will remain confidential and no one will know that you took part in the study.

What do I do if I have any further questions?

Figure A.2: The blank participant information leaflet, pages 1–4 (page 5 carries only contact details and version information).